A local proof supplement for the required Gowers inverse theorem

Current source revision, 8 November 2026. Packaging revision 3.

Target. For fixed $k \geq 3$ and $\delta > 0$, every complex one-bounded function on $[N]$ of normalized $U^{k+1}[N]$ norm at least δ correlates with a one-bounded Lipschitz test along a linear orbit on one of a fixed finite family of connected simply connected nilmanifolds of step at most k. The family and positive correlation and Lipschitz bounds are fixed before N.

Review status. The exact assembled conclusion and its current $k = 20, 39$ and proposed fixed $k = r - 1$ consumer interfaces have independent mathematical acceptance relative to the explicitly retained published inputs. Current sources incorporate the reviewed strong-family, rational-subgroup and Taylor-coordinate corrections. Historical module decisions retain their original scope. The combined source-coverage index and its Bertrand and Gaussian addenda have a separate bounded independent decision. Retained published premises remain explicit. No installation or release approval is asserted.

How to read this supplement. The assembled proof appears first. The complete supporting modules follow in the order listed below. Original page numbers restart within each module; PDF bookmarks identify them. The directory contains exact standalone TeX and PDF copies, source-use and application records, independent decisions, and an origin map. Historical author-proposal wording in a module is preserved; its later exact decision is in the review directory.

Published inputs. The seven exact published sources are registered by edition, DOI and SHA256 in SOURCE-REGISTER.json. Their private local PDF copies are separate from the proof-only archive. The unpublished GTZ erratum and draft are provenance for findings, not accepted black-box inputs. The Tointon book copy is personal-use evidence and must not be redistributed.

No.	Module	Pages
0	Assembled normalized finite-family linear inverse theorem	10
1	The normalized order-two base	5
2	Tautification and symbol shifts	4
3	The local multilinearity construction	6
4	Difference and additive-quadruple arguments	4
5	One-variable factorization and frequency linearization	8
6	Affine transfer and multivariable factorization	8
7	Fresh frequency reconstruction at each rank	9
8	The integral Hall quotient and rank comparison	9
9	The final semidirect orbit and unit completion	8
10	The finite rank iteration	4
11	Factorial multilinearization and literal symmetry	7
12	Plain additive quadruples on an interval	3
13	The symmetrization and final symbol argument	7
14	Exact same-step linearization	7
15	The normalized converse bound	3

Assembly of the precise inverse-theorem input

Author successor: /root/astra_v14_baseline_theta

7 November 2026

Status. This successor assembles the exact local proofs identified below and retains their separate independent decisions. The preceding assemblies' local splitting, selection, analytic and finite-family joins have conditional acceptance. The corrected base, degree conversions, rank iteration and final Section 12 and Section 13 units have their own scoped decisions. The local converse and plain additive-quadruple argument have also been independently accepted. The actual Sections 9–11 reconstruction has conditional acceptance. The one-variable Appendix D/F packet has bounded decisions. This successor connects its affine-offset transfer and the multivariable E.11/E.13 uses to the independently reviewed local multifactor supplier. Its upstream and downstream decisions are separate. The new composition and remaining source-coverage review must still be assessed; no release approval is presumed here. This document does not convert scoped decisions into unconditional GI or whole-source acceptance. Exact versions, decisions and open interfaces are bound in the accompanying records.

1 Exact target and conventions

The target, denoted GI(k), is the following unchanged linear formulation. For each fixed integer $k \geq 3$ and each $0 < \delta \leq 1$, there are a finite family $\mathcal{M}_{k,\delta}$ of nilmanifolds G/Γ, each with connected simply connected nilpotent group of step at most k and a discrete cocompact lattice, fixed smooth metrics, and constants $C, c > 0$, such that every 1-bounded $f : [N] \to \mathbb{C}$ with $\|f\|_{U^{k+1}[N]} \geq \delta$ correlates with a 1-bounded test F of Lipschitz constant at most C along a linear orbit:

$$\left|\mathbb{E}_{n\in[N]} f(n)F(g^n x)\right| \geq c.$$

The family and constants depend on k, δ, not on N, f or the orbit parameters. Strict source thresholds can be used with $\delta/2$ to give this weak-inequality version. No effective numerical bounds are asserted.

For $d \geq 2$, the interval norm is the norm of the zero extension to a cyclic group of sufficiently large order divided by the norm of the interval indicator, as in the published GTZ definition on p. 1232. We use the same normalization at every inverse step. On a finite abelian group A, write

$$\|v\|_{U^d(A)}^{2^d} = \mathbb{E}_{x,t_1,\dots,t_d\in A} \prod_{\omega\in\{0,1\}^d} \mathcal{C}^{|\omega|} v(x + \omega \cdot t),$$

where $\mathcal{C}$ is complex conjugation. Our correlation convention is $|\mathbb{E} f\psi|$; conjugating the test function converts the other common convention without changing its degree or bounds.

All limit nilcharacters in the assembly are internal in the explicitly specified sense of the local Lemma 13.2 and tautification units: the vertical identity holds for all limit group elements and all limit points, with the natural extension of a standard vertical character. Every constructor used

below must supply that domain, rather than only standard-point equivariance. This is an explicit application check, not a consequence of bounded Lipschitz continuity alone.

The manuscript currently requires GI(20) and GI(39). The proposed relative inverse replacement also requires GI($r-1$) for each fixed $r = \max(s, 4)$ chosen before the norm threshold and asymptotic variable. A polynomial orbit without a same-step linearization or an N-dependent family does not satisfy this target.

2 Intermediate interfaces that must be supplied

The following lists the available components and the remaining mathematical premises of the assembly argument. A local decision applies only to its stated scope; the incoming source contracts remain explicit.

1. The compatible bounded polynomial inverse theorem of order 2, proved in `gi-base-002` and independently accepted relative to its retained published forward bracket theorem and Bohr-set inputs. It gives the fixed finite family and normalized interval U^3 norm. The needed trivial-subgroup specialization of the false quadratic extension is proved locally, and the periodic connected polynomial construction replaces the affected skew-iterate route.

2. The nilcharacter approximation/selection interface: from positive correlation with a bounded degree-d nilsequence, obtain a degree-d unit-vector nilcharacter and a scalar component giving positive correlation. For a dense family of parameters, make the choices in a limit fashion and, after passage to a dense subfamily if necessary, keep the finite dimensions, complexities, bounds and positive correlation standard and uniform. Section 4 below gives the finite-net and unit-vector construction used at this interface, with conditional independent acceptance against the previous finite inverse family. Later selection applications retain their own source contracts.

3. Linearization at inverse order s: for a dense family of degree $s-1$ nilcharacters χ_h correlated with $\Delta_h f$, obtain one multidegree $(1, s-1)$ nilcharacter $\chi(h, n)$ with which $\Delta_h f$ has degree-at-most-$s-2$ correlation for many h. This is the precise Theorem 7.1 interface. The modified Proposition 7.3 is supplied by `local-proof-sol-lp02-03-r3`; the quotient and rank-lower comparison by `lp04-001`; the final orbit by the accepted Section 12 successor; and the degree conversions, unit-vector recovery, finite rank iteration and rank-zero endpoint by `rank-iteration-001`. The frozen Sections 9–11 reconstruction supplies an independently reviewed proof of the one-step frequency premise at every newly constructed rank family, conditional on its precise one-variable D and F contracts. Its theorem takes the same uniform unit-vector degree–rank family and accumulated multidegree factor as the rank iteration. It reconstructs all core/linear/independent data anew; none is inherited. Its fixed unit padding is removed by projecting the first factor of the entire LP04 matrix, so the matrix is exactly $\chi_h \otimes \overline{\zeta_h}$ required by the next rank step. The Section 12 successor supplies the same fixed coordinate isometry as the earlier version named in the frozen rank packet; its only declared change makes the zero-index groups explicit. The new binding is recorded here without rewriting that earlier freeze. All these steps retain the full-limit vertical identities and uniform bounds.

   For clarity, the invariant concerns represented families: ordinary unit-vector tests with exact global vertical identities, fixed standard bounds and finitely many standard models and characters. The initial positive-degree selection is made among these representations, by the

parameter-level selection argument for the preceding inverse order; it does not upgrade an arbitrary weak nilcharacter or preserve a prescribed symbol. Degree-to-rank conversion retains the represented function and restricts its character, so this invariant starts at (m, m), where $m = s - 1$, with accumulated factor $A = 1$.

Distinguish the fixed inclusion J that pads the incoming χ_h in Sections 9–11 from the final Section 12 inclusion I, for which $Z(h, n) = I\zeta_h(n)$ on the retained shift set. The full ordinary-top character in Section 12 is chosen before the LP04 comparison and restricts to its cancelling rank-top character. The comparison thus supplies the entire vector

$$W_h = (J\chi_h) \otimes \overline{\zeta_h}$$

on one common model of degree-rank at most $(m, R - 1)$. If $QJ = \mathrm{id}$ is the coordinate projection on the first factor, then

$$(Q \otimes \mathrm{id})W_h = \chi_h \otimes \overline{\zeta_h} =: V_h.$$

This fixed bounded linear map preserves the common model, internal parameter dependence and standard Lipschitz bound. The projection recovering ζ_h from Z is instead the one associated with I. In particular, this conclusion concerns every matrix entry, not a separately selected scalar entry.

The identity $\chi_{h,a} = \sum_b \zeta_{h,b}(\chi_{h,a}\overline{\zeta_{h,b}})$ uses the unit norm of the whole vector ζ_h and therefore uses exactly V_h in the correlation recovery. The finite ordinary restart dictionary preserves the global unit and vertical identities after internal selection; the ordinary tests chosen in the proof of E.5 are extended only after their finite selection. Tensoring the accumulated factor with Z preserves its unit norm and multidegree $(1, m)$. Sections 9–11 then reconstruct all frequency data for this new family. No ordinary degree drop is claimed before the rank endpoint: at rank zero the degree-rank axiom makes the remaining factor ordinary degree at most $m - 1$, which is then absorbed into the lower-degree test. The geometric, analytic and source premises of all these component constructions remain those of their stated contracts.

The centered fractional-part convention in this chain is $(-1/2, 1/2]$, a fixed fundamental cube permitted by Section 12. The reconstruction's acceptance remains conditional on its suppliers; the affine-offset coefficient transfer is supplied by the local multifactor lemma, as connected below.

4. Symmetrization at inverse order s: from that input obtain one degree-s nilcharacter Θ and a bounded nilsequence $\Psi(h, n)$ supported on

$$J = \{(i, j) : i + j \le s - 1\} \ \cup \ \{(i, s - i) : 2 \le i \le s\}, \tag{1}$$

such that the relevant $\chi(h, n)$ is a bounded finite linear combination of components of $\Theta(n + h) \otimes \overline{\Theta(n)} \otimes \Psi(h, n)$. This is the required Theorem 7.4 output, supplied by the locally authored Section 13 proof with its scoped independent decision. It takes the degree-at-most-$s - 2$ correlation actually supplied above; the printed strict hypothesis is not used. Its proof binds the corrected multilinearization, literal block symmetry, local Lemma 13.2 and later shift/sign calculations. It also retains the precise stronger Proposition 8.4, E.11 and E.13 applications. The precise plain Proposition 8.4 input is supplied by the accepted local `plain-quadruples-001`: select the same scalar coordinate of $\chi(h, \cdot) \otimes \phi_h$ and retain that entire lower factor. For $H \subset [-N, N]$ of density α, correlation δ and global scalar bound B, that proof gives standard positive density and correlation depending only on α, δ, B. Its output

has signs $++--$ and shifts in the second and fourth factors by $h_1 - h_4$, exactly as used in Section 13. The functions have global polynomial representations; the interval cutoff belongs to f. The proof removes its remaining bounded cutoff by an inequality, not by an asserted identity. Finite component selection and the internal version preserve standard uniform constants. This is an explicit substitution of the supplying proof for the published Proposition 8.4 invocation in the frozen Section 13 proof, lines 88–125. That earlier file remains unchanged and does not itself refer to the later local proof. Its intervals $I_1 = [-N, N]$ and $J = [N]$, scalar component, global test domain, and quadruple count on the N^3 scale are exactly the local theorem's data. The reparametrization $(h_1, h_2, h_3, h_4) = (h_0 + a, h_0 + b, h_0 + a + b, h_0)$ is therefore unchanged; the subsequent pigeonhole has $h_0 \in I_1$ and $a, b \in [-2N, 2N]$. The E.11 and E.13 inputs require the actual multivariable Appendix D.5 factorization and all-box reconstruction; a one-variable D.4 proof does not supply that premise. This obligation is supplied by the local multifactor theorem and its two independent decisions, as connected below.

5. Splitting for a union of two finite downsets: a fixed bounded nilsequence on their union is uniformly approximable, to any fixed positive error, by a finite sum of products of nilsequences supported on the two downsets, with fixed finite complexity and bounds. This is the E.4 application used explicitly in the next section.

6. The converse correlation bound: a bounded scalar function with positive correlation against a bounded degree-d nilsequence has positive $U^{d+1}[N]$ norm, with a bound for each fixed complexity and correlation. The local `converse-001` proves the required implication using the published compact cube constraint, an explicitly normalized smooth average, a uniform Fourier bound and Gowers–Cauchy–Schwarz. It uses accepted exact same-step linearization for polynomial tests. It does not import the disputed averaging-domain transition in Green–Tao 11.2 or invoke the inverse theorem being proved. Its exact source application and independent decision are bound separately.

7. Countability, up to filtered isomorphism, of the fixed rational filtered nilmanifolds of bounded degree, and equivalence of fixed smooth metrics on a compact nilmanifold. The published countability assertion is on p. 1250 with its Mal'cev-coordinate justification. Its exact scope must include the taut rational objects constructed by this project.

8. The local increment linearization theorem: an exact polynomial nilsequence of degree s has a same-step linear representation on fixed geometry with a fixed Lipschitz comparison. This is supplied by the separately authored and independently accepted $P_1 \rtimes \mathbb{R}$ unit, rather than by the defective full-P_0 suspension in Appendix C.

The increment unit supplies the exact sequence and finite-family join used here. It does not assert the stronger clause in the original Proposition C.2 about one universal projection into the original, unnormalized nilmanifold for every polynomial: bounded normalization is absorbed in the test function F_q. That distinction changes no part of the target in section 1.

The successor connections to the frozen multifactor proof are as follows. They select a supplying proof for the existing conclusion and preserve the old component files. Their independent decisions are recorded separately when available.

First, the one-variable weak factorization uses a progression $A+b[M]$ with $|A| = O(N)$, standard bounded nonzero integer b, and $M \asymp N$. Apply the local inverse-affine lemma with one variable to a real lift of the horizontal polynomial. Its input is the positive binomial coefficient bound on

$m \mapsto f(A + bm)$. In the inverse substitution, the coefficient from input degree j to positive output degree i has size $O(N^{j-i})$ and a standard denominator independent of A. One standard integer multiplier clears the integer portions; the residual $O(N^{-j})$ becomes $O(N^{-i})$. Thus the same original horizontal character, multiplied by a standard positive integer, has the required ambient coefficient bound. It remains nonzero. No bound on the constant coefficient is asserted or needed. This is the exact proof-level transfer at DF002 lines 132–145, distinct from its right-coset periodicity argument.

Second, at Section 13 lines 327–343 take the local multifactor bias theorem with dimension $s + 1$, degree s, and its scale equal to $8N$. The input is the actual reflected unit nilcharacter σ_ε and reflected convex polytope obtained by the orthant selection. Every positive progression is the full $[8N]$; the auxiliary lower-degree test is constant. Its conclusion supplies a globally represented lower-degree nilsequence agreeing with σ_ε on all of $[-8N, 8N]^{s+1}$. The inverse reflection is an integer linear substitution, so the same full-box conclusion holds for σ. This does not extend an identity from only the selected positive orthant by assumption.

Third, after the displayed diagonal pullback in Section 13, the actual unit tensor-product representative $\xi(h, n)$ of the antisymmetric symbol satisfies $s![\xi] = 0$ on $[-8N, 8N]^2$. By the definition of the zero symbol, the coordinates of $\xi^{\otimes s!}$ are lower-degree nilsequences on that box; their finite product of models is one common lower-degree model. Apply the local torsion theorem in dimension two, degree s, scale $8N$, and positive standard tensor power $s!$. It gives a globally represented degree-at-most-$(s-1)$ nilsequence equal to ξ on that same symmetric box. Only afterward restrict to $[-N, N] \times [N]$, as in Section 13 lines 345–363.

There is also a distinct converse use at Section 13 lines 298–305. For $s \geq 4$, put $d = s - 3 \geq 1$ in the accepted local converse and apply it to the uniformly selected lower-degree test correlated with the chosen coordinate of α'. This gives the positive $U^{s-2}[N]$ bound there, uniformly after the stated selections. For $s = 3$, a degree-zero nilsequence is constant: a fixed positive correlation with a bounded constant directly implies a fixed positive $|\mathbb{E}_{n\in[N]}\alpha'(n)|$: if the correlation is at least $c > 0$ and the constant has modulus at most M, this mean has modulus at least c/M. Zero extension has group mean $(N/|Q|)\mathbb{E}_{[N]}\alpha'$, and the interval indicator has U^1 quantity $N/|Q|$, so normalization gives exactly that interval mean. No degree-one test is inserted into this case.

Closure of fixed-degree nilsequences under conjugation, scalar multiplication and finite products uses the actual product nilmanifold and product filtration. The top degree is the maximum of the input degree bounds. This elementary object identification is also required when it is used in the induction below.

3 Correct splitting of the symmetrization remainder

We first give the lattice detail needed by the splitting application. It uses the integral Hall-coordinate lemma in the independently reviewed LP04 unit, not a general assertion that disjoint rational Lie subgroups automatically separate points after quotienting by a lattice.

Lemma 1 (Splitting on two finite multidegree downsets)**.** *Let $J_1, J_2 \subset \mathbb{N}^a$ be finite downsets. Every bounded Lipschitz nilsequence of multidegree supported on $J_1 \cup J_2$ is uniformly approximable, to each fixed positive error, by a finite sum of products of nilsequences supported on J_1 and J_2. The construction uses fixed geometries and bounds before the polynomial coefficients or the interval length vary, for each fixed input geometry and test bound.*

Proof. Use bounded tautification to set the polynomial's value at zero to the identity and restrict the ambient group to the group generated by its positive coordinate groups. For each positive $j \in J_1 \cup J_2$ choose a finite generating set of $\Gamma \cap G_j$. Its real generated group is G_j, by the rational

subgroup assertion. Introduce one formal letter of weight j for each such generator. Take the free real Lie algebra on these letters and discard every individual-letter multidegree whose total vector weight is outside $J_1 \cup J_2$. This is an ideal: the complement of a downset is upward closed. Every letter has positive total weight, so the surviving length is bounded by $\max_{j \in J_1 \cup J_2} |j|$.

Let U be the exponential group and Λ the group generated by the exponential letters. The Hall-coordinate lemma proves that Λ is a lattice and that all weight filtration groups are rational. Sending each letter to its chosen original lattice generator gives a filtered Lie homomorphism $\phi : U \to G$ with $\phi(\Lambda) \subseteq \Gamma$. It maps U_j onto G_j: the letters of weight exactly j already generate G_j as a real group. Lift each Taylor coefficient of the normalized original polynomial through this surjection. The multivariate Taylor theorem, B.9, makes their ordered binomial product an adapted polynomial $\widetilde{g}$ with $\phi\widetilde{g} = g$. Pulling the test back along $U/\Lambda \to G/\Gamma$ therefore preserves the sequence exactly. Its Lipschitz bound increases by a fixed geometric factor only.

For $\nu = 1, 2$ let I_ν be the connected ideal subgroup spanned by the surviving Hall words whose weight is outside J_ν. The quotient U/I_ν has degree support in J_ν. The integral Hall lemma says that its quotient lattice is obtained by deleting precisely those coordinates. No surviving Hall word is deleted by both quotients, since its weight belongs to $J_1 \cup J_2$. Consequently the map

$$U/\Lambda \longrightarrow U/(I_1\Lambda) \ \times \ U/(I_2\Lambda)$$

is injective. In detail, if two cosets have equal images, their difference has integral coordinates in each projected Hall system. The union of the retained coordinate sets is the full Hall system, so every coordinate of the difference is integral. The difference is in Λ. This checks the lattice condition as well as the Lie algebra condition.

The pullbacks of the two algebras of Lipschitz functions on these compact quotients contain the constants, are closed under conjugation, and together separate points by that injection. The complex Stone–Weierstrass theorem approximates the pulled-back test uniformly by the algebra they generate. Each element of that algebra is a finite sum of products with one factor from each quotient, since each of the two individual algebras is closed under multiplication. Composing with $\widetilde{g}$ gives the asserted splitting. For an internal bounded Lipschitz test, first approximate by one standard Lipschitz test on the fixed compact manifold, using Definition 5.1, and then perform this standard finite approximation. For a fixed uniform Lipschitz and norm bound, compactness of that class in the uniform norm supplies a finite net and thus fixed finite bounds. Only this fixed-bound form is used below. If the positive groups are trivial, the sequence is constant and the assertion follows with one constant product. □

The coordinate order is (h, n). Set

$$A = \{(0, j) : 0 \le j \le s - 1\}, \qquad B = \{(i, j) : j \le s - 2, \ i + j \le s\}.$$

Both are downsets and $A \cup B = J$. Indeed the only point of J with $j = s - 1$ is $(0, s - 1)$, covered by A. Every other point of J has $j \le s - 2$ and lies in B. Conversely a point of B of total degree s has $i \ge 2$, so belongs to the second part of (1); all its other points lie in the first part. This proves equality in both directions.

Apply interface 5 to the actual Ψ with these two downsets. A sequence supported on A is independent of h: its polynomial map has every h-direction derivative in the trivial first-coordinate group. It is a degree-at-most-$s - 1$ function $\psi(n)$ of n alone. For a factor supported on B, fixing h gives a polynomial nilsequence in n of degree at most $s - 2$, since all higher n-direction groups are trivial. If normalization of this section at fixed h is needed, the accepted bounded tautification theorem supplies it with uniform geometry and parameter dependence.

Here is the needed approximation-to-correlation connection. After uniform selection, suppose the preceding correlation is at least a standard $\epsilon_0 > 0$ on a dense set of h, and the lower-degree test factors have magnitude at most a standard M. Choose the uniform splitting error less than $\epsilon_0/(2M)$, also absorbing the fixed finite component and coefficient bounds into M. The error in each normalized correlation is then at most $\epsilon_0/2$. A finite sum remains, so one summand has a positive standard correlation on a dense subfamily by finite pigeonhole. The same finite choice of components of Θ can be made on a dense subfamily. The result is two fixed bounded scalar nilsequences θ, θ' of degree at most s such that

$$n \longmapsto f(n+h)\overline{f(n)}\theta(n+h)\theta'(n) \tag{2}$$

is degree-at-most-$s-2$ biased for a dense set of h. The B-factor is absorbed into that lower-degree test. The A-factor $\psi(n)$ is multiplied into the unshifted component forming θ'. No scalar component is divided by, or assumed to have modulus one.

The printed source on p. 1272 uses the opposite coordinate restrictions while claiming independence of h. The corrected downsets and the approximation estimate above are the explicit replacement for that step; its historical printed form is preserved in the findings record.

4 Uniform nilcharacter selection and fixed geometries

Here is the selection needed at the start of each induction step. Use the finite-family inverse statement already obtained at the previous order; the finite-family extraction below is performed after each order, before proceeding to the next. This avoids assuming that an arbitrary external assignment of nilmanifolds to parameters is an internal family.

On one fixed taut degree-d nilmanifold, the top group is central and its quotient by its lattice is a compact torus T. For each integral character η of T, a unit-vector smooth function with that vertical character exists as follows. Cover the compact quotient by T by finitely many trivializing charts. Choose smooth real functions ρ_j supported inside the charts with $\sum_j \rho_j^2 = 1$, using a smooth partition of unity and normalizing its vector by its positive Euclidean norm. In the jth trivialization define the jth component by $\rho_j(b)e(\eta(t))$, and extend it by zero off that chart. These components are smooth, have the exact character, and their squared magnitudes sum to one. The central rational quotient and its lattice are justified by the accepted quotient geometry in the local Lemma 13.2 unit. This is the explicit construction used in published Lemma 6.4.

Components of such unit-vector functions, including frequency zero, form a conjugation-stable multiplicative family: use tensor products for multiplication and conjugation for conjugation. They contain constants and separate points. Distinct points on the same T-fibre are separated by an integral character, with a nonzero component of the unit vector. Points on different fibres are separated by a smooth frequency-zero scalar unit function pulled back from the base. Stone–Weierstrass therefore gives uniform approximation of every continuous test by a finite linear combination of these components.

For the fixed norm and Lipschitz bound in the previous inverse theorem, Arzela–Ascoli gives a finite uniform net of tests. Apply the preceding approximation to each element of that net with error smaller than a fixed fraction of the positive inverse correlation. There are finitely many models and net elements, so the total list of unit-vector test maps, dimensions, frequencies, coefficient bounds and Lipschitz bounds is fixed. Pigeonholing a scalar component of an approximating sum preserves a fixed positive correlation. No component is assumed to have modulus one; the entire vector has norm one.

For each finite interval and each of its good shifts, ordinary choice selects one polynomial orbit and one item from this fixed finite list. Taking the ultralimit of these finite assignments gives an internal family. A finite pigeonhole step restricts to a dense set of shifts with one common fixed model and fixed test map if required. The unit and vertical identities hold at every representative and hence for all limit arguments, with the natural extension of the chosen standard character. This establishes the selection interface needed here from the preceding finite inverse theorem. It does not replace separate selection applications later in Sections 8–13, which remain with their actual consumers.

For the countability interface, use the rational Mal'cev-coordinate description of a lattice in a connected simply connected nilpotent Lie group, as invoked in published Section 3 and p. 1250. In each finite dimension the rational structure constants have countably many possibilities. The rational subspaces defining the finitely many filtration groups are specified by finite matrices with rational entries, again a countable choice. In the rational structure determined by the lattice, a finite list of rational logarithmic generators specifies the lattice. Thus there are at most countably many such filtered lattice data up to isomorphism. All local constructions here have finite rational filtrations and discrete cocompact lattices, so belong to this class. On each compact quotient any two fixed smooth Riemannian metrics are bi-Lipschitz equivalent: compare their positive quadratic forms on the compact unit tangent bundle. These are the exact countability and metric comparisons used in the extraction below.

5 The finite-group inequality needed for the last induction step

Lemma 2. *For bounded complex functions a, b on a finite abelian group Q and an integer $d \geq 1$,*

$$\mathbb{E}_{h\in Q}\|a(\cdot+h)b(\cdot)\|_{U^d(Q)}^{2^d} \leq \|a\|_{U^{d+1}(Q)}^{2^d}\|b\|_{U^{d+1}(Q)}^{2^d}.$$

Proof. For $t = (t_1, \dots, t_d)$ write $C_a(x;t) = \prod_\omega \mathcal{C}^{|\omega|}a(x+\omega\cdot t)$, and define C_b likewise. Expanding the left side and using the bijection $(x,h) \mapsto (x, y = x+h)$ gives

$$\mathbb{E}_t(\mathbb{E}_y C_a(y;t))(\mathbb{E}_x C_b(x;t)).$$

Its absolute value is at most $\left(\mathbb{E}_t|\mathbb{E}_y C_a(y;t)|^2\right)^{1/2}\left(\mathbb{E}_t|\mathbb{E}_x C_b(x;t)|^2\right)^{1/2}$. Each squared expression is the $(d+1)$-cube average: expand the square and substitute its second base point as the first plus a new increment. It therefore equals the corresponding U^{d+1} norm to the power 2^{d+1}. This proves the inequality. The original left side is nonnegative because it is an average of powers of Gowers seminorms. □

Choose a cyclic group Q with size between C_sN and $2C_sN$ for a fixed sufficiently large C_s, and zero-extend all functions supported on $[N]$. The normalized interval and group norms differ by the fixed indicator factor. For each fixed order that factor is bounded above by one and below by $N/|Q|$, using the finite-group Gowers–Cauchy–Schwarz inequality. Thus a fixed positive interval norm gives a fixed positive group norm, and conversely; the constants depend only on the fixed ambient ratio and order. Taking C_s larger than 2^{s+1} ensures the same ambient group is valid for all interval orders used here. The zero-extended product in (2) is exactly the group product for $|h| \leq N$; there is no wrapped overlap since the ambient size is larger than $4N$.

By interface 6, (2) has a uniformly positive $U^{s-1}[N]$ norm for a dense set of h. Let $a = f\theta$ and $b = \overline{f}\theta'$, both zero-extended. The average over all group shifts of $\|a(\cdot+h)b\|_{U^{s-1}(Q)}^{2^{s-1}}$ is bounded below by a fixed positive constant: the contributing set of interval shifts has positive density also in

Q, and the other summands are nonnegative. Lemma 2 and the fixed bound on b imply $\|a\|_{U^s(Q)}$ is positive by a standard amount, hence so is $\|f\theta\|_{U^s[N]}$. The inverse theorem at order $s-1$ now makes $f\theta$ correlate with a degree-at-most-$s-1$ nilsequence ψ. The actual product $\theta\psi$ is a degree-at-most-s nilsequence on the product filtered nilmanifold. Thus f is s-biased.

6 Starting and completing the induction

The bounded induction base is the separately frozen and reviewed `gi-base-002`. It starts from the exact forward bracket theorem, proves the affected trivial-subgroup specialization, constructs a periodic connected polynomial test and transfers its correlation directly to the normalized interval. The older base in assembly001 is superseded and remains historical. No unpublished correction is treated as an accepted black box.

For completeness the initial derivative step has the same normalization control. The cube identity on the fixed ambient cyclic group gives

$$\mathbb{E}_h \|\Delta_h f\|_{U^s(Q)}^{2^s} = \|f\|_{U^{s+1}(Q)}^{2^{s+1}}.$$

If the interval norm is positive by a standard amount, so is the right side. Since f is bounded, all summands have a fixed upper bound. The nonzero differences occur only on the shifts represented by $[-N, N]$, so a dense subset of those shifts has a uniformly positive $U^s[N]$ norm. Apply the inverse theorem of order $s-1$, then interface 2, to obtain the family required by interface 3. Interfaces 3–5 and the preceding section complete the induction step. The bounded limit-N case is included: a bounded interval has standard finite size, and any function on it is represented by a Lipschitz function along a rational circle rotation, with a standard finite Lipschitz bound. Thus the induction proves the limit polynomial inverse conclusion for every limit N, conditional on the listed intermediate interfaces and base.

We spell out the finite-family extraction rather than replace it by pointwise compactness. Enumerate the rational filtered nilmanifolds of degree at most s up to isomorphism as $M_1, M_2, \ldots$, with one fixed smooth metric on each, using interface 7. If the finite polynomial inverse statement failed at some fixed $\delta > 0$, for each j choose a 1-bounded $f_j : [N_j] \to \mathbb{C}$ with norm at least δ, but correlation at most $1/j$ against every polynomial nilsequence from the first j manifolds with 1-bounded test and Lipschitz norm at most j. Its ultralimit has positive norm, so the limit inverse conclusion gives a fixed standard filtered nilmanifold, bounded Lipschitz limit test and positive standard correlation. Normalize the test by a fixed bound to make it 1-bounded. It is isomorphic to some fixed M_{j_0}; fixed metric equivalence transfers its Lipschitz bound to a standard constant. Representatives of the polynomial and test satisfy the same bounds with one slightly larger standard constant on an ultrafilter-large set. For j larger than that constant and j_0, the selected correlation upper bound tends to zero, contradicting the positive lower bound. This yields the finite-family polynomial statement with all parameters fixed independently of N.

Finally interface 8 converts every member of the finite polynomial family to a fixed nilmanifold of step at most s, with a fixed Lipschitz bound and exactly equal sequence. A finite maximum of those bounds and the unchanged positive correlation give the exact linear target in section 1. This final join depends on the reviewed increment construction; it does not import the failed full-polynomial suspension claim.

7 What remains before this is a complete proof

The local units and their exact incoming connections are now assembled relative to their retained published inputs. The affine coefficient transfer and multivariable factorization, bias and torsion conclusions have separate independent decisions. Their substitutions and the degree-zero converse boundary are new explicit connections in this successor. The earlier composition decision is not silently extended to these new bytes.

The retained published forward bracket/Bohr, quantitative equal-side equidistribution, rational-coordinate and compact cube-constraint inputs keep their exact source scope. Related-use coverage and the complete affected-dependency audit remain separate. This packet is submitted for independent acceptance of the assembled target and its actual current consumer interfaces. It makes no release-incorporation or whole-source approval claim. The target keeps its original normalized norm, quantifier order, fixed finite family and same-step linear output.

The polynomial GI(2) base from the published bracket theorem

Author proposal: /root/astra_v14_baseline_theta

7 November 2026

Scope. This is an author proposal awaiting independent review. It derives the exact polynomial interval base from the forward bracket-quadratic theorem and Bohr regularity in Green–Tao, *An inverse theorem for the Gowers $U^3(G)$ norm*, Proceedings of the Edinburgh Mathematical Society 51 (2008), 73–153. It replaces the needed use of the false quadratic-extension assertion by an explicit finite-box proof. It also constructs the bracket tests directly on connected polynomial models, so neither Theorem 12.9 nor its erroneous skew-shift iterate parameters are a theorem premise. The forward Bohr/bracket inverse theorem and its unchanged proof inputs remain published inputs, not newly proved by this unit. The complete GI theorem and broader source-risk review remain separate.

1 Exact target and published input

The required base is this statement. For every $\delta > 0$ there are finitely many fixed rational filtered nilmanifolds of degree at most two, with connected simply connected groups, and constants $L < \infty, c > 0$ such that every complex 1-bounded $f : [N] \to \mathbb{C}$ with $\|f\|_{U^3[N]} \geq \delta$ correlates by at least c with $F(g(n)\Gamma)$ on one of them. The polynomial g is adapted, $\|F\|_\infty \leq 1$, and $\mathrm{Lip}(F) \leq L$. All geometry and constants are fixed before N and f. The interval norm is zero extension to a sufficiently large cyclic group divided by the norm of the interval indicator.

The published forward input is Theorem 10.11, pp. 124–125, especially (10.7) and (10.9). For each fixed sufficiently small $\eta > 0$, and each prime $P > 2$, a function $u : \mathbb{Z}/P\mathbb{Z} \to \mathbb{C}$ with $|u| \leq 1$ and $\|u\|_{U^3} \geq \eta$ yields a frequency set S of size at most $D(\eta)$, a proper progression Q containing a Bohr set $B(S, \rho_0)$ with $\rho_0 \geq \rho_*(\eta) > 0$, and one bracket quadratic

$$\phi(n) = c + \sum_j a_j\{\alpha_j n\} + \sum_{j,k} a_{jk}\{\alpha_j n\}\{\alpha_k n\} \pmod 1.$$

Here $\alpha_j = \xi_j/P$ represents a frequency in S, $\{t\} \in [-1/2, 1/2)$, and coefficients are unrestricted real numbers. For every nonempty $A \subseteq Q$ there is a shift h_A such that

$$|\mathbb{E}_{n\in A} u(n + h_A)\mathrm{e}(-\phi(n))| \geq b_*(\eta) > 0.$$

The actual powers and exponential constants printed in 10.11 are stronger than needed; only the stated positive/finiteness dependence on η is consumed. Its proof uses the forward Theorem 2.7, Corollary 10.6 and Proposition 10.10. The affected special case in the latter is supplied next.

2 The exact specialization replacing quadratic extension

The final PDF's Lemma 10.7, pp. 122–123, invokes quadratic extension from a subgroup. The author notice of December 2023 identifies Proposition 3.2, p. 88, as false. The later use needed

here is Proposition 10.10, where the progression is proper and $H = \{0\}$. The notice numbers these items 10.6 and 10.8 in its preprint version; we bind to the final PDF numbering and argument.

Lemma 1. *On a finite box $B = \prod_{j=1}^{d} \{0, \dots, L_j - 1\}$, any $v : B \to \mathbb{R}/\mathbb{Z}$ whose third additive differences vanish whenever the whole cube lies in B has the form*

$$v(l) = c + \sum_j b_j l_j + \sum_j d_j \binom{l_j}{2} + \sum_{j<k} e_{jk} l_j l_k.$$

Consequently a locally quadratic phase on a proper progression with trivial subgroup has the polynomial coordinate representation needed in Proposition 10.10, without extension to the ambient group.

Proof. Coordinates of length one can be omitted. Each mixed second difference is constant on its rectangular domain: every coordinate difference of it is a vanishing third difference, and the domain is connected by coordinate steps. The same holds for each available diagonal second difference when $L_j \geq 3$. Choose their constants as e_{jk} and d_j, setting $d_j = 0$ when $L_j = 2$. Subtract these quadratic terms. Every remaining first difference is constant because its differences in every available coordinate vanish. When $L_j = 2$ its domain has only one slice in direction j, so mixed differences suffice. Subtract the resulting linear function; the remainder is constant by connectedness. The coordinate map of a proper progression is injective and pulls back the local quadratic identities to the box. This proves the claim. Division by two in $\mathbb{R}/\mathbb{Z}$, if one wants the symmetric coefficient convention of (10.5), is possible; no uniqueness is used. □

In Proposition 10.10 the independent vectors supplied by Corollary 10.6 express each progression coordinate as a real linear combination of $\{\xi n/P\}$ on that progression. Substituting the lemma's polynomial gives precisely its bracket quadratic. The later localization and shift in the proof of 10.11 use this specialization, not the extension of a phase from a nontrivial subgroup. The remaining published forward inverse and Bohr-set applications are retained as explicit inputs. This does not claim to prove a corrected general version of 3.2.

3 A smooth periodic bracket test with positive correlation

Use Lemmas 8.1–8.2 to choose a regular Bohr set $B = B(S, \rho) \subseteq Q$, where ρ has a positive lower bound depending only on η and may be taken at most $1/8$. Its density in $\mathbb{Z}/P\mathbb{Z}$ is at least $b_0(\eta) > 0$. Apply the displayed (10.9) input with $A = B$ and write its shift as h.

For $d = |S| > 0$, choose a fixed $0 < \epsilon < 1/(100d)$ sufficiently small in terms of $b_*(\eta)$ and d. Let $\chi : \mathbb{R}/\mathbb{Z} \to [0, 1]$ be one on $|x| \leq \rho(1 - \epsilon)$, zero on $|x| \geq \rho(1 + \epsilon)$, and linear on the intervening strips. Its Lipschitz bound is at most $(2\epsilon\rho)^{-1}$, with an inessential larger fixed bound permitted. In particular it vanishes on a fixed neighbourhood of the fractional-part seam at $1/2$. Put

$$W(n) = \prod_{j=1}^{d} \chi(\alpha_j n)^{1+4d}, \qquad \Psi(n) = W(n)\mathrm{e}(-\phi(n)).$$

The difference $W - 1_B$ vanishes outside the regularity boundary shell $B(S, \rho(1+\epsilon)) \setminus B(S, \rho(1-\epsilon))$ and is bounded by one. Bohr regularity bounds the shell by $Cd\epsilon|B|$ with an absolute fixed C. Choose ϵ so that $Cd\epsilon \leq b_*(\eta)/2$. It follows that

$$\left| \mathbb{E}_{n \in \mathbb{Z}/P\mathbb{Z}} u(n + h)\Psi(n) \right| \geq \tfrac{1}{2} b_*(\eta) b_0(\eta) =: c_0(\eta) > 0. \tag{1}$$

When S is empty the phase is constant and this conclusion already uses a constant test; that case needs no cutoff construction.

Every factor in Ψ is periodic with period P, because $P\alpha_j$ is integral. We now represent this same function for all integers, not merely signed representatives of $\mathbb{Z}/P\mathbb{Z}$.

4 Connected polynomial models for every bracket factor

Write the coefficients below as $a = q + s$, with $q \in \mathbb{Z}$ and $|s| \le 1/2$. A linear bracket factor is

$$\chi(\alpha n)\mathrm{e}(a\{\alpha n\}) = \chi(\alpha n)\mathrm{e}(s\{\alpha n\})\mathrm{e}(q\alpha n).$$

It is the test $F(x, y) = \chi(x)\mathrm{e}(sx + y)$ along the linear polynomial $(\alpha n, q\alpha n)$ on $\mathbb{R}^2/\mathbb{Z}^2$. The seam is harmless because χ vanishes there. This test is 1-bounded and has Lipschitz bound $O(1 + \operatorname{Lip}\chi)$, uniformly in a, α.

A quadratic bracket factor satisfies the exact identity

$$\begin{aligned}&\chi(\alpha n)^2\chi(\gamma n)^2\mathrm{e}(a\{\alpha n\}\{\gamma n\})\\ &= [\chi(\alpha n)\chi(\gamma n)\mathrm{e}(s\{\alpha n\}\{\gamma n\})]\mathrm{e}(q\alpha\gamma n^2)\,[\chi(\gamma n)\mathrm{e}(-q\alpha n[\gamma n])]\,[\chi(\alpha n)\mathrm{e}(-q\gamma n[\alpha n])].\end{aligned}$$

Indeed the omitted term in expanding the integer part is $q[\alpha n][\gamma n] \in \mathbb{Z}$, where $[t] = t - \{t\}$. The first bracket is a test on a two-torus with uniformly bounded Lipschitz constant. The pure quadratic phase is a test $\mathrm{e}(z)$ along $z = q\alpha\gamma n^2$ on $\mathbb{R}/\mathbb{Z}$, using the degree-two filtration $G_0 = G_1 = G_2 = \mathbb{R}$, $G_3 = 0$.

For the remaining bracket use the connected simply connected Heisenberg group with coordinates (u, v, w) and law

$$(u, v, w)(u', v', w') = (u + u', v + v', w + w' + uv').$$

Its lattice is $\mathbb{Z}^3$ in these coordinates; its degree-two filtration has $G_0 = G_1 = G$ and $G_2 = \{(0, 0, w)\}$. On a right lattice quotient, reduce v first, then u. The resulting coordinates are

$$(\{u\}, \{v\}, w - u[v] \pmod 1).$$

Define a test on this fundamental cube by $F_H(\{u\}, \{v\}, z) = \chi(\{v\})\mathrm{e}(z)$. It is well-defined and globally Lipschitz. Across the u faces the central coordinate does not change; the central faces are periodic; across the v faces the cutoff is identically zero in a neighbourhood. Within the remaining finitely many fixed charts its Lipschitz constant is bounded by a fixed geometric multiple of $1+\operatorname{Lip}\chi$.

Evaluate this test on the polynomial $p(n) = (\beta n, \alpha n, 0)$. The exact value is

$$F_H(p(n)\Gamma) = \chi(\alpha n)\mathrm{e}(-\beta n[\alpha n]).$$

For completeness its left first difference is $(\beta h, \alpha h, -\alpha\beta hn)$, its next difference is $(0, 0, -\alpha\beta hk)$, and its third difference is the identity. These satisfy the degree-two filtration for every integer shift, so this is an adapted polynomial without any appeal to an incorrect formula for linear skew iterates. Take $\beta = q\gamma$ and interchange α, γ for the two remaining factors.

Apply the linear construction to each coefficient of $-\phi$ and the quadratic construction to all ordered pairs, inserting zero coefficients if needed. Each frequency receives one cutoff from the linear factor and $4d$ from the quadratic factors, exactly the power in W. The constant phase is harmless. Products give a fixed connected rational polynomial model of degree at most two, of dimension at most $2d + 9d^2$. The product test is 1-bounded and has a finite Lipschitz bound

depending only on d, ρ, ϵ. Arbitrarily large coefficients occur only in polynomial orbit parameters. There are finitely many geometric product types for $d \le D(\eta)$.

Thus the model represents the periodic function $\Psi(n)$ exactly for all $n \in \mathbb{Z}$. Individual polynomial factors need not be P-periodic; their product is periodic by the exact bracket identity. That distinction is used in the next change of variables.

5 The normalized interval transfer

Fix an integer $C > 16$ and choose a prime P with $CN < P < 2CN$. Zero-extend the given interval function to u on $\mathbb{Z}/P\mathbb{Z}$. The interval-indicator Gowers norm is at least its mean N/P, so $\|u\|_{U^3} \ge \delta/(2C)$. Choose once and for all a positive η at most this bound and within the smallness range of the published forward theorem. All preceding data now depend only on δ.

Using the exact periodicity of Ψ in (1), change variables to obtain

$$\left|\mathbb{E}_{n\in\mathbb{Z}/P\mathbb{Z}} u(n)\Psi(n-h)\right| \ge c_0(\eta).$$

The zero extension gives

$$\left|\mathbb{E}_{n\in[N]} f(n)\Psi(n-h)\right| \ge (P/N)c_0(\eta) > 0.$$

The translated sequence is represented by the same test on the same fixed model, with polynomial orbit $g(n-h)$. Translation preserves adapted polynomiality, either by the derivative criterion or B.5. No signed-representative split, discontinuous interval indicator or additional N-dependent geometry is introduced. Norm, Lipschitz and correlation bounds have precisely the required fixed dependence. Bertrand's postulate applies to the integer $CN > 1$, so the prime choice also covers every positive N.

6 Source comparison and bounded related-use findings

The unpublished author notice identifies the false general quadratic extension, but is used here only to locate the affected proof. The needed trivial-subgroup specialization is proved in Section 2. Its published source locator is Proposition 3.2 on p. 88, correcting the p. 89 metadata typo in assembly001's RELATED-USES record.

The actual publisher image of Example 12.3 on p. 136 gives $T(x, y) = (x + \alpha, y + \beta + mx)$ but prints an iterate omitting mnx and using $mn(n + 1)\alpha/2$. Direct iteration gives

$$T^n(x, y) = (x + n\alpha, y + n\beta + mnx + m\alpha\tbinom{n}{2}).$$

Already $n = 1$ distinguishes these formulas. The later pure quadratic choice in Lemma 12.11 inherits the printed iterate convention. Rather than silently change its parameters, Section 4 constructs all needed factors directly on connected polynomial models. This preserves the qualitative base theorem's strength and avoids those printed iterates entirely.

The cutoff's derivative bound also depends on its transition width. This proof retains the finite factors $(\epsilon\rho)^{-1}$ instead of importing the universal numerical bounds stated for those tests in Lemma 12.11. The required application needs only finite bounds after η is fixed.

The bounded audit scope is the forward chain 2.7 → 10.6 → 10.10 → 10.11, the affected trivial-subgroup use of 10.7, Bohr regularity 8.1–8.2, and the elementary construction 12.2–12.11 as consumed by the interval base. The chain notation records dependencies to inspect, not acceptance

of all their proofs. The converse half of 2.7, the finite-field-only extension use and unrelated applications are not premises of this unit. The old assembly001/base text remains frozen. This successor changes its base supplier from 12.9 to the precise bracket theorem plus local construction; its separate converse argument is unchanged.

Tautification and the shifted nilcharacter construction

Project proof draft: /root/astra_v14_baseline_theta

7 November 2026

Status. This is an author proof proposal. Its internal geometric input is the rational-subgroup and quotient argument in the separately frozen local Lemma 13.2 proof [2], which is also awaiting independent acceptance. The present unit does not assert that the complete inverse theorem has been repaired.

1 The exact change of filtration

All Lie groups and rationality conventions are those of [1, Sections 3–6]. A rational subgroup is connected, and its intersection with the ambient lattice is cocompact. The three domain filtrations used here are the degree filtration on $\mathbb{Z}^a$, the multidegree filtration on $\mathbb{Z}^a$, and the degree-rank filtration on $\mathbb{Z}$ from [1, Definition 6.17]. For limit objects, defining relations are interpreted by their natural ultrapower extension as in [1, Appendix A.2–A.3, pp. 1324–1325]. Thus vertical equivariance quantifies over every limit point of the nilmanifold and every limit element of the top group, using the extension of the standard vertical character. We do not infer this exact internal identity from an identity asserted only at standard points. For these three cases respectively put

$$H = G_1, \qquad H = \langle G_{e_1}, \dots, G_{e_a} \rangle_{\mathbb{R}}, \qquad H = G_{(1,0)}.$$

In the middle case the connected real generated subgroup equals the group generated by the connected subgroups: the latter is the connected analytic subgroup with the generated Lie algebra, and such subgroups in a simply connected nilpotent group are closed. The rational-subgroup lemma in [2, Section 2] shows that H is rational.

Theorem 1. *Suppose G/Γ is a rational filtered nilmanifold with finite downset of degrees J for one of these three domain filtrations. Every polynomial nilsequence $F(g(n)\Gamma)$ admits an exactly equal representation on $H/(H \cap \Gamma)$ with filtration*

$$G'_0 = H, \qquad G'_i = G_i \quad (i > 0).$$

For degree-rank, 0 denotes $(0,0)$. This filtration is taut and has the same positive-degree groups and no larger degree support, hence the same allowed degree bound. Its test-function Lipschitz bound depends only on the original fixed filtered nilmanifold and original Lipschitz bound, not on the polynomial coefficients. For a nilcharacter of positive degree d, the same standard vertical character on G_d is preserved. These assertions hold uniformly for limit maps and families of polynomial coefficients with fixed geometric data.

Proof. As in [2, Section 3], first restrict to G_0 and its lattice if necessary. Choose $g(0) = q\gamma$ with q in a fixed compact representative set, and set

$$h(n) = q^{-1} g(n) \gamma^{-1}, \qquad F_q(x\Gamma) = F(qx\Gamma).$$

Then $h(0) = 1$, $F_q(h(n)\Gamma) = F(g(n)\Gamma)$, and h is polynomial for the old filtration. Indeed every G_i is normal in G_0; use the constant map $g(0)^{-1}$, product closure in [1, Corollary B.4], and filtered conjugation by q. The Lipschitz bound is uniform because q lies in a fixed compact set.

We show that $h(n)$ belongs to H. In the degree case the difference $h(n)h(0)^{-1}$ is in G_1 for every $n \in \mathbb{Z}^a$, by the derivative condition. In the degree-rank case the same statement holds with $G_{(1,0)}$. In the multidegree case connect 0 to n by a finite path of positive or negative coordinate steps. At a step in the jth coordinate, $h(v \pm e_j)h(v)^{-1} \in G_{e_j}$. Multiplying these increments along the path and using $h(0) = 1$ proves the assertion. This argument applies to every integer input and puts no bound on its size.

Every original positive group is contained in H. For degree and degree-rank this follows from nesting and the least positive index. For multidegree, any positive index dominates some e_j. Consequently the displayed new groups are well-defined subgroups of H, are descending, and satisfy the commutator axioms inherited from G, including those with index zero because $H \subseteq G_0$. Their intersections with $H \cap \Gamma$ equal their intersections with Γ and are lattices. In degree-rank all the required equalities $G_{(d,0)} = G_{(d,1)}$ for $d > 0$ remain unchanged. The new group at zero is exactly the required positive group or group generated by the minimal positive groups, so the new filtration is taut.

All positive iterated derivatives of h have the same required group membership as before, and the zeroth-order membership was just checked. Thus h is polynomial for the new filtration. Restrict F_q through the fixed smooth inclusion $H/(H \cap \Gamma) \longrightarrow G/\Gamma$. The inclusion is an embedding of compact manifolds, so this costs only a fixed Lipschitz comparison factor and gives the asserted exact sequence.

For a nilcharacter let $\eta : G_d \longrightarrow \mathbb{R}$ be its standard vertical character. The proof in [2, Section 3] applies verbatim with G_d in place of its top cube group: for each $t \in G_d \cap \Gamma$, the continuous map $q \mapsto \eta(qtq^{-1})$ on connected G_0 is integer-valued by the unit-norm vertical identity at $q\Gamma$. It is therefore constant. The lattice spans G_d as a connected real group, so $\eta(qtq^{-1}) = \eta(t)$ for all $t \in G_d$ and $q \in G_0$. Hence F_q and its restriction have the same standard vertical character.

For families, choose one bounded standard representative section of the fixed compact quotient. Such a section can be formed from finitely many local smooth sections, taking the first available chart. Apply this one section to each constant coefficient. The resulting choices are limit maps when the coefficients are limit maps. Discontinuities in the parameter do not enter a Lipschitz bound in the orbit variable. All metric constants and groups remain fixed. The exact identities can be imposed on representatives on an ultrafilter-large set by the witness argument of [2, Section 7]; the ordinary proof then passes to the limit. □

This also proves equality of the classes of nilsequences under the old and taut conventions, at each finite degree support: one inclusion is the theorem and the reverse inclusion is immediate. It therefore preserves the equivalence relation defined by a lower-degree nilsequence. It is not merely an assertion that the top-degree function has a similar replacement.

Corollary 2. *In a taut degree or degree-rank filtration of degree at most (d, r) or d, the ambient group has nilpotency class at most d. In a taut multidegree filtration supported on a finite downset J, its class is at most $D = \max_{a \in J} |a|$. If the support is contained in the downset below one positive index d, the group G_d is central.*

Proof. For degree and degree-rank, brackets of j elements of the ambient first positive group belong respectively to G_j or $G_{(j,0)}$, which vanish when $j > d$. For multidegree, expand an iterated Lie bracket using the generating Lie algebras $\mathfrak{g}_{e_i}$. A bracket of length j lies in some $\mathfrak{g}_a$ with $|a| = j$, hence vanishes for $j > D$. The generated Lie algebra has the asserted nilpotency class, and so

does its simply connected group. Finally $[G_d, G_{e_i}] \subseteq G_{d+e_i} = 1$ in the multidegree case. The same argument uses the first positive group in the other two cases. Since these groups generate the ambient group, G_d is central. □

2 A complete shifted nilcharacter construction

Theorem 3. *For any of the three domain filtrations above, let χ be a nilcharacter of positive degree d. Let $i > 0$ and let u belong to the domain group at index i. Then*

$$n \longmapsto \chi(n+u) \otimes \overline{\chi(n)}$$

is a nilsequence with support strictly below d. The representing geometric data and Lipschitz bound can be fixed independently of u within this domain group. For limit u the representation and test function can be chosen in a limit fashion. This supplies the shift assertion required in [1, Lemma E.8(iv)] for the three domains used in the proof.

Proof. By Theorem 1 represent the same χ on a taut nilmanifold G/Γ, with the same degree and vertical character. Write its polynomial map as g. In the degree-rank case use $i = (1, 0)$: any nonzero element of a positive domain group requires this index, and if the given domain group is trivial then $u = 0$ also belongs to this first group. Put $U(v) = (v, 1)$ and $D(v) = (v, v)$ in G^2. For each index j let L_j be the connected subgroup generated by

$$U(G_{j+i}) \quad \text{and} \quad D(G_j).$$

These groups are rational by [2, Lemma 2], and are descending. The generator commutators have the following memberships:

$$\begin{aligned}
[D(G_j), D(G_k)] &\subseteq D(G_{j+k}),\\
[D(G_j), U(G_{k+i})] &\subseteq U(G_{j+k+i}),\\
[U(G_{j+i}), U(G_{k+i})] &\subseteq U(G_{j+k+2i}) \subseteq U(G_{j+k+i}).
\end{aligned}$$

Passing to generated Lie algebras and using BCH, as in [2, Section 4], proves $[L_j, L_k] \subseteq L_{j+k}$, including index zero. Thus L_0/Δ, with $\Delta = L_0 \cap \Gamma^2$, is a rational filtered nilmanifold of degree at most d. In the degree-rank case its additional equalities hold: $L_{(a,0)} = L_{(a,1)}$ follows from $G_{(a,0)} = G_{(a,1)}$ and $G_{(a+1,0)} = G_{(a+1,1)}$ for $a > 0$.

The difference $a_u(n) = g(n+u)g(n)^{-1}$ is polynomial for the shifted filtration $(G_{j+i})_j$. Its zeroth-order value belongs to G_i. Every further iterated derivative is an iterated derivative of g with the derivative in direction u innermost, so the defining index sum has the additional term i. Therefore $U(a_u(n))$ and $D(g(n))$ are polynomial for (L_j). By [1, Corollary B.4], so is their product

$$P_u(n) = (g(n+u), g(n)).$$

This verifies the actual orbit, not only its coordinate projections.

As $i > 0$, $G_{d+i} = 1$ and $L_d = D(G_d)$. Corollary 2 shows that G_d is central in G_0, hence L_d is central in L_0. The function

$$T(x\Gamma, y\Gamma) = F(x\Gamma) \otimes \overline{F(y\Gamma)}$$

restricted to L_0/Δ is invariant under $D(G_d)$, since its two vertical characters cancel. The rational central-quotient argument in [2, Sections 2 and 6] therefore gives a nilsequence on L_0/L_d with the exact value $T(P_u(n)\Delta)$. Its filtration group at d is trivial and all groups outside the downset below

d were already trivial. Its support is consequently strictly below d. The same argument gives a fixed Lipschitz bound on the quotient.

The quotient may not yet be taut. Apply Theorem 1 to this lower-degree nilsequence, preserving its exact values without enlarging its degree support. The groups L_j and their quotient depend on the index i but not on the element u. The map $u \mapsto P_u(0)$ is a limit map when g and u are limit data. The one fixed bounded section used in Theorem 1 then gives a limit choice of the final normalized representation, with uniform Lipschitz bounds independent of u. This proves all the assertions. □

For the erratum's needed case the domain is $\mathbb{Z}^2$ with multidegree, $d = (1, s-1)$, $i = (1, 0)$ and $u = (k, 0)$. The resulting support is contained in

$$\{a : a \leq (0, s-1)\} \ \cup \ \{a : a \leq (1, s-2)\},$$

uniformly in k. This is exactly the support needed before applying the published splitting and vertical Fourier lemmas in the proof of the new difference decomposition. Those applications, and the subsequent two Cauchy–Schwarz steps, are separate proof obligations.

3 Scope of the correction

The tautification theorem supplies the general representation repair suggested, but not fully proved, in Section 2 of the author erratum. It also proves the top centrality and nilpotency-class assertions for the taut conventions. The shifted construction above supplies the needed instance of E.8(iv) without leaving its auxiliary rationality or central quotient implicit. This unit does not claim to have audited every filtration introduced in the published proof. In particular, E.8(v), E.9, Lemma 12.1, the Section 12 quotient, and the final modified Proposition 7.3 argument still need their own exact applications or replacements.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics **176** (2012), 1231–1372.

[2] Project proof, *A local replacement for the filtration argument in GTZ Lemma 13.2*, draft 002, 7 November 2026. Manifest SHA-256:

`95771803bed9d9135430152e2e20ad450c46aaae3ef0eeba0cab33bee59fd06c`

This is an explicitly linked author proof awaiting independent acceptance.

A local replacement for the filtration argument in GTZ Lemma 13.2

Project proof draft: /root/astra_v14_baseline_theta

7 November 2026

Status. This is a proposed project proof, not an independent acceptance. It replaces only the multilinearity lemma and its auxiliary filtration construction. It does not certify the other corrections to the inverse theorem. All source and consumer bindings are recorded separately.

1 Statement and conventions

We use the nilmanifold, polynomial-map and nilcharacter conventions of [1, Sections 3, 4 and 6]. In particular all Lie groups below are connected, simply connected and nilpotent; filtration groups are connected rational Lie subgroups. A rational subgroup J means that $J \cap \Gamma$ is a lattice in J. The multidegree ordering is coordinatewise. Write $\mathbf{1} = (1, \dots, 1) \in \mathbb{N}^s$ and $|a| = \sum_j a_j$. The group G_a is trivial when $a \not\leq \mathbf{1}$. We use $\mathrm{e}(t) = \exp(2\pi i t)$ and complex conjugation componentwise in tensor products. A vector-valued nilcharacter has a unit-vector-valued Lipschitz test function and a real vertical character, as in [1, Definition 6.1]. For limit objects, every defining relation is read in its natural ultrapower extension, using [1, Appendix A, Definition A.2 and the following discussion, and Definition A.3, pp. 1324–1325]. In particular the vertical identity is required for every limit point $x \in {}^*(G/\Gamma)$ and every $t \in {}^*G_{\mathbf{1}}$, with the standard homomorphism η extended to ${}^*\eta$. This is the internal nilcharacter convention used in this proof. We do not infer this exact identity from an identity only at standard points; bounded Lipschitz continuity alone would not justify that inference.

Theorem 1 (The unchanged multilinearity statement)**.** *Let $s, m \geq 1$ be fixed standard integers. Let χ be a nilcharacter of multidegree at most $\mathbf{1}$ on $\mathbb{Z}^s$, or a limit nilcharacter with that description in the sense of [1, Sections 5–6]. For fixed homomorphisms $L_1, L_1', L_2, \dots, L_s : \mathbb{Z}^m \longrightarrow \mathbb{Z}$, the three pullbacks below are nilcharacters of degree at most s and satisfy*

$$[\chi(L_1 + L_1', L_2, \dots, L_s)]_{\mathrm{Symb}^s} = [\chi(L_1, L_2, \dots, L_s)]_{\mathrm{Symb}^s} + [\chi(L_1', L_2, \dots, L_s)]_{\mathrm{Symb}^s}.$$

The same assertion holds in each coordinate. More explicitly, each component of

$$\chi(x + x', y_2, \dots, y_s) \otimes \overline{\chi(x, y_2, \dots, y_s)} \otimes \overline{\chi(x', y_2, \dots, y_s)} \tag{1}$$

is a polynomial nilsequence of degree at most $s-1$ on $\mathbb{Z}^{s+1}$. The representing nilmanifold is fixed by the original filtered nilmanifold; its test-function Lipschitz bound is bounded in terms of that fixed data and the original Lipschitz bound, independently of the polynomial coefficients.

The symbol notation abbreviates the degree-s equivalence relation of [1, Definitions 6.6 and 6.22]. Thus the explicit assertion about (1) supplies the asserted equality, with no approximation or loss in degree. It is enough to prove the ordinary statement with uniform fixed-data bounds and then take limit sequences. We supply that uniformity below. We do not assume G_0 equals the group generated by its positive multidegree groups.

2 Subgroups and lattices used in the construction

We use the following basic nilpotent Lie-group facts recorded in [1, Section 3, pp. 1242–1243]: the exponential is a homeomorphism; the connected subgroup generated by a set is the exponential of the generated Lie algebra; and, for a finite set A, its discrete generated group is cocompact in its connected real generated group. The last assertion is applied only when A lies in an already specified discrete lattice. Here are the compactness details needed to apply these facts.

Lemma 2. *Let M be a connected simply connected nilpotent Lie group with lattice Λ. Let $J_1, \dots, J_t$ be connected rational subgroups. Then the connected subgroup generated by the J_i is closed, simply connected, and rational in M. This also applies to diagonal or coordinate embeddings of such subgroups into a finite product of M.*

Proof. First a lattice Λ_J in a connected Lie group J is finitely generated. Choose a proper left-invariant path metric, and R such that every point is within distance R of some lattice point; compactness of J/Λ_J, followed by inversion, gives such an R. Follow a path from the identity to a lattice point in steps of length at most one and choose nearby lattice points, using the prescribed endpoints. Consecutive chosen lattice points differ by an element in the compact ball of radius $2R + 1$. Its intersection with the discrete lattice is finite and generates the lattice.

The connected real subgroup generated by this finite generating set is all of J. To justify this last assertion, let U be that subgroup. It contains Λ_J, so J/U is compact. A connected subgroup of a simply connected nilpotent group is closed, and its homogeneous space is a Euclidean space. One can see the latter directly by a chain of Lie subalgebras from $\mathfrak{u}$ to $\mathfrak{j}$, each of codimension one and an ideal in the next: the normalizer of a proper subalgebra of a nilpotent Lie algebra strictly contains it, so such a chain is obtained by choosing one normalizing vector at each step. Successive exponential coordinates give the Euclidean coordinates on J/U. For completeness, the normalizer assertion follows by taking the first member of the ascending central series not contained in the subalgebra; the preceding member is contained in it, so any vector in that member but outside the subalgebra normalizes it. Hence compactness forces $U = J$.

Choose a finite generating set for each $J_i \cap \Lambda$. Their union A lies in Λ and generates all the J_i as a connected real group. Its abstract generated group is discrete because it lies in Λ. The finite-set fact from [1, p. 1243] therefore makes it a lattice in $\langle A \rangle_{\mathbb{R}}$. Consequently the larger group $\Lambda \cap \langle A \rangle_{\mathbb{R}}$ is also a lattice there. A Lie subalgebra of a nilpotent Lie algebra exponentiates to a closed simply connected subgroup, which proves the other assertions. For a diagonal or coordinate embedding, apply the same argument to the images of the finite generating sets, which lie in the product lattice. □

We will also use the elementary central-quotient consequence explicitly. If J is such a group, Δ its lattice, and C a connected central rational subgroup, then J/C is connected simply connected nilpotent and the image of Δ is a lattice. Cocompactness follows from the continuous image of J/Δ. For discreteness, if lattice cosets tend to the identity coset, choose local lifts tending to the identity and write the lattice points as those lifts times elements of C. Reduce those elements modulo the lattice $C \cap \Delta$ into a fixed compact set. The adjusted lattice points now lie in a compact subset of J, so only finitely many occur. Thus their cosets cannot accumulate at the identity. Simple connectedness follows either from exponential coordinates or the homotopy sequence with connected kernel C.

3 Bounded normalization at zero

Write $\chi(n) = F(g(n)\Gamma)$, with the data in Theorem 1. Initially restrict to the fixed subgroup G_0 and its lattice $\Gamma_0 = G_0 \cap \Gamma$, restricting F through the smooth inclusion $G_0/\Gamma_0 \longrightarrow G/\Gamma$. This only changes a fixed metric comparison constant. In the following notation $G = G_0$ and $\Gamma = \Gamma_0$.

Choose a fixed compact set of representatives for G/Γ. Write $g(0) = q\gamma$, with q in that compact set and $\gamma \in \Gamma$. Set

$$h(n) = q^{-1}g(n)\gamma^{-1} = q^{-1}(g(n)g(0)^{-1})q, \qquad F_q(z\Gamma) = F(qz\Gamma). \tag{2}$$

Then $h(0) = 1$ and $F_q(h(n)\Gamma) = F(g(n)\Gamma)$ exactly. Each G_a is normal in G_0, because $[G_0, G_a] \subseteq G_a$. The constant map with value $g(0)^{-1}$ is polynomial into the original filtration. Product closure of polynomial maps [1, Corollary B.4], followed by the filtered conjugation by q, shows that h is polynomial into the same multidegree filtration. No conjugation of the lattice is made.

The maps $z\Gamma \mapsto qz\Gamma$ are smooth with uniformly bounded derivatives for q in the fixed compact set. Thus F_q has a uniform Lipschitz bound. If $F(tz) = \mathrm{e}(\eta(t))F(z)$ for $t \in G_{\mathbf{1}}$, then

$$F_q(tz) = \mathrm{e}(\eta_q(t))F_q(z), \qquad \eta_q(t) = \eta(qtq^{-1}). \tag{3}$$

This is a homomorphism on $G_{\mathbf{1}}$. Its restriction to $G_{\mathbf{1}} \cap \Gamma$ is integer-valued: put $z = \Gamma$ in (3) and use both $F_q(t\Gamma) = F_q(\Gamma)$ and the fact that $F_q(\Gamma)$ is a unit vector. In fact $\eta_q = \eta$. For each fixed $t \in G_{\mathbf{1}} \cap \Gamma$, the continuous map $q \mapsto \eta(qtq^{-1})$ on connected G_0 has integer values, so it equals its value $\eta(t)$ at the identity. The lattice generates $G_{\mathbf{1}}$ as a connected real group, by the argument in Lemma 2. Two continuous real homomorphisms agreeing on that lattice therefore agree on $G_{\mathbf{1}}$. This proves the assertion for every $q \in G_0$. In particular the normalization preserves the same standard vertical character, including for limit choices of q.

4 The descending auxiliary filtration

For $a \in \{0,1\}^s$ with $a \neq 0$, define the following homomorphisms on G_a whenever the indicated first coordinate occurs:

$$A_a(u) = (u,u,u) \quad (a_1 = 0), \qquad B_a(u) = (u,u,1), \quad C_a(u) = (u,1,u) \quad (a_1 = 1).$$

For $1 \leq i \leq s$ let H_i be the connected subgroup of G^3 generated by these images for all a with $|a| \geq i$. Set $H_{s+1} = \{1\}$ and

$$K = H_1, \qquad K_0 = K_1 = H_1, \qquad K_i = H_i \ (i \geq 1). \tag{4}$$

Every group just defined is fixed by the original filtration, not by g. Lemma 2 makes it a connected closed simply connected rational subgroup of G^3. Put $\Delta = K \cap \Gamma^3$. The sequence (K_i) is descending by its definition.

We check the commutator law. For generators of types A, A, A, B, and A, C, the coordinatewise commutator has respectively type A, B, C and belongs to the group with index $a + b$. The weight adds. If two generators both have first index one, every possibly nontrivial coordinate commutator belongs to $G_{a+b} = \{1\}$ because $(a+b)_1 = 2$. Thus the B, B, B, C, and C, C commutators are trivial. These assertions hold for their Lie algebras as well. Bracketing generated Lie algebras and using the Jacobi identity gives

$$[\mathfrak{h}_i, \mathfrak{h}_j] \subseteq \mathfrak{h}_{i+j} \quad (i, j \geq 1).$$

Here iterated brackets remain in the indicated space: each bracket is a sum of iterated brackets of the original generators, and its total original weight is at least $i + j$. BCH, which terminates for a nilpotent group, then gives $[H_i, H_j] \subseteq H_{i+j}$. The index-zero assertions also hold since $K_0 = H_1$: for example $[K_0, K_j] \subseteq H_{j+1} \subseteq H_j = K_j$. Hence (K_i) is a rational degree filtration, is taut at zero, and has degree at most s.

The top group has a particularly simple exact description:

$$K_s = H_s = \{(uv, u, v) : u, v \in G_{\mathbf{1}}\}. \tag{5}$$

Indeed $a = \mathbf{1}$ is the only index of weight s and has first coordinate one. The group $G_{\mathbf{1}}$ is abelian since $[G_{\mathbf{1}}, G_{\mathbf{1}}] \subseteq G_{\mathbf{21}} = \{1\}$. Thus its B and C images commute and their product is exactly the displayed connected group. It is central in K, also following from the commutator law.

5 The exact polynomial orbit

Choose once and for all a total order of $\{0,1\}^s$ extending its partial order. By [1, Lemma B.9], the normalized map has the Taylor expansion

$$h(n) = \prod_{a \in \{0,1\}^s \setminus \{0\}} u_a^{n^a}, \qquad u_a \in G_a, \qquad n^a = \prod_{j=1}^{s} n_j^{a_j}. \tag{6}$$

The absent constant coefficient equals $h(0) = 1$. Define

$$P(x, x', y) = (h(x + x', y), h(x, y), h(x', y)), \qquad y = (y_2, \ldots, y_s).$$

This is exactly the ordered product of the following factors, in the order of (6). If $a_1 = 0$, its factor is

$$A_a(u_a)^{\prod_{j=2}^{s} y_j^{a_j}}.$$

If $a_1 = 1$, its factor is

$$B_a(u_a)^{x \prod_{j=2}^{s} y_j^{a_j}} C_a(u_a)^{x' \prod_{j=2}^{s} y_j^{a_j}}.$$

The latter identity is exact because the powers of the same element u_a commute. In either case the integer exponent has degree $|a|$ and the fixed group element lies in $K_{|a|}$. A map $v^{Q(n)}$ of this form is polynomial into (K_i): its j-fold derivative is $v^{\Delta^j Q(n)}$, which lies in K_j for $j \leq \deg Q$, and is the identity for $j > \deg Q$. Here derivatives commute within the cyclic one-parameter subgroup generated by v, and $K_{\deg Q} \subseteq K_j$ is used in the first case. The values also lie in K_0. Product closure [1, Corollary B.4] now proves that

$$P \in \mathrm{poly}(\mathbb{Z}_{\mathbb{N}}^{s+1} \longrightarrow K_{\mathbb{N}}). \tag{7}$$

This argument proves the assertion for all integer inputs, not just a box. In particular it verifies that the orbit lies in the normalized ambient group K, which the exact-weight construction at index zero did not ensure.

6 The lower-degree quotient and its test function

Let $\overline{K} = K/K_s$, let $\pi : K \longrightarrow \overline{K}$ be the quotient map, and let $\overline{\Delta} = \pi(\Delta)$. The central-quotient argument after Lemma 2 shows that $\overline{K}/\overline{\Delta}$ is a compact nilmanifold with connected simply connected

nilpotent group. Give it the filtration $\overline{K}_i = \pi(K_i)$. It is taut and has degree at most $s-1$. For each $i \le s$, $K_s \subseteq K_i$, so

$$\overline{\Delta} \cap \pi(K_i) = \pi(\Delta \cap K_i).$$

The right side is cocompact in $\pi(K_i)$, by the compact quotient of $K_i/(\Delta \cap K_i)$; it is discrete as a subgroup of $\overline{\Delta}$. Thus the quotient filtration is rational at every index. Equation (7) remains polynomial after applying π, directly from the derivative definition.

Define on K/Δ

$$T_q(z_1, z_2, z_3) = F_q(z_1\Gamma) \otimes \overline{F_q(z_2\Gamma)} \otimes \overline{F_q(z_3\Gamma)}.$$

It is well-defined and Lipschitz, by restriction from $(G/\Gamma)^3$. For $(uv, u, v) \in K_s$, equation (3) multiplies this tensor by

$$\mathrm{e}\big(\eta_q(uv) - \eta_q(u) - \eta_q(v)\big) = 1.$$

Therefore T_q is invariant under K_s and descends to a function $\overline{T}_q$ on $\overline{K}/\overline{\Delta}$.

For explicit control of its regularity, fix a smooth metric on K/Δ and average it over the compact central torus $K_s/(K_s \cap \Delta)$. The averaged metric and the original fixed metric are uniformly equivalent. The quotient metric makes the quotient a Riemannian submersion. A function constant on its fibers and Lipschitz upstairs is Lipschitz downstairs with the same bound for these metrics, by horizontal path lifting. All the metric comparisons are fixed by the original filtered nilmanifold. The bound for T_q is at most a fixed multiple of the bound for F_q, since its three factors have norm one. The earlier compact normalization therefore makes the Lipschitz bound for $\overline{T}_q$ uniform in g.

Finally the expression (1) is exactly

$$\overline{T}_q(\pi(P(x, x', y))\overline{\Delta}).$$

This proves the claimed degree at most $s-1$ representation.

For clarity, the individual factors also have valid degree-s representations. In one copy of G, take the connected group J_i generated by G_a for $|a| \ge i \ge 1$, and set $J_0 = J_1$. The same rationality and commutator arguments give a taut rational degree-s filtration on J_1, with $J_s = G_{\mathbf{1}}$. The Taylor factors of h and of each of its linear pullbacks are polynomial for this filtration by the same integer-polynomial argument. Restricting F_q gives a unit-vector-valued Lipschitz test function with vertical character η_q. Hence those factors are nilcharacters of degree at most s.

Substitute the homomorphism $n \mapsto (L_1(n), L_1'(n), L_2(n), \dots, L_s(n))$ into the lower-degree representation. Derivatives, or the explicit polynomial factors above, show that degree does not increase. The definition of symbol equivalence proves Theorem 1. Relabeling coordinates gives the other coordinate identities.

7 Limit data and the actual published consumers

For a limit nilcharacter the filtered nilmanifold and its metric data are standard fixed data, and the polynomial and Lipschitz test functions are limit sequences with bounded Lipschitz norm. Choose the bounded representatives q separately in each member of those sequences; this gives a limit map with values in the same compact set. All groups K_i, the lattice, the quotient and the metric constants constructed above are unchanged along the sequence. The same standard vertical character is retained by the preceding argument. Representatives for the test functions may be taken to satisfy their defining unit-norm and vertical-character identities on an ultrafilter-large set of indices: otherwise, choosing a violating point, and for the second identity a violating group element, at every bad index would contradict the exact identity of the limit function at the

resulting limit points. This uses the all-limit quantifiers just specified and the evaluation rule in [1, Definition A.3]; it would be invalid with only standard-point quantifiers. Discard those bad indices. The formulas now hold at every remaining member, and the test-function bounds are uniform. Their limit is therefore a limit polynomial nilsequence of degree at most $s-1$ in precisely the required sense. No compactness argument selects a nilmanifold depending on an unbounded polynomial coefficient.

The first published use, immediately after Lemma 13.2 on p. 1318, expands $[\widetilde{\chi}(n+h,\dots,n+h)]$ by applying the identity in each coordinate. Before any use of symmetry the exact expansion is

$$[\widetilde{\chi}(n+h,\dots,n+h)] = \sum_{\omega\in\{0,1\}^s} [\widetilde{\chi}(z_{\omega_1},\dots,z_{\omega_s})], \qquad z_0 = n,\ z_1 = h.$$

Every map used here is a fixed integer homomorphism from $\mathbb{Z}^2$. Thus the coefficient counting and subsequent symmetry step in that argument receive exactly the same multilinearity statement; no stronger inverse theorem is asserted by this local repair.

The further explicit appeals in Lemmas 13.3, 13.4 and 13.5 on pp. 1319–1322 likewise use addition, subtraction and repeated copies of their variables. These are integer homomorphisms from the displayed finite Cartesian powers of $\mathbb{Z}$, so the theorem applies without a range or normalization change. Constant shifts in those arguments still require the separate symbol-shift assertion in E.8; this proof does not discharge it. The repaired lemma supplies the multilinearity input to Theorem 7.4, then to the published inverse-theorem argument. Other corrections, including the false original Proposition 8.3 and its replacement, remain separate obligations. Source and current-manuscript consumer spans are recorded in the accompanying contracts.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics **176** (2012), 1231–1372. The frozen final published PDF is the source for the numbered results used here. The arXiv v5 correction motivates the auxiliary generator types but is not used as an accepted theorem in this proof.

Difference decomposition and the additive-quadruple estimate

/root/sol_protocol_final_review

7 November 2026

1 Inputs and conventions

We use the natural-ultrapower convention of GTZ Appendix A and the project shift theorem in tautification-003. Its acceptance remains separate. A bound $\gg 1$ is positive and standard. Write N for an unbounded limit integer, $J = [N]$ and $I = [-N, N] \cap {}^*\mathbb{Z}$. All changes of variables are integer identities at the representative indices. The author erratum guides the proof. Its Proposition 3.1 and modified Proposition 7.3 are the conclusions here, rather than assumed premises.

The shift theorem gives a lower-support sequence with fixed geometry and uniform Lipschitz bound. It gives neither a separated finite decomposition nor a correlation conclusion. Products, conjugates, and translates use product manifolds, conjugated tests, and translated polynomial orbits, respectively. A bounded internal scalar depending only on parameters is degree zero in the averaging variable.

2 Difference decomposition

Proposition 1 (LP02)**.** *For standard $s \geq 3$, let χ be a multidegree $(1, s-1)$ nilcharacter. For every standard $\epsilon > 0$ each component of $D_k(h, n) = \chi(h+k, n) \otimes \overline{\chi(h, n)}$ is approximated uniformly in k, h, n to error ϵ by*

$$\sum_{j=1}^{L} \Psi_{j,k,a_j}(n) B_{j,k}(h, n).$$

Here L is standard, Ψ_{j,k,a_j} is a component of a unit-vector degree-$(s-1)$ nilcharacter, and $B_{j,k}$ is supported on the downset below $(1, s-2)$. The families are internal with standard finite geometry lists and uniform bounds.

Proof. Apply the project shift theorem, Theorem 3, with positive index $(1, 0)$ and shift $(k, 0)$. The support is in the union of the downsets $J_0 = \downarrow (0, s-1)$ and $J_1 = \downarrow (1, s-2)$. We now give the fixed geometric split explicitly. Write G/Γ for the taut lower-support representation from the shift theorem and $J = J_0 \cup J_1$. For each positive $a \in J$ take a rational basis of $\mathfrak{g}_a$ for the rational structure of Γ. This exists because G_a is rational. Take a free rational Lie algebra with one generator of weight a for each basis vector. Kill all homogeneous words whose total weight is outside J. The excluded weights form an upward-closed set, so these words generate a homogeneous ideal. All generator weights are positive. Thus word length is bounded by $\max_{a \in J} |a| = s - 1$ in the quotient, which is finite-dimensional and nilpotent. Let $\mathfrak{u} = \bigoplus_{a \in J \setminus \{0\}} \mathfrak{u}[a]$ denote it and let U be its simply connected real group. Set $\mathfrak{u}_b = \bigoplus_{a \geq b} \mathfrak{u}[a]$. This is a descending rational multidegree filtration with the required bracket rule.

Sending generators to their basis vectors defines a rational homomorphism $\rho : U \to G$. The killed words map to zero by the filtration rule in G. It maps each positive layer onto G_b: the

generators tagged b span $\mathfrak{g}_b$, and all weights at least b map into that layer. At zero it is onto because G is taut and generated by its coordinate groups $G_{(1,0)}$ and $G_{(0,1)}$. Thus the polynomial lifts exist by published GTZ Corollary B.10, applied to this layerwise-surjective filtered map, and can be selected internally at the representative indices.

Choose a homogeneous rational basis of $\mathfrak{u}$ and let L_0 be its integer span. There is a positive integer D such that $\Lambda = \exp(DL_0)$ is a lattice and $\rho(\Lambda) \subset \Gamma$. Here are the scaling details. In this basis BCH is a finite rational polynomial with linear part addition. For two inputs in DL_0, terms of degree at least two lie in DL_0 once D clears all the coefficient denominators. Inversion negates the logarithm. Thus this set is a subgroup, discrete in exponential coordinates. It is cocompact: order the basis by increasing total weight and successively reduce each weight's logarithm coordinates modulo D. Right multiplication by an element of the corresponding homogeneous lattice leaves lower weights unchanged, and at the current weight is ordinary addition. The resulting bounded box of logarithms is a compact set of representatives. The same argument applies to every filtration layer and every homogeneous quotient. To impose $\rho(\Lambda) \subset \Gamma$, use rational Mal'cev coordinates of the second kind adapted to Γ, as in GTZ Section 3. The coordinates of $\rho(\exp x)$ are finite rational polynomials in the chosen coordinates of x, with zero constant term. Taking D divisible by their denominators makes the coordinates of $\rho(\exp(Dx))$ integral for $x \in L_0$. This gives the required lattice inclusion. Increasing D preserves the BCH and compactness arguments.

For $i = 0, 1$, quotient U by the connected subgroup with Lie algebra the sum of homogeneous weights outside J_i. These sums are ideals by the downset property. The image lattices are the exponential lattices in the surviving homogeneous bases. The pair of quotient maps separates points of U/Λ: if two points have the same image in both, the logarithm of their group difference lies in the projected lattice in each quotient. Each homogeneous weight survives in at least one quotient. Hence every coordinate of that logarithm belongs to $D\mathbb{Z}$, so the difference lies in Λ. This proves separation on the quotient spaces, rather than just intersection-triviality of the two ideals.

The pullbacks of Lipschitz tests from the two compact quotients generate a conjugation-closed algebra containing constants and separating points. Stone–Weierstrass gives a uniform approximation by finite sums of products of these tests. The lifted tests of the shift family have a common Lipschitz and sup bound on this fixed compact manifold, since the map to G/Γ is fixed and smooth. A standard finite $\epsilon/4$ net for this bounded equicontinuous class gives finitely many standard product lists, each chosen to tolerance $\epsilon/4$. Select the first net entry within $\epsilon/4$ using an internal sup-norm comparison at representative indices. Use a smaller net radius if needed. Thus the splitting error is at most $\epsilon/2$, uniformly in k, h, n. Pad the lists with zero factors to a standard common length.

In the J_0 quotient all surviving generators have zero first weight and strictly positive second weight. The restricted one-variable degree filtration is therefore taut: its degree-zero and degree-one groups coincide. The highest layer is central by the bracket law and the degree bound. The orbit is independent of h, since the first-variable derivative lies in the trivial first-coordinate group. Restrict to $h = 0$ and apply published Lemma E.5 with this taut one-variable geometry of degree $s-1$. Use the ordinary functions constructed in its proof on printed p. 1353. For each of the finitely many standard J_0 tests, fix ordinary Lipschitz unit-vector tests with global vertical frequency and ordinary top-invariant coefficient functions. Extend these functions to the ultrapower after making these finite choices. The resulting tests have exact unit norm and equivariance at all limit points and all limit top-group elements, and internal selection from the finite lists preserves both identities. Its test decomposition gives linear functionals of unit-vector nilcharacters times degree-at-most-$(s-2)$ tests. Expand the functionals in coordinates and absorb their coefficients and lower-degree factors into the J_1 factors. The product still has support J_1. Choose individual Fourier tolerances after the finite lists and their sup bounds so that the added error is at most $\epsilon/2$. On fixing h, k, the lower factor has degree at most $s-2$ in n by the $(0,1)$ derivative index. All geometry lists are fixed and

finite, and all orbit data and choices are internal. □

3 The two averages

Proposition 2 (LP03)**.** *Under the hypotheses of GTZ Theorem 7.2, for standard $s \geq 3$, there are $\gg N^3$ additive quadruples in H for which*

$$\chi_{h_1}(n) \otimes \chi_{h_2}(n + h_1 - h_4) \otimes \overline{\chi_{h_3}(n)} \otimes \overline{\chi_{h_4}(n + h_1 - h_4)}$$

is $(s-2)$-biased on J. The witnessing tests have standard bounded complexity and positive standard correlation.

Proof. Extend f by zero outside J. GTZ A.11 and A.12 select the lower-degree correlation tests internally on the dense internal set $H \subset I$. The relation includes a selected positive lower bound $1/j$ with standard integer j. Select global representatives of the lower-degree tests and of the restricted nilcharacters by the same internal selection, storing the representations as part of the witness. Only their original values on J enter the unsmoothed identities. The bounded-complexity descriptions are a countable union of internal sets. The boundedness principle A.5 gives uniform standard complexity and a common positive lower bound. Expand the bounded-dimensional tensor contraction in coordinates. Retain a positive-density subset of H with a fixed witnessing coordinate pair by the standard finite pigeonhole. Write U_h, V_h for the conjugates of those coordinates of $\chi(h, \cdot)$ and χ_h. Then there are scalar lower-degree tests c_h such that

$$\left|\mathbb{E}_{n \in J} f(n + h)\overline{f(n)}U_h(n)V_h(n)c_h(n)\right| \geq \eta > 0$$

on the retained set, with standard uniform bounds. Align the phase by multiplying c_h by a unit scalar. Set $c_h = 0$ outside that set and extend V_h boundedly there. This is an internal choice. The real average over $h \in I$ is $\gg 1$.

Apply Cauchy–Schwarz in n to eliminate $\overline{f(n)}$. It gives

$$\mathbb{E}_{n \in J} \left|\mathbb{E}_{h \in I} f(n + h)U_h(n)V_h(n)c_h(n)\right|^2 \gg 1.$$

Expand the square and put $h' = h + k$, with $K = [-2N, 2N] \cap {}^*\mathbb{Z}$. The exact new parameter cutoff is $1_{h+k \in I}$, degree zero in n. On $I \times K$ the new normalized average is $|I|/|K|$ times the old one. With $c_{h,k} = 1_{h+k \in I} c_h \overline{c_{h+k}}$ this gives

$$\left|\mathbb{E}_{h \in I, k \in K, n \in J} f(n + h)\overline{f(n + h + k)}U_h(n)\overline{U_{h+k}(n)}V_h(n)\overline{V_{h+k}(n)}c_{h,k}(n)\right| \gg 1.$$

The U product is a component of D_k. Take LP02 with error less than half the correlation divided by a common bound for the other factors. A term of the standard finite sum retains a positive standard correlation. Absorb its lower factor into $c_{h,k}$.

For its surviving coordinate $\Psi_{k,a}(n)$ use the identity

$$\Psi_{k,a}(n) = \sum_b [\Psi_{k,a}(n)\overline{\Psi_{k,b}(n + h)}]\Psi_{k,b}(n + h).$$

The identity uses the unit vector norm of $\Psi_k(n+h)$. Individual coordinates need not have modulus one. The bracket is a component of the conjugate transpose of the shift tensor. The one-variable degree case of the project shift theorem gives degree at most $s-2$, uniformly and internally in h, k, using the finite geometries of LP02. Select a surviving term by the same finite averaging argument.

Absorb the bracket into $d_{h,k}$ and write the remaining coordinate as $P_k(n+h)$. The lower test is still exactly zero unless both $h, h+k$ are retained shifts.

Set $m = n+h$ and $M = [-2N, 2N] \cap {}^*\mathbb{Z}$. The change to the full M average, with the exact cutoff, multiplies the normalized correlation by $N/|M|$. We have

$$\left|\mathbb{E}_{m\in M,k\in K} f(m)\overline{f(m+k)}P_k(m)\mathbb{E}_{h\in I}1_J(m-h)V_h(m-h)\overline{V_{h+k}(m-h)}d_{h,k}(m-h)\right| \gg 1.$$

The outer factor depends only on (m,k) and is bounded. The second Cauchy–Schwarz in (m,k) gives $\gg 1$ for the mean squared modulus of the inner average. Expand with independent $h, h' \in I$. Put

$$(h_1,h_2,h_3,h_4) = (h, h'+k, h+k, h'), \qquad n = m-h.$$

The shift substitution is injective, with inverse $(h,h',k) = (h_1, h_4, h_3-h_1)$. It has $h_1+h_2 = h_3+h_4$. The first cutoff restricts n to J, where $m = n+h \in M$ automatically. The second is $1_J(n+h_1-h_4)$. The four factors in the new average are

$$V_{h_1}(n)V_{h_2}(n+h_1-h_4)\overline{V_{h_3}(n)V_{h_4}(n+h_1-h_4)}.$$

The lower factor is $d_{h,k}(n)\overline{d_{h',k}(n+h_1-h_4)}$ times the remaining cutoff. The ratio $N/|M|$ is comparable to one. The parameter vanishing is exact unless all four shifts are retained. This proof keeps zero extension throughout.

To absorb the cutoff, choose fixed $0 < \lambda < 1/4$. On the circle $\mathbb{R}/16\mathbb{Z}$ take a piecewise linear test equal to one on $[\lambda, 1-\lambda]$, zero off $[-\lambda, 1+\lambda]$ in $[-8,8)$, between zero and one, with Lipschitz bound $O(1/\lambda)$. Evaluate it at $(n+h_1-h_4)/N$. This is a one-step nilsequence in n, with fixed geometry and internal orbit. Its argument lies in $[-2,3]$ and does not wrap. The normalized mean absolute error against the cutoff is $O(\lambda + 1/N)$ uniformly in all shifts: only two endpoint bands of total $O(\lambda N+1)$ integers differ. Multiplying by bounded factors changes only the constant. Choose λ after the positive correlation bound so the error loses at most half. Since $s \geq 3$, the one-step test is absorbed into degree at most $s-2$. Parameter vanishing persists.

If the outer correlation has modulus at least $\beta > 0$ and each inner average has modulus at most B, at least a fraction $\beta/(2B)$ of the inner averages have modulus at least $\beta/2$. The shift triple box has size comparable to N^3. By injectivity, these yield $\gg N^3$ additive quadruples in H. Conjugating the scalar correlations undoes the initial choice of V_h and preserves lower degree. The resulting scalar correlations are coordinates of the claimed tensor, giving bias on the original interval J. □

4 Scope

The published proof inputs used here are GTZ Appendix A, Corollary B.10, E.5, the correlation conventions, and the stated hypotheses of Theorem 7.2. The project shift input and these exact applications await separate nonauthor review. This unit supplies the additive-quadruple result, not the subsequent frequency analysis, the other affected symbol rules, or the final GI theorem.

One-variable factorization and approximate additive linearization

Author proposal: operator-local agent /root/astra_v14_baseline_compiler

Protocol 1.4; exact source contracts for the Sections 9–11 repair

This is a new connecting proof, pending independent review. It does not approve itself, the complete inverse theorem, or any release. The earlier Section 13 packet is unchanged. We use the final published sources [GTZ], [GT12], [GT14], and [GT08] identified in the accompanying source record. Only one-variable factorization is asserted here.

1 Objects, source differences, and the required outputs

All nilpotent Lie groups below are standard, connected and simply connected; all lattices and filtrations are standard and rational. Subgroups described as rational are connected, and their intersection with the ambient lattice is cocompact in that subgroup. We use the natural ultrapowers of these objects, with the internal-function convention of [GTZ, Appendix A]. A bounded quantity is bounded by some standard constant. In the finite combinatorial proof all constants depend only on the stated positive density and error constants. Passing that finite proof to an ultrapower therefore preserves uniformity and internal choices. Braces denote the representative in $(-1/2, 1/2]$.

An $\mathbb{N}$-filtration $G_\bullet$ may have $G_1 \neq G_0 = G$. Its positive groups satisfy $[G_i, G_j] \subseteq G_{i+j}$ and are rational. The polynomial convention is the ordinary degree filtration on the source. In fixed adapted Mal'cev coordinates, write the coordinates of a polynomial map as a finite polynomial in n or $\binom{n}{j}$. Coordinate degree is bounded in terms of the fixed filtered group. For a smooth factor, "small Taylor coefficients" means that its constant coefficient is bounded and its coefficient of index $j \geq 1$ is $O(N^{-j})$, in the group Taylor convention of [GTZ, Lemma B.9]. Equivalent coordinate bounds are justified below.

The statement of [GTZ, D.2] cannot be used as a literal equivalence: on $\mathbb{R}/\mathbb{Z}$, $g(n) = n/N$ is equidistributed on $[N]$ but is not equidistributed on $[\lfloor N/2 \rfloor]$. For example its first Fourier average on that half interval tends to $2i/\pi$, whereas its full-interval average tends to zero. Its nontrivial horizontal character is $1/N$-Lipschitz. We instead use the one-variable weak factorization [GT12, Proposition 9.2], whose proof uses the quantitative theorem on an actual dense progression. The published correction [GT14, pp.1175–1176] explicitly retains the one-variable results. Its minor corrections on pp.1182–1183 are part of this source choice.

The properness requirement printed in the second bullet of [GTZ, D.4] is removed. Only a weak-factorization step that starts with failed total equidistribution forces a proper subgroup. Also, a rational periodic factor is not asserted to take values in Γ.

Theorem 1 (One-variable factorization, DF-D4)**.** *Let N be unbounded and let $g \in {}^*\mathrm{poly}(\mathbb{Z}, G_\bullet)$. There are a standard connected rational subgroup $P \leq G$ and internal polynomial maps $\epsilon, p, \gamma \in {}^*\mathrm{poly}(\mathbb{Z}, G_\bullet)$ such that*

$$g(n) = \epsilon(n)p(n)\gamma(n) \qquad (n \in {}^*\mathbb{Z}).$$

The map p takes values in ${}^*P$, with its induced filtration. On $[N]$, ϵ is bounded, is $O(1/N)$-Lipschitz, and has the small Taylor coefficient bounds above. There is a standard discrete subgroup $\Lambda \geq \Gamma$

with $\gamma(n) \in {}^*\Lambda$ for every n, and a standard integer $q > 0$ with $\gamma(n+q)\Gamma = \gamma(n)\Gamma$ for every n. For every standard finite-index subgroup $\Delta \le P \cap \Gamma$, $p(n)\Delta$ is totally equidistributed in P/Δ on $[N]$. The subgroup P is allowed to equal G or to be trivial.

Theorem 2 (Approximate four-function linearization, DF-F1)**.** *Let N be unbounded, let $j \ge 1$ be standard, and put $e = N^{-j}$. Let $H \subseteq [-N, N] \cap {}^*\mathbb{Z}$ be internal, and let $\xi_i : H \to {}^*\mathbb{T}$, $1 \le i \le 4$, be internal functions. Suppose that for at least cN^3 ordered quadruples in H^4 satisfying $h_1 + h_2 = h_3 + h_4$ one has*

$$\|\xi_1(h_1) + \xi_2(h_2) + \xi_3(h_3) + \xi_4(h_4) - a\|_{\mathbb{T}} \le Ce,$$

where $c > 0$ and $C < \infty$ are standard and $a \in {}^*\mathbb{T}$. For any specified index i there are standard $K < \infty$, $c' > 0$, an internal $H' \subseteq H$ with $|H'| \ge c'N$, internal $\alpha_k \in {}^*\mathbb{T}$, $\beta_k \in {}^*\mathbb{R}$, and $\delta \in {}^*\mathbb{T}$ such that

$$\xi_i(h) = \delta + \sum_{k=1}^{K} \{\alpha_k h\}\beta_k + O(e) \quad (h \in H'). \tag{F}$$

The implicit constant is standard and uniform over H'. The bracket sum restricted to H' is an exact Freiman homomorphism: it preserves every relation $h_1 + h_2 = h_3 + h_4$ with all four points in H'. No initial Freiman property, boundedness of the real β_k, or regularity of the input functions is assumed. All choices may be made internally.

2 Weak factorization and coefficient bounds

Lemma 3 (Weak step, DF-D3)**.** *Suppose first that $G_1 = G$. If $g(n)\Gamma$ is not totally equidistributed on $[N]$, it admits the factorization of Theorem 1 with p taking values in a standard proper connected rational subgroup, without the final equidistribution requirement. All other bounds and the discrete-overgroup and period assertions hold.*

Proof. Failed total equidistribution supplies an internal dense progression and a standard Lipschitz test with noninfinitesimal discrepancy. Thus for some standard $0 < \delta < 1/2$ it fails total δ-equidistribution in the quantitative convention of [GT12]. Reduce δ further to include the rationality constant of the fixed Mal'cev basis. Apply the one-variable case of [GT12, Proposition 9.2, pp.522–523] with this δ, the actual degree of $G_\bullet$, and the actual dimension of G. Its bounds are standard. The same argument works for an internal Lipschitz test with a standard norm bound: a finite uniform net in the standard Lipschitz ball first supplies a standard test with positive discrepancy. This uses compactness of G/Γ.

Here are the stronger bounds extracted from that source's construction, including the distinctions needed for later use. Write $g(0) = c\lambda$ with c in a fixed bounded fundamental set and $\lambda \in {}^*\Gamma$. Then $\widetilde{g} = c^{-1}g\lambda^{-1}$ has value 1 at zero. Left translation by bounded c changes a Lipschitz norm by a standard factor, and right multiplication by λ^{-1} changes no point of G/Γ. Thus the same failed-total-equidistribution argument applies to $\widetilde{g}$, with a possibly smaller standard δ.

The source's horizontal-character step gives a nonzero standard integer vector k with

$$\psi(\widetilde{g}(n)) = \sum_{r \ge 1} t_r \binom{n}{r}, \qquad \|k \cdot t_r\|_{\mathbb{T}} = O(N^{-r}). \tag{D1}$$

Here $k \cdot \psi$ is the real lift of the horizontal homomorphism; in particular it kills all commutators. Standardness of k follows because the quantitative bound leaves finitely many integer vectors. The source obtains (D1) by applying its Theorem 2.9 to the witnessing progression and undoing that

progression by its Lemma 8.4. This use does not assert the false single-scale equivalence described above.

Choose integers a_r nearest to $k \cdot t_r$. The coordinate support condition in [GT12, Lemma 6.7] says that t_r lies in the coordinate subspace for G_r. If k vanishes on that subspace, take $a_r = 0$ and $v_r = w_r = 0$. Otherwise fix a rational vector u_r in that subspace with $k \cdot u_r = 1$, and put

$$v_r = a_r u_r, \qquad w_r = (k \cdot t_r - a_r)u_r.$$

Define ϵ_0 and γ_0 by Mal'cev coordinates

$$\psi(\epsilon_0(n)) = \sum_r w_r \binom{n}{r}, \qquad \psi(\gamma_0(n)) = \sum_r v_r \binom{n}{r}, \qquad p = \epsilon_0^{-1}\widetilde{g}\gamma_0^{-1}. \tag{D2}$$

The same coordinate-support lemma and closure of polynomial maps under products show that these are polynomial for the original filtration. The real horizontal lift vanishes identically on p, since its coefficient at every index is $k\cdot t_r-(k\cdot t_r-a_r)-a_r = 0$. Consequently p lies in the connected kernel of that real homomorphism. This is a standard proper rational subgroup. A kernel of the circle-valued homomorphism alone would not establish connectedness; (D2) supplies the real zero required here.

Each w_r is $O(N^{-r})$. The vectors v_r have a common standard denominator, but their numerators need not be bounded. This denominator statement is all that is required. By [GT12, A.11–A.12], γ_0 is uniformly rational and periodic modulo Γ with a standard period. The more explicit common discrete overgroup argument below also applies. Set $\epsilon = c\epsilon_0$ and $\gamma = \gamma_0\lambda$. These are the correct ordered factors of the original g. Multiplication by bounded c preserves the stated small-coordinate bounds, and the added λ lies in the original lattice. The next lemma verifies all closure and Taylor assertions used here and in the descent. □

Lemma 4 (Closure of the controlled factors)**.** *In a fixed rational nilpotent group, finite products of bounded polynomial maps with coefficient bounds $O(N^{-r})$ retain those bounds and are $O(1/N)$-Lipschitz on $[N]$. These bounds imply the group Taylor bounds in Theorem 1. A finite collection of rational lattices commensurable with Γ, together with sets of points whose rational coordinates have a fixed common denominator, is contained in a standard discrete subgroup $\Lambda \geq \Gamma$. Any fixed-degree polynomial map into this group is periodic modulo Γ with a standard period.*

Proof. Coordinate products, inverses, and changes between first-kind and Mal'cev coordinates are fixed polynomials by [GT12, A.2–A.3]. In a coordinate polynomial $\sum a_r n^r$, the bounds $a_r = O(N^{-r})$ are equivalent to bounded coefficients after substituting $n = Nt$. Products of such polynomials retain that property. Converting back to the binomial basis gives at index r a fixed linear combination of coefficients of monomial indices $\ell \geq r$, hence $O(N^{-r})$. Constant coefficients stay bounded. On the fixed compact coordinate range for $0 \leq t \leq 1$, the group-coordinate maps have bounded derivatives; differentiation in n contributes N^{-1}. This proves the Lipschitz assertion, with any fixed standard metric.

For clarity about the group Taylor convention, one may assign formal weight r to each coefficient of n^r in the logarithm of a coordinate polynomial. The recursive Taylor extraction by $g(0)$, then $g_1^{\binom{n}{1}}$, then $g_2^{\binom{n}{2}}$, and so on uses only fixed BCH polynomials. Its coefficient $\log g_r$ contains no monomial of formal weight less than r. Indeed after removing indices below r, each surviving polynomial term of formal weight w has ordinary degree at most w. The residual group map is the identity at $0, \ldots, r-1$, as a formal identity in its input coefficients. Its logarithmic terms of weight $w < r$ are therefore polynomials of degree less than r with r zeros, and vanish. BCH

products add weights, so this assertion is preserved at each extraction step. The remaining finitely many monomials have weight at least r, and are $O(N^{-r})$. This proves the group Taylor assertion, including under products and fixed rational embeddings of subgroups.

For the discrete-overgroup assertion, use a rational basis adapted to the lower central series and let its weights be $w_i \geq 1$. All points of each given lattice have logarithmic coordinates with bounded denominators, as do the specified coordinate sets, by the fixed polynomial coordinate changes. Choose a standard integer D divisible by these denominators and by the denominators of all coefficients in the finite BCH formula in this basis. Increase D for the finitely many basis conversions if necessary. The set

$$L = \sum_i D^{-(2w_i-1)}\mathbb{Z}X_i$$

is closed under BCH. For a nonlinear term with $t \geq 2$ inputs of weights $w_{i_1}, \ldots, w_{i_t}$ contributing to output weight w_k, one has $w_k \geq \sum w_{i_a}$. Therefore the difference of denominator exponents is

$$(2w_k - 1) - \sum_a (2w_{i_a} - 1) \geq t - 1 \geq 1.$$

The extra power of D clears that BCH coefficient. Linear terms and negation also preserve L. Thus $\exp L$ is a standard discrete subgroup containing all the required points and Γ. It is cocompact, and $[\exp L : \Gamma]$ is finite. This argument also proves commensurability claims needed when a lattice of a rational subgroup is enlarged inside G.

For the period in the required right-coset convention, write the Mal'cev coordinate vector of $\gamma(n)$ as a polynomial of fixed degree D_0. Its values have a common standard coordinate denominator, since $\gamma(n) \in {}^*\Lambda$. Interpolating at $0, \ldots, D_0$ shows that its coefficients have a common standard denominator too. Their numerators may be arbitrary internal integers. By the rational polynomial formulas for multiplication and inversion, the vector

$$R(n, r) = \psi\big(\gamma(n)^{-1}\gamma(n + r)\big)$$

is a polynomial in the two integer variables n, r with a common standard coefficient denominator A. The identity $R(n, 0) = 0$ is a polynomial identity, so every monomial has a positive power of r. For the standard integer $q = A$, every coordinate of $R(n, q)$ is an integer for every internal integer n. Adapted Mal'cev coordinates identify Γ with the integer-coordinate points. Hence $\gamma(n)^{-1}\gamma(n + q) \in {}^*\Gamma$, exactly the assertion $\gamma(n + q)\Gamma = \gamma(n)\Gamma$. This proves the conclusion of [GT12, A.12] directly; its printed left quotient $\gamma(n + q)\gamma(n)^{-1}$ is not the quotient used here. □

3 Minimal subgroup descent, with all finite covers retained

Proof of Theorem 1. First address the filtration convention. If G_1 is proper, write $g(0) = c\lambda$ as above and set $p = c^{-1}g\lambda^{-1}$. Then $p(0) = 1$ and

$$p(n) = \lambda\big(g(0)^{-1}g(n)\big)\lambda^{-1} \in G_1.$$

The groups G_r are normal in G_0, so conjugation preserves them. The induced filtration on G_1 has zeroth and first group both G_1, and the positive groups remain G_r. Hence the one-variable result with the convention $G_0 = G_1$ applies there. This also shows directly that a controlled representation exists: $\epsilon = c$, $\gamma = \lambda$, and the represented subgroup is G_1. If $G_1 = G$, the identity-factor representation is already enough. Trivial groups cause no exception.

Call a subgroup P a representing subgroup if a factorization with all conclusions of Theorem 1 except total equidistribution exists. Choose one with least dimension among the nonempty collection of standard rational representing subgroups. This uses a least element of the finite set $\{0, \dots, \dim G\}$, not an infinite iteration or a uniform choice over all finite covers.

Suppose that for a standard finite-index $\Delta \le P \cap \Gamma$ the orbit $p\Delta$ is not totally equidistributed. The induced filtration of P is rational for Δ, since passing to a finite-index lattice does not change its rational Lie algebra. If $P \cap G_1$ is proper in P, the normalization just given already represents p in that proper subgroup, with a bounded constant on the left and a Δ-element on the right. Otherwise apply Lemma 3 in P/Δ to obtain

$$p = \epsilon_2 p_2 \gamma_2$$

with p_2 in a standard proper rational $P_2 < P$. Rationality relative to Δ is rationality relative to $P \cap \Gamma$, so P_2 is rational in G. The actual new factorization is

$$g = (\epsilon\epsilon_2) p_2 (\gamma_2 \gamma). \tag{D3}$$

The order of both products is necessary in a noncommutative group. Lemma 4 proves the small-coefficient assertions for the left product, in the ambient coordinates. For the right product, the rational lattice coordinates of γ_2 relative to Δ have common standard denominators in the original rational structure. The same lemma supplies a single discrete overgroup containing its values, the old overgroup, and Γ. It also supplies the standard period modulo Γ. Thus (D3) is a representation by the lower-dimensional subgroup P_2, contrary to minimality.

No choice of Δ was imposed in that minimality argument. Consequently the chosen p is totally equidistributed for every standard finite-index $\Delta \le P \cap \Gamma$, with the same factors. This is the required quantifier order. The argument only produces standard constants for this particular internal orbit and fixed standard geometry. It does not assert a prescribed effective bound uniform over all internal orbits. □

4 A finite combinatorial proof of the linearization contract

We prove a finite assertion with constants depending only on c, C and then transfer. Take $N \ge 1$, $0 < e \le 1$, and suppose the hypotheses of Theorem 2 hold in finite sets. Put $M = \lceil 1/e \rceil$ and $Q = 100N+1$. Round each $\xi_i(h)$ in the circle to $y_i(h) \in M^{-1}\mathbb{Z}/\mathbb{Z}$, with error at most $1/(2M) \le e/2$. Since $Me \le 2$, the approximate sum can take only a bounded number, depending on C, of values in this grid. For at least $c_1 N^3$ of the original quadruples its rounded value is a fixed a_0 in that grid. Here $c_1 > 0$ depends only on c, C.

Embed $[-N, N]$ in $B = \mathbb{Z}/Q\mathbb{Z}$ and identify the target grid with $A = \mathbb{Z}/M\mathbb{Z}$. In $B \times A$ form the four sets

$$A_1 = \{(h, y_1(h))\}, \quad A_2 = \{(h, y_2(h))\}, \quad A_3 = \{(-h, y_3(h))\}, \quad A_4 = \{(-h, y_4(h))\}. \tag{F1}$$

The signed first coordinates are essential. The selected quadruples now give $c_1 N^3$ representations of $(0, a_0)$ in $A_1 + A_2 + A_3 + A_4$. The initial additive relation has no wraparound, since $4N < Q$. Fourier inversion and Holder's inequality in this finite abelian group give

$$r_{A_1+A_2+A_3+A_4}(0, a_0) \le \prod_{i=1}^{4} E(A_i)^{1/4}, \qquad E(A_i) = |\{x_1 + x_2 = x_3 + x_4 : x_a \in A_i\}|.$$

For completeness, with unnormalized Fourier transforms the inverse formula has factor $|B \times A|^{-1}$; Holder bounds the sum of four transforms by the fourth moments, each of which equals $|B \times A|E(A_i)$. Those factors cancel. Each $E(A_i) \le |H|^3 \le 27N^3$. It follows that $E(A_1) \ge c_2 N^3$ for a positive $c_2 = c_2(c, C)$. The same reasoning can isolate any A_i; reflecting its first coordinate back if necessary gives the graph of the specified y_i. We henceforth call that function y and its graph A_1.

Apply [GT08, Theorems 5.2 and 5.3, p.95] to A_1 in the actual finite group $B \times A$. Their hypotheses follow from the energy bound and $|A_1| \le 3N$. They give a subset Γ' of its graph and a constant $K_0 = K_0(c, C)$ such that

$$|\Gamma'| \ge N/K_0, \qquad |9\Gamma' - 8\Gamma'| \le K_0 N. \tag{F2}$$

We use these published theorems themselves, not the special correlation premise in [GT08, Proposition 5.4].

Graph refinement with the actual cyclic target

Let $V = \{a \in A : (0, a) \in 8\Gamma' - 8\Gamma'\}$. The map $\Gamma' \times V \to 9\Gamma' - 8\Gamma'$ given by addition is injective: equality determines the first graph coordinate, then its unique second coordinate, and finally a. Hence $|V| \le K_0^2$. There is a set S_A of at most $O(1 + \log K_0)$ characters of A for which

$$V \cap \{a : \|\chi(a)\|_{\mathbb{T}} < 1/4 \ \forall \chi \in S_A\} = \{0\}.$$

Indeed, for any nonzero a a uniformly random character takes values uniformly among the roots of unity of its order; at least one third of those values have distance at least $1/4$ from zero. Choose enough independent characters that the union bound over $V \setminus \{0\}$ is less than one. This proves the required separation without identifying A with the dual of B, an identification that is unavailable here.

Partition each coordinate of the character vector on A into intervals of length at most $1/100$. A cell contains a subset $\Gamma'' \subseteq \Gamma'$ of size $c_3 N$. In any element of $8\Gamma'' - 8\Gamma''$, every selected character of the second coordinate has a real representative of absolute value less than $16/100 < 1/4$ when its first coordinate is zero. It follows that the only element of this difference set with first coordinate zero is $(0, 0)$. Therefore $4\Gamma'' - 4\Gamma''$ is a graph. Write H'' for the projection of Γ'', and let

$$\zeta : 2H'' - 2H'' \longrightarrow A$$

be the function whose graph is $2\Gamma'' - 2\Gamma''$. It is an exact Freiman homomorphism on this domain: two sums of two of its graph points lie in $4\Gamma'' - 4\Gamma''$, where the first coordinate determines the second. Also $\zeta(0) = 0$. Thus $\zeta(x + y) = \zeta(x) + \zeta(y)$ whenever $x, y, x + y$ lie in its domain. This proves precisely the generalization of [GT08, Lemma 9.2] needed here.

A proper progression and an explicit linear bracket formula

The set H'' has density at least $c_3/101$ in B. By [GT08, Lemma 6.4, pp.97–98] there is a bounded set of characters S with $B(S, 1/4) \subseteq 2H'' - 2H''$. Add the faithful character $x \mapsto x/Q$ to S, and shrink the radius to $\rho = 1/16$. Write $d = |S|$, a bounded integer. [GT08, Lemma 10.4, p.121] supplies a proper centered progression

$$P = \Big\{ \sum_{a=1}^{r} l_a v_a : |l_a| < L_a \Big\} \subseteq B(S, \rho), \qquad |P| \ge c_4 Q,$$

with $r \le d$, and linearly independent vectors $\Phi(v_a) = (\{\chi(v_a)\})_{\chi\in S}$. The coset subgroup in that lemma is zero because S contains a faithful character. Its proof also gives

$$\Phi\left(\sum_a l_a v_a\right) = \sum_a l_a \Phi(v_a) \quad \text{on } P. \tag{F3}$$

This is the lift of a lattice progression inside the real cube, not an unjustified inference from reduction modulo one. Delete inactive generators with no nonzero integer coefficient allowed; all remaining v_a lie in P. Every partial sum along a coordinate path to a point of this centered box stays in P. The local additivity of ζ consequently gives

$$\zeta\left(\sum_a l_a v_a\right) = \sum_a l_a \zeta(v_a).$$

Choose real lifts b_a of $\zeta(v_a)$ and real dual vectors u_a to the independent vectors $\Phi(v_a)$ on their span. Using (F3), define $b_\chi = \sum_a b_a (u_a)_\chi$. Then, for every $p \in P$,

$$\zeta(p) = \sum_{\chi\in S} b_\chi \{\chi(p)\} \pmod 1. \tag{F4}$$

This is a linear calculation. It uses neither the quadratic-extension theorem nor the quadratic classification lemma in [GT08]. The author notice concerning that extension theorem is therefore not a new assumption here.

Translation, fixed carries, and the Freiman restriction

Averaging translates of P in B gives an $x \in B$ with $|H'' \cap (x + P)| \ge |H''||P|/Q \ge c_5 N$. For $h = x + p$ and $h_0 = x + p_0$ in this intersection, the graph relation gives $y(h) - y(h_0) = \zeta(p - p_0)$, since $\Gamma'' - \Gamma''$ is contained in $2\Gamma'' - 2\Gamma''$. Both p, p_0 lie in the domain of ζ. Its Freiman identity applied to $(p - p_0) + p_0 = p + 0$ gives

$$y(h) = \delta_x + \zeta(p).$$

For each $\chi \in S$, write $\chi(h) = \alpha_\chi h$ on the cyclic group. The difference

$$\{\chi(h - x)\} - \{\chi(h)\} + \{\chi(x)\}$$

is an integer in a fixed set of at most three values. Restrict to a cell on which all these carries are fixed. This costs a factor at most 3^d and, by (F4), gives

$$y(h) = \delta + \sum_{\chi\in S} b_\chi \{\alpha_\chi h\}.$$

Finally partition each $\{\alpha_\chi h\}$ into a fixed number of intervals of length at most $1/10$, and retain a largest cell H'. For any integer relation $h_1 + h_2 = h_3 + h_4$ in H', the corresponding sum of four fractional parts, with signs $+, +, -, -$, is an integer of absolute value at most $1/5 < 1$. It is therefore zero in $\mathbb{R}$ for each character. The real weighted sum is an exact Freiman homomorphism on H'. Undoing the rounding changes the displayed value by at most $e/2$. All density losses depend only on c, C and all sets and choices are finite. This proves the finite assertion, and transfer with $e = N^{-j}$ proves Theorem 2. A bounded rank can be made a fixed standard rank by passing to its ultrapower value or by adding zero terms.

5 Consumer boundary and related-use scope

Theorem 1 supplies exactly the one-variable factorization consumed by the Sections 9–11 repair. That repair still must derive its nonzero coset integral and its degree-rank horizontal inclusion for the actual Taylor quotients. In particular, it may multiply a character by a standard denominator to kill the rational periodic contribution; it may not silently erase that contribution. No such application is accepted by this packet.

Theorem 2 applies separately at each of the finitely many required indices j, including $j = 1$, and to any chosen one of the four functions. The number of applications is standard. Dense refinements and all losses remain standard only when that finite iteration is recorded by the consumer. The theorem supplies no global Freiman extension beyond H'.

The related-use audit is bounded by uses of the single-scale equivalence, properness in D.4, composition of factorization factors, deletion of rational Taylor contributions, and the graph/Bohr/bracket joins in F.1. The multi-variable D.5 and its consumers in Appendix E require their own exact application review. They are not inferred from the one-variable theorem here. Likewise, this packet does not accept Gaussian's pending Sections 9–11 argument or any downstream assembled inverse theorem.

References

[1] [GTZ] B. Green, T. Tao and T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$-norm, *Annals of Mathematics* 176 (2012), 1231–1372. DOI: 10.4007/annals.2012.176.2.11.

[2] [GT12] B. Green and T. Tao, The quantitative behaviour of polynomial orbits on nilmanifolds, *Annals of Mathematics* 175 (2012), 465–540. DOI: 10.4007/annals.2012.175.2.2.

[3] [GT14] B. Green and T. Tao, On the quantitative distribution of polynomial nilsequences—erratum, *Annals of Mathematics* 179 (2014), 1175–1183. DOI: 10.4007/annals.2014.179.3.8.

[4] [GT08] B. Green and T. Tao, An inverse theorem for the Gowers $U^3(G)$ norm, *Proceedings of the Edinburgh Mathematical Society* 51 (2008), 73–153. DOI: 10.1017/S0013091505000325.

Equal-scale factorization and the bias and torsion joins

Author proposal: /root/astra_v14_baseline_compiler

Protocol 1.4; separate successor to the one-variable D/F packet

This packet does not revise the frozen D/F or Section 13 proofs. It supplies a proposed connecting proof for the previously open D.5-to-E.11/E.13 dependency. All acceptance and incorporation decisions remain independent. The only quantitative equidistribution input is the *final published* equal-side theorem in [GT14, pp.1176–1177], with the published minor corrections to [GT12]. No unpublished arbitrary-side extension is imported.

1 Conventions and exact statements

Fix standard positive integers k, d and an unbounded integer N. All groups, rational structures, lattices, coordinate bases and finite dimensions are standard, with natural ultrapower extensions. A nilpotent Lie group is connected and simply connected. The rational degree filtration $G_\bullet$ has finite length; the source $\mathbb{Z}^k$ has its total-degree filtration. Multiindices have nonnegative integer entries, and $|\alpha| = \sum_i \alpha_i$. All vertical identities for a limit nilcharacter hold at every limit group element and point, as in the accepted tautification-003 convention. We do not substitute equivariance only at standard points.

In fixed Mal'cev coordinates, small polynomial coefficients mean bounded constant coefficient and coefficient $O(N^{-|\alpha|})$ at positive index α in either the monomial or product-binomial basis. The constants are standard and may depend on the actual fixed geometric data and orbit. Total equidistribution on $[N]^k$ means equidistribution on every product of dense arithmetic progressions in $[N]$, with standard Lipschitz tests. It also applies to internal tests with a common standard Lipschitz bound: finite uniform nets in a compact Lipschitz ball reduce those tests to standard ones, with the desired accuracy chosen before taking the limit.

Theorem 1 (Equal-scale factorization, MF-D4)**.** *For $g \in {}^*\mathrm{poly}(\mathbb{Z}^k, G_\bullet)$ there are a standard connected rational subgroup $P \le G$ and internal polynomial maps ϵ, p, γ such that*

$$g(n) = \epsilon(n)p(n)\gamma(n) \qquad (n \in {}^*\mathbb{Z}^k). \tag{1}$$

The map p lies in P with its induced filtration, and is totally equidistributed on $[N]^k$ in P/Δ for every standard finite-index $\Delta \le P \cap \Gamma$, with the same p and P. The coefficients of ϵ are small as specified above; it is bounded and $O(1/N)$-Lipschitz on every fixed standard dilate $[-AN, AN]^k$. Its group Taylor coefficient at positive index α is $O(N^{-|\alpha|})$ as well. There is a standard discrete subgroup $\Lambda \ge \Gamma$ containing all values of γ, and a standard $q > 0$ such that

$$\gamma(n + qe_i)\Gamma = \gamma(n)\Gamma \qquad (n \in {}^*\mathbb{Z}^k, 1 \le i \le k). \tag{2}$$

There is no properness requirement on P, and no claim that γ is Γ-valued. All factors in (1) are defined on the full integer lattice.

Write $I = [-N, N] \cap {}^*\mathbb{Z}$.

Theorem 2 (Actual bias and torsion outputs, MF-E11/E13)**.** *Let $d \geq 2$ be standard, and let χ be a globally represented unit vector nilcharacter of degree d on ${}^*\mathbb{Z}^k$. If a scalar coordinate of χ has nonzero correlation with a degree at most $d-1$ nilsequence after multiplication by the indicator of a convex polytope in I^k and a product of dense progressions in $[N]$, then χ agrees on all of I^k with a globally represented degree-at-most-$(d-1)$ vector nilsequence. The same conclusion holds if $\chi^{\otimes a}$ agrees on I^k with a degree-at-most-$(d-1)$ vector nilsequence for a standard integer $a \geq 1$.*

These assertions give exactly the lower-degree conclusions of [GTZ, E.11] and [GTZ, E.13] on their required symmetric box. They include finite scalar-coordinate reductions for vector correlations. They do not require an extrapolation from an unrelated subprogression or the deletion of a rational Taylor coefficient.

2 A weighted inverse-affine estimate

The following estimate also supplies the large-offset transition behind [GT12, Proposition 9.2]. Merely inserting an offset of size N into the height bound of its Lemma 8.4 would not give the required standard bound.

Lemma 3 (Large offsets, bounded steps, MF-AFFINE)**.** *Let $f(n)$ be a real polynomial of fixed total degree at most D in k variables. Let A_i be internal integers with $|A_i| = O(N)$ and let b_i be nonzero integers of bounded absolute value. Put $\widetilde{f}(m) = f(A_1 + b_1 m_1, \dots, A_k + b_k m_k)$. Suppose every positive product-binomial coefficient $\widetilde{f}_\beta$ satisfies $\|\widetilde{f}_\beta\|_{\mathbb{T}} = O(N^{-|\beta|})$. There is a standard positive integer r, depending only on D, k and the step bounds, such that every positive coefficient of rf has the same bound $O(N^{-|\alpha|})$ modulo one.*

Proof. Invert the substitution over $\mathbb{R}$ and expand

$$\prod_i \binom{(n_i - A_i)/b_i}{\beta_i} = \sum_{\alpha \leq \beta} c_{\alpha\beta} \binom{n}{\alpha}. \tag{3}$$

Expanding each falling factorial first in monomials shows that the denominators divide a fixed integer formed from $\beta_i!$ and $b_i^{\beta_i}$. Conversion of ordinary powers to binomial coefficients uses integer Stirling coefficients. Thus one standard r clears all denominators, independently of the integer numerators A_i. The coefficient of n^ν before the last conversion is a polynomial in the A_i of total degree at most $|\beta| - |\nu|$. Consequently

$$|c_{\alpha\beta}| = O(N^{|\beta|-|\alpha|}), \qquad r c_{\alpha\beta} \in {}^*\mathbb{Z}. \tag{4}$$

The case $\beta = 0$ contributes only to $\alpha = 0$. For each $\beta > 0$ write $\widetilde{f}_\beta = z_\beta + e_\beta$, where z_β is an integer and $e_\beta = O(N^{-|\beta|})$. In the coefficient of index $\alpha > 0$ in rf, the contributions $r c_{\alpha\beta} z_\beta$ are integers, while

$$\sum_{\beta \geq \alpha} r c_{\alpha\beta} e_\beta = O(N^{-|\alpha|}). \tag{5}$$

There are only a fixed number of indices. This proves the claim. Negative steps cause no change after taking a positive common denominator. The uncontrolled constant coefficient causes no positive-index error. □

3 From a dense rectangle to the published equal-side theorem

Lemma 4 (Weak factorization in finitely many parameters)**.** *Suppose $G_0 = G_1 = G$. If $g\Gamma$ is not totally equidistributed on $[N]^k$, it has a factorization (1) with p in a standard proper connected rational subgroup, retaining all conclusions of Theorem 1 except the final equidistribution condition.*

Proof. Failure supplies a product of progressions $P_i = \{a_i + b_i t : 1 \le t \le L_i\} \subseteq [N]$ with $L_i \ge \delta N$, standard $\delta > 0$, and a bounded Lipschitz test with noninfinitesimal discrepancy. Their integer steps obey $1 \le b_i \le 2/\delta$ and their offsets are $O(N)$. Subtract the Haar mean, normalize the test and rotate a constant complex phase so that the real mean discrepancy is positive.

In the index rectangle $\prod[L_i]$, partition each axis into blocks of a common length $L = \lfloor \eta N \rfloor$, leaving a final remainder of length less than L. Choose the standard $\eta > 0$ small compared with the discrepancy, δ, k and the test norm. The discarded proportion is at most $\sum_i L/L_i \le k\eta/\delta + o(1)$, so its contribution is less than one quarter of that discrepancy. At least one full cube of side L has a positive standard discrepancy. On this cube the polynomial is $\widetilde{g}(m) = g(A_1 + b_1 m_1, \dots, A_k + b_k m_k)$ with $|A_i| = O(N)$.

Apply the exact equal-side theorem of [GT14, pp.1176–1177] at length L, using a standard error parameter below this discrepancy and below the reciprocal rationality bound of the fixed adapted basis. The result gives a nonzero bounded horizontal integer character ξ for which all positive coefficients of the real horizontal lift of $\xi\widetilde{g}$ have circle norm $O(L^{-|\alpha|}) = O(N^{-|\alpha|})$. Only finitely many such characters are possible, so the selected character is standard. Lemma 3 replaces it by a nonzero standard multiple and gives the same coefficient bounds for ξg on the original variables. This is the precise rectangle-to-equal-cube join; the original unequal-side theorem is not used.

Normalize $g(0) = c\lambda$ with bounded c and $\lambda \in {}^*\Gamma$ and apply the preceding argument to $c^{-1}g\lambda^{-1}$, whose failure is preserved by bounded left translation. Its coordinates are $\sum_{\alpha>0} t_\alpha \binom{n}{\alpha}$, with t_α supported in $G_{|\alpha|}$ by [GT12, Lemma 6.7]. Write the real horizontal lift as $u \cdot \psi$, with u a fixed integer vector. Choose integers a_α nearest to $u \cdot t_\alpha$. On a support where u is nonzero choose a fixed rational vector v_α of that support with $u \cdot v_\alpha = 1$, and define

$$w_\alpha = (u \cdot t_\alpha - a_\alpha)v_\alpha, \quad z_\alpha = a_\alpha v_\alpha, \quad \psi(\epsilon_0(n)) = \sum w_\alpha \binom{n}{\alpha}, \quad \psi(\gamma_0(n)) = \sum z_\alpha \binom{n}{\alpha}. \tag{6}$$

If u vanishes on a support, set all these vectors and its integer to zero. Coordinate support and polynomial group closure make these maps polynomial for the original filtration. The real lift is identically zero on $p = \epsilon_0^{-1} c^{-1} g \lambda^{-1} \gamma_0^{-1}$. Thus p lies in its standard proper connected rational kernel. The original factors are $\epsilon = c\epsilon_0$ and $\gamma = \gamma_0 \lambda$. The w_α have precisely the required small bounds; the z_α have a common standard denominator, without a numerator bound. Controlled products and periods are checked next. □

4 Controlled products, periods, and descent

Scale all variables by $n_i = Nt_i$. Fixed coordinate polynomials with small coefficients become polynomials in t with bounded coefficients. The fixed BCH and coordinate-change polynomials preserve this property under finite products, inverses and fixed rational subgroup embeddings. Conversion between monomial and product-binomial bases is triangular in componentwise indices, so the coefficient of $\binom{n}{\alpha}$ is $O(N^{-|\alpha|})$. Differentiating the scaled coordinate maps on the compact box $[-A, A]^k$ gives the claimed $O(1/N)$ Lipschitz bound on every fixed dilate. A real-coordinate polynomial extension of ϵ has the same bounds, which will be used below.

These estimates also imply the group Taylor bounds. Assign formal weight $|\alpha|$ to each logarithmic coefficient of n^α and weight zero to the bounded constant term. Group Taylor extraction in any fixed admissible order uses fixed BCH polynomials. Inductively every term of formal weight w has coordinate polynomial total degree at most w; removing a Taylor factor at index β preserves this property because its binomial power has degree $|\beta|$. The coefficient at index α cannot contain a term of formal weight less than $|\alpha|$: such a term has total degree less than $|\alpha|$ and has zero coefficient at that index. This is the same triangular Taylor extraction as [GTZ, B.9], now with product-binomial indices. Each surviving BCH monomial contributes at least $N^{-|\alpha|}$ decay. In particular no estimate is inferred solely from separate degree in individual variables.

The common discrete overgroup argument is independent of the number of parameters. In a rational lower-central basis of weights w_i, choose a standard D clearing every coordinate denominator and every fixed BCH coefficient. As in DF002, the additive lattice $\sum_i D^{-(2w_i-1)}\mathbb{Z}X_i$ is BCH-closed: a nonlinear term with t inputs has output denominator exponent exceeding the input sum by at least $t-1$. Its exponential is a standard discrete overgroup $\Lambda \geq \Gamma$ containing the rational factors and any finite product of them.

Here is the full multivariable period argument, in the right-coset direction. Each coordinate of $\gamma(n)$ is a polynomial of fixed degree whose values have a common standard denominator. Tensor-product interpolation on a fixed integer grid gives a standard common denominator for all its coefficients. For every i the vector

$$R_i(n,r) = \psi\big(\gamma(n)^{-1}\gamma(n+re_i)\big)$$

is a rational coefficient polynomial with a common standard denominator. It is identically zero when $r=0$. Choose a standard positive q divisible by all these denominators. Each monomial then has a positive power of r, so $R_i(n,q)$ is integral for every internal integer vector n. This is exactly (2). This proof does not use the wrongly oriented quotient in the printed proof of [GT12, A.12]. The corresponding one-variable sublemma of DF002 has a separate independent root review.

To finish Theorem 1, first normalize the constant term and restrict to G_1 if $G_1 \neq G$. The identity

$$c^{-1}g(n)\lambda^{-1} = \lambda\big(g(0)^{-1}g(n)\big)\lambda^{-1} \in G_1$$

holds in any number of variables, since first derivatives lie in G_1 and the positive groups are normal in G_0. Thus a controlled representation exists with a taut positive filtration. Choose a standard rational representing subgroup of least dimension. For any standard finite-index lattice $\Delta \leq P \cap \Gamma$, failure of total equidistribution in P/Δ permits Lemma 4, after the same normalization if its induced first group is proper. It supplies $p = \epsilon_2 p_2 \gamma_2$ in a proper rational $P_2 < P$. The correct new identity is

$$g = (\epsilon\epsilon_2)p_2(\gamma_2\gamma).$$

The preceding controlled-product and discrete-overgroup arguments verify every required bound and period. Rationality relative to Δ is the same rational structure as relative to $P \cap \Gamma$. This contradicts minimal dimension. It proves total equidistribution for every standard finite-index cover with the same factors. Only a minimum among $0, \ldots, \dim G$ was selected; no infinite sequence of refinements or dependence of the chosen factors on the later cover is hidden here.

5 One quotient for all periodic residue classes

We now work in a taut ambient degree-d group G, so G_d is central. Tautification-003, Theorem 1, supplies this exact representation for each nilsequence used below with the same positive-degree groups and the same standard vertical character. Products of these representations remain taut.

Apply Theorem 1, and choose a standard representative $\lambda_\rho \in \Lambda$ for each value of the right coset of $\gamma(n)$ on a residue class $n \equiv \rho \pmod q$, $\rho \in (\mathbb{Z}/q\mathbb{Z})^k$. Such representatives can be standard: Λ/Γ is a fixed finite set, and periodicity makes the coset constant on each class. Let

$$\Gamma_0 = \bigcap_{\lambda \in \Lambda} \lambda \Gamma \lambda^{-1}, \qquad \Delta = P \cap \Gamma_0. \tag{7}$$

The first intersection is the kernel of the action on the finite coset set, so it is a standard normal finite-index subgroup of Λ contained in Γ. Thus Δ is finite index in $P \cap \Gamma$, and p is totally equidistributed on P/Δ. For every ρ the function

$$x \longmapsto F(zx\lambda_\rho \Gamma) \tag{8}$$

is well-defined on this *same* quotient: for $\delta \in \Delta$, $\lambda_\rho^{-1}\delta\lambda_\rho \in \Gamma$. No deletion of a rational horizontal coefficient is involved.

Lemma 5 (A nonzero integral from the actual cutoff)**.** *Let F be an internal bounded Lipschitz scalar test of standard norm on G/Γ. If*

$$|\mathbb{E}_{n \in I^k} 1_\Omega(n) 1_{P_1 \times \cdots \times P_k}(n) F(g(n)\Gamma)| > c > 0 \tag{9}$$

for a convex polytope $\Omega \subseteq I^k$ and dense progressions $P_i \subseteq [N]$, then for some residue ρ and bounded $z = \epsilon(n_0)$ one has

$$\left| \int_{P/\Delta} F(zx\lambda_\rho \Gamma)\, d\mu(x) \right| > c' > 0. \tag{10}$$

Here c' is standard and positive. The factors and quotient are those just constructed for the same actual g.

Proof. Partition the progression product into its finitely many nonempty residue classes modulo q. Each intersection is again a product of dense progressions, by the elementary congruence calculation for each axis. Their steps are bounded by the old step times q. Rotate a common complex phase in (9); one class retains a real sum at least $c_0 N^k$ for a standard $c_0 > 0$. Fix this lower bound before selecting a mesh.

Use index coordinates on that progression product. The inverse image of Ω intersected with the index rectangle is convex and lies in a box of side $O(N)$. Partition it by cubes of side $\lfloor \eta N \rfloor$. The cubes meeting its boundary have at most $O_k(\eta N^k) + O_k(N^{k-1})$ integer points, with constants also depending on the fixed step bounds. Indeed place disjoint unit cubes at those integer points and enclose them in the $O_k(\eta N + 1)$ neighborhood of the convex boundary; apply the published boundary-volume estimate [GT10, Corollary A.2, p.1810]. For a convex set with empty interior, first thicken by an arbitrarily small ball and then take the decreasing limit of its boundary neighborhoods. Choose standard η small enough that the discarded boundary contribution is below $c_0 N^k/4$. The total number of points in the full cubes is at most N^k. Some cube contained in the convex set therefore gives a dense progression product R whose phase-adjusted average is at least $3c_0/4$.

The choice of η can also ensure that $|F(\epsilon(n)x\lambda_\rho\Gamma) - F(\epsilon(n_0)x\lambda_\rho\Gamma)|$ is below $c_0/4$ for all $n, n_0 \in R$ and all $x \in P/\Delta$. This follows from the $O(1/N)$ variation of ϵ, its bounded image, the fixed standard representative λ_ρ, and compactness of P/Δ. Choose the mesh only after these standard constants and the original positive discrepancy are known. For $n \in R$, (1) and the fixed right coset give $F(g(n)\Gamma) = F(\epsilon(n)p(n)\lambda_\rho\Gamma)$. Freeze ϵ at n_0. Total equidistribution of the same p in P/Δ on R identifies the resulting average with the integral in (10), up to an infinitesimal error. Internal-test uniformity follows from the finite Lipschitz net specified at the start. This proves (10). □

6 Exact reconstruction on the symmetric box

Let $G = G^{\chi} \times G^{0}$ be the taut product of a degree-d nilcharacter representation and any degree-at-most-$(d-1)$ auxiliary representation. The latter has trivial degree-d group. Write F_{χ} for the original unit vector test, and let η be its real standard vertical character. Set

$$K = P_d = P \cap G_d.$$

We justify the rational intersection with the original lattice explicitly. Fix a rational Mal'cev basis for G/Γ. A connected subgroup H with compact $H/(H \cap \Gamma)$ has rational Lie algebra in this basis. Indeed, [GT12, Lemma A.2] makes the coordinates of $\log \gamma$ rational for $\gamma \in \Gamma$. The Lie algebra $\mathfrak{l}$ generated by $\log(H \cap \Gamma)$ therefore has a rational basis, by selecting finitely many independent iterated brackets. If $\mathfrak{l}$ were proper in $\mathfrak{h}$, put it in a maximal proper subalgebra $\mathfrak{m}$. The last lower central term not contained in $\mathfrak{m}$ supplies a vector outside $\mathfrak{m}$ normalizing it. Thus maximality makes $\mathfrak{m}$ an ideal, and $\mathfrak{h}/\mathfrak{m}$ has dimension one, since otherwise a line would be a proper nonzero subalgebra. The quotient functional exponentiates by BCH to a surjection $H \to \mathbb{R}$ killing $H \cap \Gamma$, contradicting compactness. Hence $\mathfrak{l} = \mathfrak{h}$. Apply this argument to P and G_d. Their Lie-algebra intersection has a rational basis by rational linear algebra. The global exponential homeomorphism and BCH identify K with its exponential, so K is closed and connected. Equip K with its rational lower central filtration. Choose a standard Q bounding the finitely many rational coordinates. [GT12, Definition 2.5 and Proposition A.10] then give a Mal'cev basis for $K/(K \cap \Gamma)$, so this actual intersection is a lattice. The trivial subgroup case is immediate. Since G_d is central in the taut ambient group, K is central too. The coset map from $(K \cap \Gamma)/(K \cap \Delta)$ to $(P \cap \Gamma)/\Delta$ is injective. Thus $K \cap \Delta$ has finite index in $K \cap \Gamma$ and is a lattice in K, without requiring normality of Δ. The central-quotient argument in LP13-002, Section 2, now applies with its rationality premise established: the image $\overline{\Delta}$ of Δ is a lattice in $\overline{P} = P/K$. The quotient filtration kills $P_d = K$ and hence has degree at most $d-1$. All these objects depend only on the fixed standard geometry.

Suppose that

$$\eta(\mathrm{pr}_{\chi} K) = 0. \tag{11}$$

For each ρ and bounded real-coordinate parameter $t \in [-1/5, 1/5]^k$, define, initially on P/Δ,

$$H_{\rho}(x,t) = F_{\chi}(\widetilde{\epsilon}_{\chi}(10Nt)\, x_{\chi}\, (\lambda_{\rho})_{\chi} \Gamma_{\chi}), \tag{12}$$

where $\widetilde{\epsilon}$ is the real-coordinate polynomial extension. This is well-defined by (7). It is invariant under K by the all-limit vertical identity and (11); centrality permits its element to pass across $\widetilde{\epsilon}$ and λ_{ρ}. Hence (12) descends to $\overline{P}/\overline{\Delta}$. The scaled-coefficient estimates imply a standard Lipschitz bound in t, uniformly in N. Smooth multiplication on fixed compact ranges gives the same conclusion in x. Invariance descends this bound to the quotient metric; any fixed standard Riemannian metric is equivalent on this compact nilmanifold.

Choose a fixed Lipschitz function $B(t)$ on $\mathbb{T}^k$ equal to one on $[-1/10, 1/10]^k$ and zero outside $(-1/5, 1/5)^k$. Using centered representatives inside that chart, define $\widetilde{H}_{\rho}(x,t) = B(t) H_{\rho}(x,t)$ there and zero outside. It is a globally bounded Lipschitz vector function. The zero collar makes the definition independent of the chart boundary. The functions are limit functions with a standard bound, since their coordinate-level construction has uniform bounds before taking the ultrapower.

For each $\rho \in (\mathbb{Z}/q\mathbb{Z})^k$ choose a standard Lipschitz function b_{ρ} on $\mathbb{T}^k$ with $b_{\rho}(\sigma/q) = 1_{\rho=\sigma}$. For example use products of triangular bumps of radius $1/q$; for $q = 1$ use the constant function one. The vector function

$$\mathcal{H}(x,t,z) = \sum_{\rho} b_{\rho}(z) \widetilde{H}_{\rho}(x,t) \tag{13}$$

is globally bounded Lipschitz with a standard bound. Its orbit is

$$n \longmapsto (\overline{p}(n)\overline{\Delta}, n/(10N),\, n/q). \tag{14}$$

This orbit has degree at most $d-1$: its first component has that degree and both tori have degree one, with $1 \le d-1$. For every $n \in I^k$, exactly the correct residue value is sampled in (13), and $B(n/(10N)) = 1$. The identities (1), (2), and (12) give

$$\chi(n) = \mathcal{H}((14)). \tag{15}$$

This is an equality of the actual vector functions on the entire symmetric box. It is not an extrapolation of an equality on just one progression. No unit condition is imposed on $\mathcal{H}$ away from the orbit; a lower-degree vector nilsequence does not require one.

7 Bias and torsion applications on the same objects

Proof of Theorem 2. For the bias assertion, choose the scalar coordinate and the lower-degree scalar test occurring in the actual nonzero correlation. If a vector correlation is given, a fixed finite coordinate choice supplies one such scalar nonzero average; its density loss is standard. Form the taut product representation $G^{\chi} \times G^0$ and the product test $F = (F_\chi)_i \overline{F_0}$, using the sign in that correlation. It has vertical character η on the degree-d group because G_d^0 is trivial. Apply Theorem 1 and Lemma 5 to this actual product orbit and test, obtaining a nonzero integral (10).

Translation of Haar measure on P/Δ by any $u \in K$ preserves that integral. The integrand is multiplied by $e(\eta(\mathrm{pr}_\chi\, u))$, since K is central in the ambient taut group. The integral is nonzero, so this phase is one for every standard $u \in K$. The real homomorphism η is continuous and K is connected; its image is therefore a connected subset of $\mathbb{Z}$, hence zero. Its natural ultrapower extension vanishes on every limit element too. This is (11). Equations (12)–(15), with the *original* vector test F_χ, now prove the asserted all-box lower-degree representation.

For the torsion assertion, let Ψ be the given degree-at-most-$(d-1)$ vector nilsequence equal to $\chi^{\otimes a}$ on I^k. Put its finitely many coordinates into one common product nilmanifold of that degree. Since χ is a unit vector, pointwise on $[N]^k$ one has

$$\sum_j (\chi^{\otimes a})_j(n)\overline{\Psi_j(n)} = 1.$$

At least one scalar term has a nonzero standard average, with lower bound at least the reciprocal of the fixed tensor dimension in absolute value. Use that term as the test on the product of the original χ geometry and the auxiliary geometry of Ψ. Its top vertical character is $a\eta$. Lemma 5 applies with the full positive cube and no nontrivial cutoff. The same Haar argument gives $a\eta(\mathrm{pr}_\chi\, K) = 0$ as a real homomorphism. As $a \ge 1$, (11) follows. The same reconstruction (12)–(15) uses the original F_χ, not its tensor power, and proves the required conclusion.

In the symbol formulation of [GTZ, E.13], $a[\chi] = 0$ means exactly that the tensor power is equivalent to the constant nilcharacter. By the definition of equivalence, its coordinates are lower-degree nilsequences on the specified box. This is precisely the Ψ premise just used, with a finite common representation. No torsion-free conclusion in degree one is asserted; the time and residue torus factors explain the requirement $d \ge 2$. □

8 Boundaries of this proposal

This proof supplies the finite-parameter factorization actually needed by E.11/E.13 and their all-symmetric-box reconstruction. It does not assert the false horizontal Taylor closeness printed in D.5. Gaussian's separate one-variable D.6 argument must retain its denominator correction in the degree-rank horizontal quotient. The inverse-affine Lemma 3 also validates the large-offset coefficient transition in the one-variable weak-factorization source. That is a related use requiring an explicit link to this frozen version; the earlier DF002 bytes and earlier decisions are preserved. The Section 13 application still needs an independent decision binding its exact E.11/E.13 objects and domains to this proposal, and the complete inverse theorem still needs its other separately assigned components.

References

[1] [GT14] B. Green and T. Tao, On the quantitative distribution of polynomial nilsequences—erratum, Ann. of Math. 179 (2014), 1175–1183. DOI: 10.4007/annals.2014.179.3.8.

[2] [GT12] B. Green and T. Tao, The quantitative behaviour of polynomial orbits on nilmanifolds, Ann. of Math. 175 (2012), 465–540. DOI: 10.4007/annals.2012.175.2.2.

[3] [GT10] B. Green and T. Tao, Linear equations in primes, Ann. of Math. 171 (2010), 1753–1850. DOI: 10.4007/annals.2010.171.1753.

[4] [GTZ] B. Green, T. Tao and T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$-norm, Ann. of Math. 176 (2012), 1231–1372. DOI: 10.4007/annals.2012.176.2.11.

Sections 9–11 at every rank step

Author proposal: /root/astra_v14_baseline_gaussian

7 November 2026

Status. This is a source-to-consumer project reconstruction, not independent acceptance. It separates two analytic supplier contracts, F and D, from the algebraic, selection and representation joins proved here. Their separate reconstruction and review remain required. The universal group uses the explicit fixed-rank weighted presentation below. No final inverse theorem is used as a premise.

1 The family at an arbitrary rank

Fix a standard $s \geq 3$, $m = s - 1$ and $1 \leq R \leq m$, and an unbounded limit integer N. All groups, lattices, metrics and finite dimensions below are standard unless a natural ultrapower is displayed or understood. Braces use $(-1/2, 1/2]$ throughout this packet. The Section 12 constructor allows either fixed half-open fundamental cube; use this same choice there. Let $H \subseteq [-N, N] \cap {}^*\mathbb{Z}$ be dense and internal. The input consists of a bounded function f on $[N]$, a global unit-vector multidegree $(1, m)$ nilcharacter $A(h, n)$, and an internal family of unit-vector degree–rank (m, R) nilcharacters χ_h. The latter have a fixed finite list of rational models and standard uniform Lipschitz bounds. For every $h \in H$, $\Delta_h f$ has uniformly positive degree-at-most-$m-1$ correlation with $A(h, \cdot) \otimes \chi_h$, with bounded internal witnessing tests. Every vertical identity is on the full natural extension of its group and compact quotient, not merely their standard points.

This is exactly the new input in rank-iteration001, Section 3. In particular no core/petal representation or independence statement is inherited from the preceding rank. They will be reconstructed. The corrected LP03 gives, after any dense internal restriction of H, many additive quadruples $h_1 + h_2 = h_3 + h_4$ for which

$$\chi_{h_1}(n) \otimes \chi_{h_2}(n + h_1 - h_4) \otimes \overline{\chi_{h_3}(n)} \otimes \overline{\chi_{h_4}(n + h_1 - h_4)} \tag{1}$$

is $(m-1)$-biased, with standard uniform witnesses. This hereditary statement means reapplying LP03 to the restricted correlation hypothesis. It does not assert that a previously chosen dense set of quadruples survives restriction.

Analytic contract F. For internal functions $X_1, \ldots, X_4 : H \to {}^*\mathbb{T}$ and $\epsilon = N^{-j}$, if $X_1(h_1) + \cdots + X_4(h_4) = O(\epsilon)$ on many additive quadruples, there are a dense internal H', a standard finite K, and internal $\alpha_k \in {}^*\mathbb{T}$, $\beta_k \in {}^*\mathbb{R}$, $\delta \in {}^*\mathbb{T}$ such that

$$X_4(h) = \sum_{k=1}^{K} \{\alpha_k h\} \beta_k + \delta + O(\epsilon) \quad (h \in H').$$

The errors have a common standard bound. A permutation of the four positions changes which function is recovered. This is the precise Appendix F.1 input; the signed-graph and no-wrap argument is a separate analytic supplier.

Analytic contract D. For a polynomial sequence in one variable in a fixed rational filtered nilmanifold there is a factorization $g = \varepsilon g'\gamma$ with ε bounded, Lipschitz $O(1/N)$ and positive-index Taylor coefficients $O(N^{-i})$; g' takes values in a connected rational subgroup Q and is totally equidistributed on every standard finite cover of $Q/(Q \cap \Gamma)$; and γ takes values in a fixed discrete overgroup $\widehat{\Gamma} \supseteq \Gamma$ and is periodic modulo Γ with standard period. The subgroup Q is allowed to equal G. All factors remain adapted to the ordinary degree filtration. Only this one-variable contract is needed. Its quantitative published supplier and corrected proof are separate.

Theorem 1. *Subject to these two analytic contracts and the stated LP03 input, a dense internal subfamily has the following representation. There are fixed standard dimension blocks C, L, B, fixed filtered lattice-compatible maps from a weighted universal group U, and one integral top character η on $U_{(m,R)}$. Up to a fixed coordinate isometry ι,*

$$\iota\chi_h(n) = F_h(g_h(n)\Gamma, g_{0,h}(n)\Gamma_0) \qquad (n \in [N]).$$

The lower model has degree–rank at most $(m, R-1)$. The functions F_h are unit vectors with uniform standard bounds and exact top-character law; they need not be equal as functions. The horizontal coefficients are fixed core frequencies, linear petals $\{\alpha_a h\}\beta_a$ for $a \in L$, and independent petals for $a \in B$. The linear petals are Freiman homomorphisms on the retained set. For almost every additive quadruple, the core, all four independent-petal blocks and any three linear-petal blocks are independent modulo $O(N^{-i})$ separately at each weight i.

For every weight-m, length-R iterated commutator w, the real number $\eta(w)$ is zero if w contains two petals, or contains one letter from B. These are precisely the annihilations required by LP04. They hold for every new rank family satisfying the initial hypotheses, without inherited frequency information. The construction of the final two-variable test is the separate Section 12 join.

2 Selection and the corrected universal representation

We use the following elementary internal-choice rule, which also makes the countability uses precise. For an increasing internal sequence of witness sets S_k, suppose each member of an internal parameter set has a witness in some standard S_k. The least successful index is an internal integer. It is uniformly bounded: otherwise the internal nonempty sets on which it exceeds $1, 2, \dots$ have a common element by countable saturation, contrary to pointwise standard boundedness. At representative indices choose the first witness in a fixed choice order. This gives internal choices with one standard bound. Countable saturation here follows by diagonalizing nested finite intersections: at representative index v, choose a witness for the largest $k \le v$ whose first k conditions hold there. This k tends to infinity along the ultrafilter. The unsupported diagonal $x_{v,v}$ alone would not establish this conclusion.

We also need the "almost all" form. For nested internal bad sets B_k in a hyperfinite set X, if every B_k has infinitesimal relative size, the same diagonal construction chooses an unbounded K for which B_K has infinitesimal relative size and contains every standard B_k. Consequently, if their countable union is not contained in a sparse internal set, some B_k is dense. Applying finite pigeonhole inside that B_k fixes an integer coefficient vector and a standard error bound. This is the exact quantifier step used for failed frequency independence below.

First restrict to one of the fixed models for χ_h. Bounded tautification and normalization use their separately proved project interfaces. In a fixed model, only finitely many integral vertical characters can occur under a common Lipschitz bound. Indeed, along a fixed unit-speed top-torus direction X, the identity $F(\exp(tX)x) = \mathrm{e}(t\eta(X))F(x)$ and $\|F(x)\| = 1$ give $2\pi|\eta(X)| \le C_X \operatorname{Lip}(F)$ by taking

small t. On an integral basis this bounds all integer character coordinates. Finite pigeonhole therefore fixes the character.

For each weight i, choose finitely many elements of $\Gamma \cap G_{(i,0)}$ whose logarithms generate its rational Lie algebra. Such elements are obtained by clearing denominators in an adapted rational Mal'cev basis; their real generated subgroup is $G_{(i,0)}$. Use one formal letter of weight i for each. Define U by killing all Lie words of weight greater than m and all weight-m words of length greater than R. Set

$$U_{(d,r)} = \exp \operatorname{span}\{\text{surviving words of weight} > d, \text{ or weight} = d \text{ and length} \geq \max(r, 1)\}$$

for positive d, with $U_{(0,0)} = U$. This is the exact LP04 weighted presentation before the additional petal relations. Its Hall-coordinate lemma supplies the lattice generated by the letters and its filtered intersections. The upward-closed word conditions prove every bracket inclusion. In particular $U_{(d,0)} = U_{(d,1)}$.

Mapping each letter to its chosen lattice element gives a filtered homomorphism $\phi : U \to G$. The killed words vanish by the degree–rank bracket law in G; the map sends the generated lattice into Γ. It is onto each $G_{(i,0)}$. In $\mathrm{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$ the images of the weight-i letters span the real vector space and generate a full sublattice. Thus the induced map of compact horizontal tori is onto. Select internal preimages of $\mathrm{Taylor}_i(g_h\Gamma)$ to get frequencies $\xi_{h,i,a} \in {}^*\mathbb{T}$. Pull the fixed top character back by ϕ and initially take G_0 trivial. This proves the exact Definition 9.11 representation, including lattice compatibility and all-limit equivariance, for every new family.

Here and below horizontal Taylor coefficients are homomorphisms on the polynomial-sequence group, are unchanged by translating n, and for $c \in G_{(i,1)}$ the sequence $c^{\binom{n}{i}}$ has only the ith horizontal coefficient c and zero others. These facts use the iterated-difference commutator rule B.7: all noncommutative correction terms have rank at least two and disappear modulo $G_{(i,2)}$; an extra difference has weight above i. Binomial differences give the last assertion. Polynomial products use B.4. No claim that arbitrary Taylor factors commute is used.

If $\xi_{i,a} = \sum_b c_{i,a,b}\xi'_{i,b}$ with standard integers $c_{i,a,b}$, map the new weight-i letter $e'_{i,b}$ to $\prod_a e_{i,a}^{c_{i,a,b}}$ in a fixed order. The universal relations and lattice inclusion hold, giving $\psi : U' \to U$. In the abelian horizontal quotient this is exactly the indicated integer matrix. Replace ϕ by $\phi\psi$ and η by $\eta\psi$; the original orbit and test are unchanged. When these matrices depend on h, the preceding bounded-choice rule and finite pigeonhole fix them on a dense internal subfamily before this step.

3 Frequency regularization with the actual quantifiers

Fix one weight and put $\epsilon = N^{-i}$. A partial description consists of fixed core frequencies, independent-candidate petals b_h, bracket-linear petals l_h, fixed rational frequencies, and $O(\epsilon)$ errors, with all original frequencies standard integer combinations of them. Every list is standard finite, and every family is internal. Dense restriction fixes the integer matrices and error bounds. Put all original frequencies in b_h to obtain an initial partial description.

Minimize lexicographically the numbers of b petals, then l petals, then core frequencies, among all such partial descriptions on dense subsets. Bracket-linear petals are required to be individually Freiman homomorphisms. This costs only a finite dense restriction: place each $\{\alpha h\}$ in one of a fixed finite collection of intervals of width less than $1/4$. For an additive quadruple the difference of the four fractional parts is an integer of absolute value less than one, hence zero. Multiplication by any real β preserves that identity. The intervals may be anywhere in the fundamental cube; none is required to surround zero.

If independence of the core, four b blocks and the first three l blocks fails on more than a sparse set of quadruples, the bad-set rule above fixes one nonzero standard integer relation with error $O(\epsilon)$ on many quadruples. There are three exhaustive cases.

If it contains only core coefficients, solve for a core with nonzero coefficient a. Choose internal ath roots of the other core frequencies. Division introduces a residue in the finite group $a^{-1}\mathbb{Z}/\mathbb{Z}$ and an $O(\epsilon)$ error. Move them to the rational and error lists. This reduces the core count without changing either petal count.

If the relation involves l but no b, it is a relation on three free parameters. Fix two by averaging over the many triples. On a dense set of the remaining parameter, solve for one l coefficient. Divide the remaining bracket-linear coefficients as real numbers β/a, and root the core coefficients. Freeze the finite rational carry and add the values at the two fixed parameters to the core. A further fixed-box restriction restores the Freiman laws for the divided bracket-linear terms. This reduces the l count without changing the b count.

If a b coefficient occurs, use a symmetry of $h_1 + h_2 = h_3 + h_4$ to place a nonzero one in position four. The Freiman laws allow elimination of the fourth l block after this permutation. Group the relation as $X_1(h_1) + \cdots + X_4(h_4) = O(\epsilon)$, with X_4 containing that b coefficient and fixed core terms. Contract F expresses X_4 as finitely many bracket-linear terms plus a constant and error. Solve for the chosen b coefficient, using roots of the other b and core frequencies and dividing the real bracket-linear coefficients. Freeze the finite rational carry, enlarge the other lists, and refine into fixed boxes as above. The original frequencies are still integer combinations of the new lists, because the old frequencies are integer multiples of their chosen roots. The b count has decreased. Each case contradicts the lexicographic minimum.

Thus the required independence holds for almost all quadruples. Apply this argument successively for the finitely many weights $1, \ldots, m$. Dense restrictions preserve previously obtained independence: their bad sets remain $o(N^3)$, while the retained parameter set has standard positive density. Standard integer relations, rational carries, error constants, finite dimensions and fixed model choices are selected internally with the explicit bounded-choice rule. This proves the regularization step of 10.5–10.7 conditional only on contract F, with its needed family uniformity.

4 Removing rational and small coordinates without losing unit norm

Apply the integer change-of-basis construction to the regularized frequency lists. We now remove each rational or small coordinate in turn. The remaining top character and lattice map must be updated; forgetting them would not give a new representation.

For a small coordinate $\xi = O(N^{-i})$, choose its small real lift and put $c = \phi(e_{i,a})^{\xi} \in G_{(i,1)}$. Set $g'_h(n) = c^{-\binom{n}{i}} g_h(n)$. B.4 proves polynomiality and horizontal homomorphism shows that exactly the selected coordinate has been removed. On $[N]$, $c^{\binom{n}{i}}$ is a bounded smooth function of $u = n/(2N)$. Indeed its logarithm is $\binom{2Nu}{i}\xi \log \phi(e_{i,a})$; its coefficients and all derivatives on a fixed compact interval are bounded because $N^i\xi = O(1)$. Extend this logarithm to the unit circle by using its real formula on the lift $(-1/4, 3/4)$ and multiplying by a fixed smooth cutoff equal to one on $[0, 1/2]$ and zero near the lift's endpoints. Let $c_h(u)$ be its exponential. Replace the test by

$$F'_h(x, x_0, u) = F_h(c_h(u)x, x_0).$$

It is globally a unit vector and uniformly Lipschitz. The top image $\phi(U_{(m,R)})$ is central, so the same exact vertical law holds everywhere. Add the circle orbit $n/(2N)$ to the lower model. It has

degree one, which is at most $(m, R-1)$ because $m \geq 2$. Thus this is an exact representation, not an approximation to the original sequence.

For a rational coordinate $q \in \mathbb{Q}/\mathbb{Z}$, put $\gamma = \phi(e_{i,a})^q$ and $g'_h(n) = g_h(n)\gamma^{-\binom{n}{i}}$. Again only that horizontal coordinate is removed. The element γ is rational relative to Γ. The subgroup $\widehat{\Gamma} = \langle \Gamma, \gamma \rangle$ is discrete: in rational Mal'cev coordinates the finite nilpotent collection formulas express every word as a polynomial in integer exponents of a finite set of generators and their finitely many commutators; clearing their fixed rational denominators gives a discrete coordinate grid. Since Γ is cocompact, its index in $\widehat{\Gamma}$ is finite. Its normal core $\Gamma' = \bigcap_{\delta \in \widehat{\Gamma}} \delta \Gamma \delta^{-1}$ is therefore a finite-index subgroup, normalized by γ.

Choose a standard b with $\gamma^b \in \Gamma$ and a standard period q_0 such that $\binom{n+q_0}{i} \equiv \binom{n}{i} \pmod{b}$. For example a sufficiently divisible multiple of $b\,i!$ works, by the integer binomial difference formula. Put $\gamma_r = \gamma^{\binom{r}{i}}$ for $r \pmod{q_0}$. Then

$$F_h(g_h(n)\Gamma, x_0) = F_h(g'_h(n)\gamma_r\Gamma, x_0) \quad (n \equiv r \pmod{q_0}).$$

The test $W_{r,h}(x\Gamma', x_0) = F_h(x\gamma_r\Gamma, x_0)$ is well-defined, unit, and has the same top-frequency law. Also $W_{0,h}(x\Gamma', x_0) = F_h(x\Gamma, x_0)$ is such a global unit test.

To interpolate the finitely many residue tests, choose disjoint small arcs around r/q_0 in $\mathbb{T}$. In the rth arc choose a smooth angle θ_r which is zero at r/q_0 and $\pi/2$ near its boundary. Set $a_r = \cos\theta_r$ there and zero outside; set $b = \sin\theta_r$ there and $b = 1$ outside all arcs. The supports of the a_r are disjoint and $\sum_r a_r^2 + b^2 = 1$. The test

$$\left(\sum_r a_r(u) W_{r,h}(x, x_0),\ b(u) W_{0,h}(x, x_0) \right)$$

is globally Lipschitz, unit and has the same vertical character. At the orbit $u = n/q_0$ it is exactly $(\chi_h(n), 0)$. This fixed zero padding is explicit; we do not assert that an arbitrary scalar interpolation stays unit or that the old dimension has been retained. Correlation and subsequent matrix comparisons recover the original family by the fixed coordinate projection.

The map from the remaining universal group must now land in the covering lattice Γ'. Choose a standard M for which every remaining lattice generator's Mth power lies in Γ'. Send each new letter to that Mth power, and divide its horizontal frequency by M. Pull back the character by the same map. For linear petals choose the division as $\{\alpha h\}(\beta/M)$; for core coordinates choose fixed roots and for independent coordinates choose internal roots. Repeating this finite process produces a common standard denominator. It is not silently dropped, as in the displayed conclusion of the printed proof of 10.10.

Independence is preserved: multiply an alleged standard integer relation between divided frequencies by the common denominator to contradict the old independence. After dividing linear petals, refine again into the fixed short boxes to make their real fractional-part sums exactly additive, hence their divided values Freiman. This is valid at arbitrary box locations. All rational covers, maps, dimensions, and periods are fixed on a dense internal subfamily before use. Small-coordinate curve bounds are uniform by the internal bounded-choice rule. A finite composition of paddings is one fixed coordinate isometry. We have proved the precise representation part of Theorem 1, without replacing the varying F_h by a single equal function through an approximation argument.

5 The horizontal consequence of factorization, with denominators

Here we derive the needed D.6 conclusion from contract D. View a degree–rank group as ordinarily filtered by $G_i = G_{(i,0)}$. We retain the finer horizontal spaces $\mathrm{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$. For a polynomial orbit $g\Gamma$ put

$$\Xi_i = \{\lambda \in \widehat{\mathrm{Horiz}_i(G)/\Gamma} : \lambda(\mathrm{Taylor}_i(g\Gamma)) = O(N^{-i}) \pmod 1\}.$$

Characters here have standard integer coordinates. Their real linear lifts define $\Xi_i^{\perp}$ inside the real horizontal vector space.

Suppose F is a bounded Lipschitz test with nonzero standard average. Take one scalar component if needed, and factor $g = \varepsilon g'\gamma$. On a progression of the standard periodic modulus, choose a bounded standard representative $\gamma_P \in \widehat{\Gamma}$ of the constant coset of γ. Set $Q_P = \gamma_P^{-1} Q \gamma_P$ and $g_P = \gamma_P^{-1} g' \gamma_P$. The finite-cover equidistribution clause in D, applied to the common finite-index lattice, makes g_P totally equidistributed in $Q_P/(Q_P \cap \Gamma)$ on every retained dense progression. Rationality of γ_P gives rationality of Q_P.

Partition into those residue progressions and then into intervals of length δN for a sufficiently small standard δ. Averaging selects a piece with an average bounded below by the original fixed threshold. On that piece freeze $\varepsilon(n)\gamma_P$ at one point. The uniform Lipschitz error is $O(\delta)$, so equidistribution gives

$$\left| \int_{P/(P\cap\Gamma)} F(\varepsilon_0 x)\, d\mu(x) \right| > c > 0 \tag{2}$$

for the connected rational $P = Q_P$ and bounded ε_0. This also holds for an internal test with a standard Lipschitz bound: compactness gives a finite standard uniform net in that Lipschitz ball. Apply total equidistribution to each net entry and then let the fixed net error tend to zero. Thus no unjustified standardness of the selected F_h or of the frozen bounded translate is being used.

We claim $\pi_i(P \cap G_i) \supseteq \Xi_i^{\perp}$. Let λ be an integer character annihilating the rational space $\pi_i(P \cap G_i)$. Then $\lambda(\mathrm{Taylor}_i(g_P)) = 0$. Conjugation changes no horizontal coefficient since $[G, G_i] \subseteq G_{(i,2)}$. Hence the horizontal homomorphism identity gives

$$\lambda(\mathrm{Taylor}_i(g)) = \lambda(\mathrm{Taylor}_i(\varepsilon)) + \lambda(\mathrm{Taylor}_i(\gamma)) \pmod 1.$$

The first term is $O(N^{-i})$. The second is rational of bounded standard denominator: derivatives of γ lie in $\widehat{\Gamma} \cap G_i$, whose horizontal image is a finite-index over-lattice of that of $\Gamma \cap G_i$. Thus for some standard positive q, $q\lambda \in \Xi_i$. Every annihilator of the rational image has this property. Taking real annihilators proves the claim. In particular no false assertion that γ takes values in Γ or has zero horizontal coefficient is needed.

6 Applying the coset conclusion to the actual quadruple

After all the preceding dense restrictions, reapply hereditary LP03. There are $\gg N^3$ biased quadruples. The exceptional sets for frequency independence have size $o(N^3)$. Permutations preserving the additive quadruple equation make the omitted linear-petal position any of the four; the union of the four exceptional sets is still sparse. Fix a biased quadruple with all four required independence statements.

Use the representations of its four nilcharacters and the lower bias test to form a product group

$$\mathcal{G} = G^{(1)} \times G^{(2)} \times G^{(3)} \times G^{(4)} \times G^{(0)}.$$

The first four groups have degree–rank at most (m, R); the last has degree at most $(m, R-1)$, including all four lower representation models and the ordinary degree-at-most-$m-1$ bias witness. The stronger strict inequality printed for this last group is unnecessary and is not assumed. Translations in n preserve horizontal Taylor coefficients by their homomorphism/difference identity. Zero extensions in (1) introduce only an interval-intersection mask. Approximate this mask in normalized L^1 to error δ by a bounded Lipschitz circle test in $n/(cN)$, with fixed large c and Lipschitz constant $O(1/\delta)$. Its two transition strips have total length $O(\delta N)$; multiplying by the uniformly bounded remaining tests controls the average error. Choosing δ below the fixed bias threshold preserves positive bias. This circle has degree one and belongs in $G^{(0)}$ because $m \geq 2$.

The resulting product test has a nonzero scalar average and exact vertical law

$$F(\phi_1(t_1)x_1, \dots, \phi_4(t_4)x_4, x_0) = \mathrm{e}(\eta(t_1) + \eta(t_2) - \eta(t_3) - \eta(t_4))F(x_1, \dots, x_4, x_0).$$

All factors, including the cutoff, are uniformly Lipschitz on a fixed standard product geometry. Apply the preceding factorization consequence to obtain a connected rational $P \subseteq \mathcal{G}$ and the nonzero coset integral (2).

7 Rational duality and the two commutator conclusions

For each weight i, let $V_{123,i}$ be the real space spanned by the three-coordinate diagonal images of core letters and by each individual coordinate image of every petal letter in positions $1, 2, 3$. Define $V_{124,i}$ likewise. The projection of $\pi_i(P \cap \mathcal{G}_i)$ to these three factors contains the indicated V.

Here is the exact duality argument. If containment failed, rationality gives a rational linear functional that vanishes on the projected space but not on V. Extend it by zero in the omitted factors and clear denominators to make it an integer character λ. Because the full space contains $\Xi_i^{\perp}$, this character is in the rational span of Ξ_i. After multiplication by another standard integer it belongs to Ξ_i itself. Evaluating it on the Taylor coefficient gives a relation

$$\sum_{a \in C} (c_{1a} + c_{2a} + c_{3a})\xi_{C,a} + \sum_{k=1}^{3} \sum_{a \in L \cup B} c_{ka}\xi_{h_k,a} = O(N^{-i}).$$

The selected independence forces every petal coefficient and each displayed core sum to vanish. That says the character vanishes on V, a contradiction. The 124 proof is identical. This uses rational real subspaces, not an unjustified identification of an arbitrary real annihilator with one integral member of Ξ_i.

Similarly the projection onto all four main factors contains the space $V_{B,i}$ generated by diagonal core images and by the individual-coordinate images of every letter in B. In the analogous integer relation eliminate the fourth L coefficient using $\xi_{h_4,a} = \xi_{h_1,a} + \xi_{h_2,a} - \xi_{h_3,a}$ for $a \in L$. Independence makes the four B coefficients vanish separately and the sum of core coefficients vanish. This is exactly annihilation of $V_{B,i}$. No numerical range for B is borrowed from the conflicting printed order of the L and B blocks.

Let w be any bracketing of R letters of weights summing to m. First suppose it has two petal occurrences. For one occurrence lift its first-coordinate image, with zero second and third coordinates, from $V_{123,i}$ to $P \cap \mathcal{G}_i$. For the second lift its first-coordinate image with zero second and fourth coordinates from $V_{124,i}$. Lift the remaining letters so that their first horizontal coordinate is the required one. Take the same iterated commutator of these lifts.

In the first factor it equals $\phi_1(w)$ exactly. Replacing one lift by its horizontal error inserts an element of $G_{(i,2)}$ in place of $G_{(i,1)}$, raising the total rank to at least $R+1$ at weight m; every such

term vanishes. Higher BCH terms have larger weight and vanish as well. In factor two, either selected petal lift has such an error; factor three has the first, and factor four the second. Therefore those coordinates are the identity. In factor zero, an R-fold commutator of the prescribed weights lies in its trivial (m, R) group. We obtain

$$(\phi_1(w), 1, 1, 1, 1) \in P. \tag{*}$$

This reasoning applies to each position of a repeated letter and to every bracketing, not only a single left-nested commutator.

If w has a B letter, use $V_{B,i}$ to lift that occurrence with all other main coordinates zero horizontally. Lift the remaining occurrences by their first coordinates and repeat the same rank-error calculation. This proves (*) as well. It works for $R = 1$; the other top coordinates then already vanish, and the lower model has trivial weight-m group.

Finally (*) gives a real-zero conclusion. The element is central in $\mathcal{G}$. Since P is a connected subgroup of a simply connected nilpotent group, its entire one-parameter subgroup belongs to P. Left translation by its tth power leaves Haar measure on $P/(P \cap \Gamma)$ invariant and commutes with the bounded coset translate. The vertical law multiplies the nonzero integral by $\mathrm{e}(t\eta(w))$. Hence $\mathrm{e}(t\eta(w)) = 1$ for every real t, so $\eta(w) = 0$ as a real number. Using only $t = 1$ would give an integer condition and would be insufficient. This proves both annihilations in Theorem 1.

8 The LP04 and rank-family joins

The output has the exact LP04 hypotheses: the corrected universal weighted group, an integral top character with all required real-zero annihilations, fixed lattice-compatible ϕ, matched horizontal Taylor tori, and a unit representation with a lower degree–rank factor. Its petal quotient kills the whole ideal generated by the bad words at every weight; the vanishing just proved concerns the top restriction needed to descend the character. These distinct roles are not identified.

The Section 12 join chooses its stronger integral extension to the ordinary top before invoking LP04 comparison, and gives one global multidegree $(1, m)$ nilcharacter equal to a fixed coordinate embedding of ζ_h. LP04 then supplies the whole matrix $(\iota\chi_h) \otimes \overline{\zeta_h}$ with strictly lower degree–rank. Projecting the first tensor factor through the fixed coordinate projection gives $\chi_h \otimes \overline{\zeta_h}$ with the same bound and exact values. Thus the unit padding used in rational removal does not weaken the actual matrix input required by rank-iteration001.

All selections, roots, covers, maps and tests were made internally on representatives; each retained dimension, character and bound is standard. There are finitely many weights and coordinates at any one rank, so all density and correlation losses remain positive standard numbers. The next rank family from rank-iteration001 again has exactly the hypotheses of Section 1. Repeating the present reconstruction at that new rank uses its own new frequency data. No independence is transported through a Stone–Weierstrass approximation or assumed for the newly selected test. There are m rank steps, so the same finite-loss conclusion holds across the full iteration. This is the required source-to-consumer compatibility, conditional on the two separately supplied analytic contracts and the independent acceptance of the project proof units used here. Representation equality is required on $[N]$. Whenever a global sequence is used, take the displayed polynomial model as its extension. Small-frequency removal need not preserve a previous extension outside $[N]$; the interval mask in the quadruple argument is precisely what makes that change harmless.

9 Bounded correction and consistency record

The source scope is final GTZ Sections 9–11, the immediate pp. 1307–1309 consumer, the D.4–D.6 interface, B.4/B.7, and the selection statements in Appendix A. The attached record pins exact passages and the current project consumers. Concrete changes are:

1. State the fixed rank R explicitly in Definition 9.1's second truncation relation, where the printed formula writes r without the fixed rank subscript. The printed weight cutoff $\sum i_j \geq s$ is already equivalent to $\sum i_j > m$ and is not an error. Rendering the source rather than relying on text extraction establishes this distinction.

2. Retain the common division from 10.9 in 10.10 and restore Freiman laws after real coefficient division. Fixed arbitrary-location boxes suffice.

3. Rational removal uses a finite cover and explicit same-character unit completion with a fixed isometry, rather than unproved unit interpolation.

4. Keep varying uniformly Lipschitz F_h; total boundedness does not make them literally equal as claimed in the immediate p. 1308 narrative.

5. Use named core/linear/independent blocks, because p. 1302 reverses the order used in the statement of 10.10 and Theorem 11.1.

6. Treat the repeated clauses (iii),(iv) of Theorem 11.1 as duplicates; there are two distinct annihilation conclusions.

7. Use degree–rank at most $(m, R-1)$ for the lower product model, not an unneeded strict inequality below that index.

8. Retain the rational Taylor term in D.5 and clear its denominator in the annihilator argument. The factorization subgroup may equal the ambient group. The separate analytic supplier addresses the D.2 total-equidistribution issue and F.1 signed-graph/no-wrap details.

9. The real-zero character argument uses all real powers in the connected subgroup. One integral phase identity alone is insufficient.

10. Reapply LP03 after each dense restriction; intersect many biased quadruples with almost-all independence, not two unrelated dense sets.

These are authored repairs and scoped dependencies, not a declaration that all historical versions or the entire inverse theorem have been checked. The original published source and all earlier frozen packets are preserved.

GTZ-LP-04: the petal quotient and degree–rank construction

Author proposal: /root/astra_v14_assertions_arithmetic

7 November 2026

Status and scope. This is a proposed project proof under protocol 1.4, awaiting independent review. It proves the algebraic quotient, its exact lattice and character statements, and the comparison needed by [1, Lemma 12.1], conditional on the explicitly stated frequency input. It does not prove that input or the entire inverse theorem. The erratum [5] motivated the repair but is not a theorem premise. The related-use audit at the end distinguishes completed connecting arguments from other proof units. The rational-subgroup and tautification lemmas [3, 4] are separately versioned project proofs awaiting independent acceptance.

1 Indices, free groups, and the actual objects

Fix integers $m \geq 1$, $1 \leq R \leq m$, and finite disjoint sets of letters C_i, L_i, B_i for $1 \leq i \leq m$. Their union is $\mathcal{A}$. The letters in C_i are core, those in L_i retained linear petals, and those in B_i discarded petals. Every letter in these three sets has weight i. A bracket word has length ℓ equal to its number of entries, weight w equal to the sum of their weights, and a multidegree recording the number of occurrences of each individual letter. Repeated entries count repeatedly. All bracketings are allowed, including length-one words. Write p for the number of petal entries, and b for the number of discarded entries. We use $[x, y] = x^{-1}y^{-1}xy$, real powers $x^t = \exp(t \log x)$, and $\mathrm{e}(t) = \exp(2\pi i t)$.

Let $\mathcal{F}$ be the free real Lie algebra on symbols X_a, $a \in \mathcal{A}$. Its usual length grading and its individual-letter multigrading are retained. Take the homogeneous ideal spanned by all words for which

$$w > m \quad \text{or} \quad (w = m \text{ and } \ell > R), \tag{1}$$

and let $\mathfrak{u}$ be the quotient. These forbidden multidegrees are closed under addition of any nonzero multidegree, so their span is indeed an ideal. All surviving words have length at most m. Thus $\mathfrak{u}$ is finite-dimensional and nilpotent. Let $U = \exp \mathfrak{u}$, with multiplication given by the finite BCH polynomial, and set $e_a = \exp X_a$. The group is connected and simply connected. These are the weighted universal objects intended in [1, Definition 9.1, p. 1279], with the top rank in the relation fixed to be R, not a running rank index.

Use $\mathrm{DR} = \{(d, r) : d, r \in \mathbb{N},\ 0 \leq r \leq d\}$ with lexicographic order. Put $U_{(0,0)} = U$. For $d > 0$ let

$$\mathfrak{u}_{(d,r)} = \operatorname{span}_{\mathbb{R}}\{W : w(W) > d \text{ or} [w(W) = d,\ \ell(W) \geq \max(r, 1)]\}, \qquad U_{(d,r)} = \exp \mathfrak{u}_{(d,r)}. \tag{2}$$

In particular the first alternative has *no* length restriction. For ranks larger than d use the notation $U_{(d,r)} = U_{(d+1,0)}$ only as the next-index convention; do not put such pairs into the index monoid. Write $U_d = U_{(d,0)}$. The top group is

$$T = U_{(m,R)} = \exp \operatorname{span}_{\mathbb{R}}\{W : w(W) = m,\ \ell(W) = R\}.$$

Lemma 1. *The groups in* (2) *form a taut degree–rank filtration of degree at most* (m, R)*. Each is connected and normal in* U*, and* T *is central.*

Proof. Bracketing words adds weights, lengths, and individual-letter multidegrees. Each displayed span is an ideal, since bracketing with a nonzero word increases weight. It is a Lie subalgebra and BCH therefore identifies its exponential with a connected subgroup. Nesting follows directly from lexicographic order: passing to a larger degree removes all lower-weight words, and at fixed degree passing to a larger rank increases the length cutoff. For positive indices $(d, r), (e, t)$, a bracket of two allowed words either has weight greater than $d + e$, or has weight $d + e$ and length at least $\max(r, 1) + \max(t, 1) \geq \max(r + t, 1)$. Hence the Lie bracket lies in $\mathfrak{u}_{(d+e,r+t)}$. The BCH commutator formula has this bracket and further brackets, all in the same ideal, which proves the group commutator condition. At index zero use normality. Every generator has weight at least one, so $U_{(1,0)} = U$; also $U_{(d,0)} = U_{(d,1)}$ for all $d > 0$. The truncation gives the claimed degree support. Bracketing a top word with any generator raises its weight above m, proving centrality of T. □

2 An integral coordinate lemma

Here is the integral premise used, with its scope separated from the subsequent quotient argument. Hall collection says that in a free nilpotent group on a finite ordered alphabet, every element has a unique collected expression in the basic commutators with integer exponents; the length-j basic commutators give a basis of its jth lower-central quotient. This published statement is recorded in [2, Definition 5.4.3, Theorem 5.4.5 and the final paragraph of §5.4, pp. 96–98]. The weight in that reference is bracket *length*, not the additional weight w used here.

The corresponding basic Lie brackets form a homogeneous basis of the free nilpotent Lie algebra. To spell out the connection, antisymmetry and Jacobi reorder a bracket $[c_i, c_j]$ until $i > j$, and when $c_i = [c_s, c_t]$ until $j \geq t$. Jacobi replaces the offending bracket by brackets with the smaller inner entry moved outward. Induction on total length and then the finite Hall ordering gives spanning by basic brackets. Independence follows by mapping the free Lie algebra to the real tensor of the lower-central graded Lie ring of the free group: the bracket is induced by the group commutator, and Hall's basis assertion makes the images of those basic brackets independent in each length. This also respects the individual letter multigrading because the reordering identities preserve every count.

Lemma 2 (Upward-closed monomial quotients)**.** *In a free nilpotent Lie algebra of class at most m, discard an upward-closed set of nonzero individual-letter multidegrees. Let V be the exponential of the quotient and Λ the discrete group generated by the surviving* $\exp X_a$*. The surviving Hall commutators give unique real coordinates*

$$(t_1, \ldots, t_v) \longmapsto c_1^{t_1} \cdots c_v^{t_v}$$

on V, and Λ is exactly the set with all $t_i \in \mathbb{Z}$. It is a lattice. A further upward-closed set of multidegrees defines a rational connected normal subgroup V_S. The quotient map q between two such monomial quotients obeys, for each such S,

$$q(\Lambda \cap V_S) = q(\Lambda) \cap q(V_S). \tag{3}$$

Proof. Order basic commutators by increasing length. For a basic group commutator c, BCH gives $\log c = W_c +$ terms of strictly greater length, where W_c is the corresponding basic Lie bracket. Each correction uses all the letters of c and possibly more occurrences of them. A discarded basic word therefore vanishes in the quotient, including every BCH correction. The surviving leading brackets are a basis, since the discarded Lie ideal is the direct sum of precisely the discarded multigraded components. Consequently the logarithm of the displayed coordinate map is triangular by length

with diagonal entries one. Solving successively in each length gives a polynomial inverse and proves unique real coordinates.

Hall collection expresses every word in the original $\exp X_a$ and their inverses as a product of basic group commutators with integer exponents. After quotienting, discarded factors are identities, so it gives the claimed integer coordinates. Conversely every surviving basic commutator is itself a word in the generators, which proves the other inclusion. In particular the subgroup is discrete. Suffix spans in the length-ordered basis are ideals, since every nonzero bracket increases length. Right multiplication by c_i^{-n} changes coordinate i by $-n$, leaves all earlier coordinates unchanged, and can change only later coordinates. Reduce the coordinates successively modulo integers. Every right coset has a representative with coordinates in $[0,1]^v$, a compact set, proving cocompactness.

For V_S, quotient once more by its multigraded ideal. The same triangular coordinate argument says that an element belongs to V_S exactly when all coordinates outside S vanish. Thus its lattice consists of the integer coordinates on the basic commutators in S, and the same compact-coordinate argument proves rationality. A quotient deletes discarded basic factors and keeps the remaining ones in their original order. Membership in $q(V_S)$ forces all retained coordinates outside S to vanish. An element of $q(\Lambda) \cap q(V_S)$ therefore lifts as the identical integer product of retained basic commutators in S. This proves (3). □

The truncation (1) is upward-closed, as are all the thresholds (2). The lemma proves that $\Gamma = \langle e_a : a \in \mathcal{A} \rangle_{\mathbb{Z}}$ is a lattice in U, and every $U_{(d,r)}$ is rational. It also proves the lattice statement below without treating it as a general property of arbitrary rational quotients.

3 The corrected kernel and the descended character

Call a word bad when $p \geq 2$ or $b \geq 1$. Let $I \subseteq \mathfrak{u}$ be the span of all bad Lie words of every weight. Let G^* be the connected real subgroup generated by all bad iterated *group* commutators in the e_a, again of every weight and including length one. Equivalently, the real generated subgroup contains every real power of each such commutator. This convention is essential: a discrete generated subgroup would not be the kernel of a map of simply connected Lie groups.

Let $\widetilde{\mathfrak{u}}$ be the free Lie algebra on $C_i \cup L_i$ with relations (1) and the vanishing of all words with at least two petal entries. Give $\widetilde{G} = \exp \widetilde{\mathfrak{u}}$ the filtration (2) on its surviving words. Put $\widetilde{\Gamma} = \langle \widetilde{e}_a : a \in C_i \cup L_i \rangle_{\mathbb{Z}}$. Define $q : U \to \widetilde{G}$ by $e_a \mapsto \widetilde{e}_a$ on retained letters and $e_a \mapsto 1$ on discarded letters.

Theorem 3. *The homomorphism q is well-defined and surjective, and*

$$\ker q = G^* = \exp I.$$

Thus G^* is connected, rational and normal; it need not be central. For every degree–rank index a, including zero,

$$q(U_a) = \widetilde{G}_a \simeq U_a/(G^* \cap U_a), \tag{4}$$

$$q(\Gamma \cap U_a) = \widetilde{\Gamma} \cap \widetilde{G}_a. \tag{5}$$

Suppose $\eta : T \to \mathbb{R}$ is a continuous homomorphism integral on $T \cap \Gamma$, and for every bad group commutator c of weight m and length R one has $\eta(c) = 0$. Then there is a unique continuous $\widetilde{\eta} : \widetilde{G}_{(m,R)} \to \mathbb{R}$ satisfying

$$\widetilde{\eta}(q(t)) = \eta(t) \quad (t \in T), \qquad \widetilde{\eta}(\widetilde{\Gamma} \cap \widetilde{G}_{(m,R)}) \subseteq \mathbb{Z}.$$

Proof. Bad multidegrees are upward-closed, so I is an ideal. The universal property of the free Lie algebra identifies $\mathfrak{u}/I$ with $\widetilde{\mathfrak{u}}$: killing discarded letters kills every word involving one, and killing all words with two petals gives exactly the other defining relations. There are no additional relations because both presentations impose the same homogeneous ideal on the same free Lie algebra. Exponentiation of this surjective Lie map gives q, and its kernel is $\exp I$ by the global exponential diffeomorphisms.

BCH puts the logarithm of every bad group commutator in I. Conversely take the bad Hall commutators. Their logarithms have the bad basic Lie brackets as leading terms, with only higher-length bad terms. The triangular matrix on the finite-dimensional bad subspace is invertible. Their logarithms therefore span I, and the connected real subgroup they generate is $\exp I$. This proves the asserted description of G^*, not just its normal closure. All these spaces are spanned by rational basic words. Lemma 2 proves rationality and $q(\Gamma) = \widetilde{\Gamma}$.

The Lie map deletes bad multigraded components and preserves weight and length of every surviving word. Applying (2) before or after the map therefore gives identical spaces. Exponentiating proves (4), and Lemma 2 proves (5).

The direct multigrading gives

$$I \cap \log T = \operatorname{span}_{\mathbb{R}}\{W : W \text{ bad},\ w(W) = m,\ \ell(W) = R\}.$$

There is no cancellation between a bad multidegree and a good one. At weight m the BCH correction terms in a group commutator have larger weight and vanish. Thus the bad group commutators of weight m and length R are precisely the exponentials of their Lie brackets. The assumption on η, and real linearity of a continuous homomorphism on the vector group T, prove that η vanishes on $G^* \cap T$. It consequently descends through (4) at the top index. Finally (5) lifts every top lattice point downstairs to one upstairs, where η is integral. This proves the last assertion. □

Why the omitted lower weights matter. With one core letter x and one discarded letter b, both of weight one, and $m = 3, R = 3$, the quotient kills b itself. A subgroup generated only by bad weight-three words cannot contain b. Moreover $[b, x]$ survives in U, so the actual kernel is not central. This example tests both false shortcuts.

The precise analytic input. In the notation of [1], take $m = s - 1$, $R = r_*$, $|C_i| = D_{*,i}$, $|L_i| = D'_{\mathrm{lin},i}$ and $|B_i| = D'_{\mathrm{ind},i}$. Use the *final output* of Lemma 10.10, after rational-core and small frequencies have been removed by Lemma 10.9. Then the two hypotheses on η in Theorem 3 are exactly Theorem 11.1(i),(ii): all weight-m, length-R words with two petals or one beyond-linear petal are annihilated. The repeated (iii),(iv) add nothing. For a bracketing convention restricted to left-normed words, Jacobi expresses any other bracket of these same entries as a linear combination of such words; their petal counts and total weight are unchanged. At the central top layer this is also an exact identity of logarithms, so the vanishing extends to all bracketings used here.

Theorem 11.1 requires the hypotheses of Theorem 7.2, the limit family of degree–rank (m, R) nilcharacters, and the sunflower representation with its common standard dimension vector and vertical frequency. Its proof uses the additive-quadruple bias conclusion of Proposition 7.3 and the stated independence from Lemma 10.10. In the repaired project, that bias conclusion must come from LP-03, not the false original Proposition 8.3. No use of Theorem 1.3 is made to supply it. This unit assumes these inputs; merely partitioning generators does not prove the required vanishing.

4 Vertical test functions and petal actions

Lemma 4. *For the descended integral character there is a finite-dimensional unit-vector-valued smooth function on $\widetilde{G}/\widetilde{\Gamma}$ of vertical frequency $\widetilde{\eta}$. Its dimension and Lipschitz bound depend only on the fixed filtered group, lattice, metric and character.*

Proof. The central torus $\widetilde{T}/(\widetilde{T}\cap\widetilde{\Gamma})$ acts freely. Choose finitely many local smooth sections of the resulting principal torus bundle and smooth real functions a_j subordinate to their domains, with $\sum_j a_j^2 > 0$ everywhere on the base. Normalize them so $\sum_j a_j^2 = 1$. In a local trivialization (z,t) put the jth component equal to $a_j(z)\mathrm{e}(\widetilde{\eta}(t))$, extending by zero outside its support. The support stays inside the chart, so the extension is smooth. Transition functions change only the component's phase. The vector has norm one, and all components have the same vertical frequency. Compactness gives a finite Lipschitz bound. If the torus or character is trivial the same construction, or the constant function, applies. □

The quotient Lie algebra is a direct sum $\mathfrak{c}\oplus P$, where $\mathfrak{c}$ consists of words with no petals and P of words with one. Here P is an abelian ideal: a bracket of two such words has two petals and is zero. Thus $\widetilde{G}_{\mathrm{petal}} = \exp P$ is abelian, normal and rational by Lemma 2. Decompose $P = \bigoplus_{a\in L} P_a$ by the unique petal letter. Core conjugation preserves each P_a. For a tuple $t = (t_a) \in \mathbb{R}^L$, the map fixing core generators and sending X_a to $t_a X_a$ on linear petals is a Lie endomorphism M_t: it preserves all defining ideals. It scales P_a by t_a, so on the petal group, written multiplicatively,

$$M_t M_u = M_{tu}, \qquad M_t(p)M_u(p) = M_{t+u}(p), \qquad M_t(gpg^{-1}) = gM_t(p)g^{-1}.$$

The last identity follows because M_t commutes with every core adjoint action and conjugation by P acts trivially on P. This proves the precise identities used in (12.8)–(12.10), without treating M_t as an automorphism when some $t_a = 0$. For integral tuples t, M_t preserves $\widetilde{\Gamma}$, because it sends each generator to an integer power. It also preserves the petal lattice by its subgroup invariance. These are the lattice facts needed by the subsequent shear semidirect products.

5 The comparison filtration and corrected Lemma 12.1

We record sufficient input explicitly. Let G/Λ, G^0/Λ^0 be fixed taut rational degree–rank nilmanifolds of degrees at most (m,R) and strictly below (m,R), respectively. Let $\phi : U \to G$ be filtered and send Γ into Λ. Suppose a_h, c_h are polynomial sequences and F_h are unit-vector-valued uniformly Lipschitz functions on $G/\Lambda \times G^0/\Lambda^0$ such that

$$F_h(\phi(t)x, z) = \mathrm{e}(\eta(t))F_h(x,z) \quad (t \in T). \tag{6}$$

Let $E_{h,i} = \prod_{a\in C_i\cup L_i\cup B_i} e_a^{\xi_{h,a}}$ in a fixed order, and suppose the horizontal Taylor coefficient of $a_h\Lambda$ is that of $\phi(E_{h,i})$ in $G_{(i,1)}/G_{(i,2)}$ modulo the horizontal lattice. Let

$$b_h(n) = \prod_{i,a\in C_i} \widetilde{e}_a^{\xi_{h,a}\binom{n}{i}} \prod_{i,a\in L_i} \widetilde{e}_a^{\xi_{h,a}\binom{n}{i}}.$$

These are exactly the core and retained-petal factors in (12.3)–(12.5). Using the function from Lemma 4, set

$$\chi_h(n) = F_h(a_h(n)\Lambda, c_h(n)\Lambda^0), \qquad \widetilde{\chi}_h(n) = \widetilde{F}(b_h(n)\widetilde{\Gamma}).$$

Proposition 5. *Under these hypotheses, every component of $\chi_h \otimes \overline{\widetilde{\chi_h}}$ is a nilsequence of degree–rank strictly below (m, R). The representing filtered nilmanifold is fixed independently of h and the coefficients; its Lipschitz bounds depend only on the fixed data and original bounds.*

Proof. For $d > 0$ put $\bar{r} = \max(r, 1)$ and let $\sigma(d, \bar{r})$ denote the successor of $(d, \bar{r})$ in the positive degree–rank list. In the product $G \times G^0 \times \widetilde{G}$, let $H_{(d,r)}$ be the connected subgroup generated by

$$J_{(d,r)} = G_{\sigma(d,\bar{r})} \times G^0_{(d,\bar{r})} \times \widetilde{G}_{\sigma(d,\bar{r})},$$
$$D(U_{(d,\bar{r})}) = \{(\phi(u), 1, q(u)) : u \in U_{(d,\bar{r})}\}.$$

Set $H_0 = H_{(1,0)}$. This specifies the zero group; it is not an arbitrary full product. All generator subgroups are rational. For the diagonal image, the image of the appropriate lattice generates its real subgroup by the Hall coordinates and $\phi(\Gamma) \subseteq \Lambda$. The rational-generated subgroup lemma of [3, Section 2] applies.

Nesting is direct. For positive indices i, j, commutators of two diagonal generators lie in $D(U_{i+j})$. A commutator involving a J_i generator has its first and third coordinates in the successor group at $i+j$, since $\sigma(i)+j \succeq \sigma(i+j)$, with the convention replacing rank zero by one. Its middle coordinate is in G^0_{i+j}. The same argument works for two J generators. Taking generated Lie algebras and BCH gives $[H_i, H_j] \subseteq H_{i+j}$. At zero use $H_0 = H_{(1,0)} = H_{(1,1)}$ and nesting. Thus this is a taut rational degree–rank filtration. Its top group is exactly

$$H_{(m,R)} = D(T), \tag{7}$$

because every group in $J_{(m,R)}$ is trivial.

We next check actual polynomial membership, including constant coefficients and the distinction between a horizontal vector and its torus class. Write $a_h(0) = v_h \lambda_h$ with v_h in a fixed compact representative set and $\lambda_h \in \Lambda$. Replace $a_h(n)$ by $v_h^{-1} a_h(n) \lambda_h^{-1}$, and replace $F_h(x, z)$ by $F_h(v_h x, z)$. Do the same in the second coordinate. Values of χ_h are unchanged, the new sequences start at the identity, and Lipschitz bounds stay uniform. Since G is taut, conjugation by v_h acts trivially on every $G_{(i,1)}/G_{(i,2)}$: its commutator is in $G_{(i+1,1)} \subseteq G_{(i,2)}$. The horizontal torus conditions remain unchanged. Equation (6) remains unchanged since the top group of the taut group is central. Tautify the original targets before constructing the total-frequency representation, using [4] and then [1, Lemma 9.12]; no conjugated, potentially irrational replacement of ϕ is being silently used here.

For each i, choose $\lambda_{h,i} \in \Lambda \cap G_i$ lifting the horizontal lattice difference between the new a_h and $\phi(E_{h,i})$. Such a lift exists by the definition of the horizontal lattice. Multiply $a_h(n)$ on the right by the inverse of $\prod_i \lambda_{h,i}^{\binom{n}{i}}$. This does not change its orbit. The factor is polynomial by the Taylor criterion [1, Lemma B.9]. The horizontal Taylor map is a homomorphism [1, Definition 9.6 and the argument following it], so now the horizontal equalities hold in the vector groups themselves.

Taylor-expand $a_h(n) = \prod_{i \geq 1} a_{h,i}^{\binom{n}{i}}$. Then $a_{h,i} \phi(E_{h,i})^{-1} \in G_{(i,2)}$. Hence $(a_{h,i}, 1, q(E_{h,i})) \in H_{(i,1)}$. The Taylor criterion and product closure [1, Lemma B.9 and Corollary B.4] show that

$$n \mapsto \left(a_h(n), 1, \prod_i q(E_{h,i})^{\binom{n}{i}} \right)$$

is polynomial for H. Both its last coordinate and b_h have the same horizontal Taylor coefficients, since multiplication of factors is addition in each horizontal quotient and discarded generators have image one. Their ratio is polynomial and all its horizontal Taylor coefficients vanish. Its Taylor

coefficient at index i therefore lies in $\widetilde{G}_{(i,2)}$. These groups, with zeroth group $\widetilde{G}_{(1,2)}$, form a degree filtration by the same commutator check. The ratio starts at one; Lemma B.9 now shows that it is polynomial for this smaller filtration. Embedded in the third coordinate it is polynomial in H. The second-coordinate map $(1, c_h(n), 1)$ is also polynomial in H. Product closure proves that the exact triple $(a_h(n), c_h(n), b_h(n))$ is polynomial in H.

Restrict $F_h \otimes \widetilde{\widetilde{F}}$ to $H/(H \cap (\Lambda \times \Lambda^0 \times \widetilde{\Gamma}))$. On $D(t)$ its vertical multiplier is $\mathrm{e}(\eta(t) - \widetilde{\eta}(q(t))) = 1$ for every $t \in T$. It therefore descends through the central rational subgroup (7). The central quotient and lattice argument in [3, Sections 2 and 6] applies here: this kernel is contained in every nontrivial indexed subgroup, unlike the earlier kernel G^*. The top group becomes trivial, so the quotient has degree–rank strictly below (m, R). Restriction, quotient charts, and the compact translations cost only constants from the fixed geometry. This proves the proposition. □

When $R = 1$, strictly below $(m, 1)$ means ordinary degree at most $m - 1$. For $R > 1$ it means degree–rank at most $(m, R - 1)$, not necessarily degree at most $m - 1$. The first sentence of the published proof of Lemma 12.1 claims the latter unnecessarily strong conclusion. The proved former conclusion is exactly what the following rank-induction step consumes.

6 Uniformity, limits, and the remaining Section 12 interfaces

Every group, ideal, Hall list, lattice and quotient above depends only on m, R and the fixed finite labelled dimension data. All formulas are finite polynomials, Lie-group maps, or choices from fixed lattice and compact representative sections. A fixed ordering chooses the first lift of a lattice class; a fixed finite atlas chooses a bounded representative. Applying these standard choices to representative sequences gives limit maps. The derivative and vertical identities are required on their full ultrapower domains, as in [4]; identities only at standard points are insufficient. Thus no bound depends on N, on h, or on the real polynomial coefficients. The character is fixed before choosing a vertical test function. No assertion of a bound uniform over unbounded characters is made. Taking the dense subfamily with fixed geometric data and character is the separate selection step in Lemma 10.10 and Section 12, using Appendix A.

The ordinary degree filtration used to build the final semidirect product is $\widetilde{G}_d = \widetilde{G}_{(d,0)}$. The petal ideal and scalar-action identities proved above give precisely the normality, commutativity, and integral stability used on pp. 1312–1314. In particular a bracket with two petal entries is zero at *every* weight. This assertion would fail if one quotiented only by the old top-weight kernel.

There is a separate unit-norm issue at the cutoff in (12.11) and the paragraph following it. A compactly supported scalar cutoff times a unit-vector function is not itself unit-vector-valued. Here is an exact completion for that use. On the final compact nilmanifold choose a smooth vertical-equivariant local unit vector V on the torus-coordinate chart, agreeing with $\widetilde{F}(g\widetilde{\Gamma})$ there. Choose a smooth function θ of the torus coordinates equal to zero on the closed inner box and equal to $\pi/2$ outside a slightly larger box still inside the chart. These coordinates are unchanged by the top vertical action. Construct a global unit-vector function V_0 of the same vertical character by Lemma 4, applied to the final top torus. Then

$$(\cos\theta\, V,\ \sin\theta\, V_0)$$

is smooth, unit-vector-valued, has the required vertical character, and equals $(V, 0)$ on the inner box. The first component extends by zero, since its support stays inside the chart. Choose θ constant near both boundaries, so both extensions are smooth. This preserves the required values for the dense parameter set after shrinking to the inner box. Its geometry and bounds are fixed

before N. The final semidirect-product filtration and polynomial orbit must still be checked at their displayed definitions; they are not consequences of the character descent alone.

7 Bounded audit of affected uses

The failure mechanism is restriction of a generated ideal or filtration to too few weights, followed by an unjustified quotient or lattice identification. The audit scope is the universal degree–rank definitions and the consumers of their commutator/Taylor quotients in Sections 9–12. It includes the zero-index and repeated normalization assertions in those passages.

1. Definition 9.1, p. 1279: use the fixed $R = r_*$ in the top relation, and count the ℓ actual entries. Equations (1) and (2), including all higher weights, give the intended object. Lemmas 1 and 2 prove its filtration and lattice, rather than inheriting typographical indices.

2. Definition 9.6, pp. 1281–1282: use $\mathrm{Horiz}_i = G_{(i,1)}/G_{(i,2)}$, and $G_{(1,2)} = G_{(2,0)}$. The vector/torus distinction is discharged in Proposition 5, not by replacing a congruence by equality.

3. Definition 9.11 and Lemma 9.12, pp. 1284–1287: their G_i means $G_{(i,0)}$. First tautify, then choose lattice generators and construct the filtered map. This order preserves a genuine lattice map; the proof does not assume arbitrary real conjugation preserves the lattice.

4. Lemma 9.14 and Corollary 9.15, pp. 1287–1288, are explicitly stated there not to be used later. Their comparison-filtration argument has the same zero-index and horizontal-lattice issues. The construction in Proposition 5 supplies that mechanism with q trivial when the character is zero. Their broader reuse requires this version, not the unqualified original paragraph.

5. Lemma 10.8, p. 1297: the vertical domain is $U_{(m,R)}$, not the larger ordinary degree group U_m. This unit consistently defines vertical frequencies only on that top group. The source argument's other linear-algebra steps are not newly reproved here.

6. Lemma 10.10 and Theorem 11.1, pp. 1299–1307: use the final core/linear/independent decomposition and the exact two vanishing assertions. The bias and independence premises remain named upstream inputs. The repeated (iii),(iv) of Theorem 11.1 are redundant. Reordering linear and independent petals requires relabelling every associated index, not changing which generators are annihilated.

7. Quotient construction, p. 1309: Theorem 3 proves the full kernel, every filtered quotient, and the top-lattice saturation needed for integral vertical descent. The kernel is not asserted central.

8. Lemma 12.1, pp. 1309–1311: Proposition 5 proves the rank drop with actual constant normalization, horizontal lattice lifts, exact polynomial orbit and conjugate vertical cancellation. The degree-only strengthening is not used or claimed.

9. Petal actions, pp. 1311–1314: the proved abelian ideal and its individual-petal splitting verify (12.8)–(12.10), including zero and negative scaling parameters and integral lattice stability. The displayed final multidegree filtration, chart gluing, and polynomial orbit remain explicit downstream construction checks for LP-00/LP-07. The unit-norm cutoff completion above removes the identified local norm obstruction.

This list is an authored bounded audit, not independent source-first coverage or acceptance of Sections 9–12. It does not classify unrelated errors outside this mechanism. The separate project proofs and the assembled inverse theorem must be independently reviewed at their exact versions.

References

[1] B. Green, T. Tao and T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$-norm, Annals of Mathematics 176 (2012), 1231–1372. https://doi.org/10.4007/annals.2012.176.2.11.

[2] M. C. H. Tointon, Introduction to Approximate Groups, LMS Student Texts 94, Cambridge University Press, 2020, §5.4, pp. 96–98. https://doi.org/10.1017/9781108652865. The author-posted published-book PDF is pinned in the source manifest; its author page lists corrections, none affecting this interface.

[3] Project proof by /root/astra_v14_baseline_theta, Lemma 13.2 repair, version lemma13-2-002, 7 November 2026. Exact manifest and proof hashes are in the accompanying source-use record.

[4] Same author, Tautification and the shifted nilcharacter construction, version tautification-003, 7 November 2026.

[5] B. Green, T. Tao and T. Ziegler, Erratum for the inverse theorem, six-page author document, pp. 5–6. Attribution and problem identification only; no correction theorem is imported from this unpublished document.

The Section 12 semidirect-product join

Author proposal: /root/astra_v14_baseline_gaussian

7 November 2026

Scope. This project proof supplies the final geometric construction on published GTZ pp. 1313–1315 from the explicit petal quotient and comparison interface of LP04. It is an author proposal, not independent acceptance. The analytic frequency input, rank iteration and other inverse theorem interfaces remain separate. We neither invoke the final inverse theorem to prove an intermediate construction nor import the unpublished erratum as a theorem. All quotients below are right lattice quotients.

1 Exact input and output

Fix the data of LP04: integers $m \geq 1$, $1 \leq R \leq m$, fixed finite sets of core and retained petal letters with positive integer weights, and the weighted free nilpotent quotient

$$G = \exp \mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{c} \oplus \mathfrak{p}.$$

The homogeneous relations discard all words of weight greater than m, all weight-m words of length greater than R, and all words containing at least two petals. The last relation holds at every weight. Thus $\mathfrak{p}$ is an abelian ideal. Write $P = \exp \mathfrak{p}$, $G_j = G_{(j,0)}$, $P_j = P \cap G_j$, $G_0 = G_1 = G$, and $P_0 = P_1 = P$. Let Γ be the lattice generated by the exponentials of the letters, and $\Lambda = P \cap \Gamma$. LP04's Hall-coordinate lemma supplies the exact integer lattice coordinates and rationality of G_j, P_j. Put $T = G_{(m,R)}$, and let $\eta : T \to \mathbb{R}$ be the descended continuous character integral on $T \cap \Gamma$.

There are ℓ retained petal letters. For $t \in V = \mathbb{R}^\ell$, let $M_t : P \to P$ scale the summand with unique petal letter a by t_a. We use the following proved LP04 identities, for all real parameters:

$$M_t(p)M_u(p) = M_{t+u}(p), \qquad M_t(gpg^{-1}) = gM_t(p)g^{-1}, \qquad M_t(P_j) \subseteq P_j. \tag{1}$$

For integral z, $M_z(\Lambda) \subseteq \Lambda$. These are endomorphisms; M_0 is not asserted to be invertible.

The family to be represented has the form

$$b_h(n) = g_0(n)M_{\{\alpha h\}}(g_1(n)), \qquad \widetilde{\chi}_h(n) = \widetilde{F}(b_h(n)\Gamma). \tag{2}$$

Here $g_0 : \mathbb{Z} \to G$ and $g_1 : \mathbb{Z} \to P$ are polynomial for the ordinary weight filtrations; $\alpha \in \mathbb{R}^\ell$ is unrestricted. Braces denote one fixed half-open fundamental cube of $V/\mathbb{Z}^\ell$, for example $[-1/2, 1/2)^\ell$. The factors used in GTZ are ordered products $e_a^{\xi_a \binom{n}{i}}$ for core letters of weight i, and the commuting product $\prod_{a \text{ petal}} e_a^{\beta_a \binom{n}{w(a)}}$ for g_1. Core coefficients are independent of h. Identifying an arbitrary input family with this specific factorization is an upstream obligation; it is not inferred from the names of the factors. For the actual coefficients in (12.3)–(12.4), the algebraic identification is exact: $M_t(\exp(uX_a)) = \exp(t_a u X_a)$, and M_t is a homomorphism on the abelian petal group.

Hence applying $M_{\{\alpha h\}}$ to the displayed g_1 gives precisely the product in (12.4). No reordering past a core factor is involved. The hypotheses that produce these coefficient data are still the separate analytic input.

The function $\widetilde{F}$ is a choice made below, before applying the LP04 comparison. It is not an arbitrary previously fixed function having only T-equivariance. This order of choices is necessary.

Theorem 1. *One can choose $\widetilde{F}$ on the fixed G/Γ, with unit vector norm and the required T-frequency η, such that the following holds. There is a fixed connected simply connected nilpotent group H, a lattice Δ, and a rational taut $\mathbb{N}^2$-filtration supported on the downset below $(1, m)$. The nilpotency class is at most $m + 1$. There is a finite list of fixed smooth unit-vector functions on H/Δ, all with the same top vertical character, with fixed finite Lipschitz bounds.*

For every α, g_0, g_1 as above and every finite nonempty parameter set $S \subseteq \mathbb{Z}$, some $S' \subseteq S$ has $|S'| \geq |S|/J$, where the fixed integer J depends only on ℓ, and one function F' in this list satisfies

$$F'(O(h, n)) = (\widetilde{\chi}_h(n), 0) \quad (h \in S',\ n \in \mathbb{Z}). \tag{3}$$

Here O is a polynomial orbit of multidegree at most $(1, m)$. The zero padding is a fixed isometric coordinate inclusion, so projection onto the original coordinates gives literal equality with $\widetilde{\chi}_h$. No scalar component is divided by.

All constants, manifolds and functions are independent of α, the polynomial coefficients, S and N. The conclusion transfers to the full internal ultrapower domains and preserves a dense set of limit parameters. It supplies this construction and its LP04 comparison join; it does not by itself establish the other premises of Theorem 7.1 or GI.

2 Extend the character before choosing the test

Lemma 2. *The character η extends to a continuous character $\bar{\eta} : G_m \to \mathbb{R}$ integral on $G_m \cap \Gamma$.*

Proof. The weight-m group is central. Its surviving Hall commutators have weight m and lengths at most R. At this weight the logarithm of each group commutator is exactly its Lie bracket: every correction has greater weight and vanishes. By LP04's exact Hall coordinates, these logarithms form a real basis and their integer span is precisely the logarithm of $G_m \cap \Gamma$. The subset of length R is a basis of T, and its integer span is $T \cap \Gamma$. Assign to those basis elements the values of η, and assign zero to all remaining weight-m basis elements. These assigned values are integers. Linear extension on this central vector group proves the assertion. The construction depends only on the fixed Hall ordering and η. □

The central torus $G_m/(G_m \cap \Gamma)$ acts freely on G/Γ. For any integral character of this torus there is a smooth equivariant unit-vector function of finite dimension. Here is the construction we use twice. Choose finitely many local smooth sections of the torus quotient and real smooth functions a_j supported in their domains with $\sum a_j^2 > 0$ everywhere. Such functions can be obtained by smooth Euclidean bumps in a finite collection of relatively compact local charts whose smaller charts cover the compact base. Replace a_j by $a_j/(\sum a_i^2)^{1/2}$. In the jth local trivialization (x, z) define the jth component to be $a_j(x)\mathrm{e}(\bar{\eta}(z))$ and extend it by zero. The support lies inside its chart. Thus each component is globally smooth, has the stated vertical transformation law, and the squared vector norm is $\sum a_j^2 = 1$. Local sections can be seen directly: choose a linear complement to the central Lie subspace. Centrality gives $\exp(X + Z) = \exp X \exp Z$, so the complement gives smooth local lifts from the group quotient. The image lattice is discrete by the Hall coordinates with the central

weight-m coordinates deleted. Small coordinate neighborhoods with disjoint lattice translates give the same local lifts on the compact quotient.

Choose this function as $\widetilde{F}$. It has the original frequency η on T and now also the required frequency on all of G_m. Every cancellation in LP04's comparison uses only the restriction to T, so this choice preserves that proof. The comparison does not require a specific previously selected $\widetilde{F}$.

Adverse test. T can be strictly smaller than P_m. For example, with $m = R = 2$ and just one petal generator of weight two, T is trivial whereas $G_m = P \simeq \mathbb{R}$. The unit-vector function $(\cos 2\pi x, \sin 2\pi x)$ on $\mathbb{R}/\mathbb{Z}$ satisfies the vacuous T-law but has no scalar vertical character for all translations. Consequently T-equivariance alone cannot be cited as the final multidegree frequency. The integral extension and new choice above are the repair, not an assertion that every old test already had the stronger property.

3 Both semidirect products and their signs

Set $K = G \ltimes P$ with multiplication

$$(g, p)(g', p') = (gg',\ (g')^{-1} p g'\, p'). \tag{4}$$

Associativity follows either directly from conjugation or from the injective homomorphism $(g, p) \mapsto (g, gp)$ into $G \times G$. Its image is the closed subgroup of pairs agreeing modulo P. The inverse is $(g^{-1}, gp^{-1}g^{-1})$. Its underlying manifold is $G \times P$, so K is connected and simply connected.

Define $\rho_t(g, p) = (gM_t(p), p)$. To check multiplication, applying ρ_t to (4) gives first component $gM_t(p)g'M_t(p')$, by (1). Multiplying the two images has the same first component. Its second is

$$M_t(p')^{-1}(g')^{-1} p g' M_t(p') p' = (g')^{-1} p g' p',$$

since all terms surrounding p' are in the abelian group P. Thus ρ_t is a homomorphism. Further, $\rho_t \rho_u = \rho_{t+u}$ and $\rho_0 = \mathrm{id}$, so ρ_t is an automorphism with inverse ρ_{-t}. This conclusion uses the additive shear action, not invertibility of M_t.

Set $H = V \ltimes_\rho K$ with the right-action convention

$$(t, k)(t', k') = (t + t', \rho_{t'}(k) k'). \tag{5}$$

In three coordinates this reads

$$(t, g, p)(t', g', p') = (t + t',\ gM_{t'}(p)g',\ (g')^{-1} p g' p').$$

The action identity proves associativity, and the inverse of (t, k) is $(-t, \rho_{-t}(k^{-1}))$. Smoothness follows from the polynomial BCH formulas and the linear maps on $\mathfrak{p}$. The manifold $V \times G \times P$ is connected and simply connected.

For a complete nilpotence and filtration check, use Lie coordinates $(t, X, Y) \in V \oplus \mathfrak{g} \oplus \mathfrak{p}$. The bracket is

$$[(t, X, Y), (t', X', Y')] = \big(0,\ [X, X'] + M_{t'}Y - M_t Y',\ [X, Y'] + [Y, X']\big). \tag{6}$$

Indeed the embedding of K has derivative $(X, Y) \mapsto (X, X + Y)$, giving the two bracket terms of K. The derivative of ρ_t is $(X, Y) \mapsto (X + M_t Y, Y)$. Conjugation by $(t, 1)$ on K is ρ_{-t} in convention (5), giving the signs displayed.

Assign bidegree $(0, i)$ to core words of weight i in the X copy, $(1, i)$ to petal words of weight i in the X copy, $(0, i)$ to petal words of weight i in the Y copy, and $(1, 0)$ to V. Every nonzero term

of (6) has the sum of the input bidegrees. For example, $[V, Y_i]$ lies in the first petal copy of degree $(1, i)$, and brackets involving two petal words vanish. All weights are at most m and the first degree is at most one. Each generator has positive total degree, so every bracket of length greater than $m+1$ vanishes. This proves nilpotence of H of class at most $m+1$.

4 The complete filtration and the lattice

Define $H_{a,j} = 1$ for every $a > 1$ and every $j \geq 0$. The remaining positive-j formulas, together with the zero-index groups, are

$$\begin{aligned} H_{0,0} &= H, & H_{a,j} &= 1 \quad (a > 1), \\ H_{1,0} &= \{(t, p, 1) : t \in V, p \in P\}, & H_{1,j} &= \{(0, p, 1) : p \in P_j\}, \\ H_{0,j} &= \{(0, g, p) : g \in G_j, p \in P_j\}. \end{aligned}$$

For $j > m$ these groups are trivial. These definitions are also the sums of the homogeneous Lie spaces whose bidegrees dominate the index. Thus the groups are connected, descending and normal; brackets lie in the sum-index subgroup by (6), and the BCH commutator terms do also. To spell out the boundary cases, two $H_{1,0}$ elements commute, $[H_{1,0}, H_{0,j}] \subseteq H_{1,j}$, and $[H_{1,i}, H_{0,j}] \subseteq H_{1,i+j}$. Brackets with index zero are covered by normality. The groups $H_{1,0}$ and $H_{0,1} = K$ generate H, which proves tautness. In particular

$$Z_* := H_{1,m} = \{(0, p, 1) : p \in P_m\} \tag{7}$$

is central. Importantly $H_{0,1}$ contains the full $G_1 = G$ in its first group coordinate, not merely P.

Let $\Lambda_K = \Gamma \ltimes \Lambda$ with law (4). Since P is normal, Γ normalizes Λ. Therefore this is a subgroup, including inverses. For $z \in \mathbb{Z}^\ell$, both ρ_z and ρ_{-z} preserve Λ_K, by the integral LP04 action. Consequently

$$\Delta = \mathbb{Z}^\ell \ltimes_\rho \Lambda_K = \{(z, \gamma, \lambda) : z \in \mathbb{Z}^\ell, \gamma \in \Gamma, \lambda \in \Lambda\}$$

is a subgroup of H. It is discrete in the product manifold topology.

Here is explicit cocompactness. Right-multiply (t, k) by $(-z, 1)$, where z reduces t into $[0, 1]^\ell$. This gives $(t - z, \rho_{-z}k)$. For an arbitrary $(g, p) \in K$, choose $\gamma \in \Gamma$ with $g\gamma^{-1}$ in a fixed compact representative set for G/Γ. Right multiplication by $(\gamma^{-1}, 1)$ gives $(g\gamma^{-1}, \gamma p \gamma^{-1})$. Now reduce the second coordinate into a fixed compact representative set for P/Λ by right multiplication by $(1, \lambda^{-1})$. The resulting compact product set represents every right coset of Λ_K. Applying this after the t reduction gives a compact product set representing every Δ coset. No bound on the intermediate conjugate or on z is needed.

The same argument inside $G_j \ltimes P_j$, using the already proved lattices $\Gamma \cap G_j$ and $\Lambda \cap P_j$, proves rationality of $H_{0,j}$. The intersections for $H_{1,j}$ are exactly $\Lambda \cap P_j$, and for $H_{1,0}$ exactly $\mathbb{Z}^\ell \times \Lambda$. These are cocompact by LP04 and the product construction. This checks all filtered lattice intersections, including zero and trivial groups. The topology, filtration, lattice and smooth metric on the compact quotient are fixed before any orbit coefficients are chosen.

5 Polynomial orbit and exact reduction

Define

$$O(h, n) = (0, g_0(n), g_1(n))(\alpha h, 1, 1)\Delta. \tag{8}$$

We verify polynomiality in the full group before taking the quotient. The source filtration on $D = \mathbb{Z}^2$ is $D_{0,0} = \mathbb{Z}^2$, $D_{1,0} = \mathbb{Z} \times \{0\}$, $D_{0,1} = \{0\} \times \mathbb{Z}$, and $D_{a,j} = \{0\}$ at every other nonzero index. It is an $\mathbb{N}^2$-filtered abelian group. The embeddings $g \mapsto (0, g, 1)$ and $p \mapsto (0, 1, p)$ are homomorphisms taking the weight-j groups to $H_{0,j}$. Each independent-of-h factor is therefore polynomial on $\mathbb{Z}^2$ with its coordinate multidegree filtration: h differences vanish, and n differences have the asserted $H_{0,j}$ membership. The map $(h, n) \mapsto (\alpha h, 1, 1)$ is a homomorphism in h with values in $H_{1,0}$, has zero second h difference, and is constant in n. Thus it too is polynomial for this same filtration. Published Corollary B.4, GTZ p. 1334, says that polynomial maps between the same index-filtered groups are closed under pointwise multiplication. Here both source and target are the precise $\mathbb{N}^2$-filtered groups just checked. Applying it to these three factors proves polynomiality of (8), including all mixed derivatives and all integer inputs. This is the only external polynomial-product theorem used in this join.

For the explicit inputs preceding Theorem 1, their univariate polynomiality can also be checked before this use. An r-fold difference of $e_a^{c\binom{n}{i}}$ is $e_a^{c\Delta^r\binom{n}{i}}$ because all its values lie in one abelian one-parameter subgroup. It vanishes for $r > i$; for $r \le i$ it lies in $G_i \subseteq G_r$. This holds for arbitrary integer increments, including negative ones. Products are polynomial by the same Corollary B.4; for g_1 all factors already commute in P. No Taylor expansion with a potentially unjustified centrality hypothesis is imported for this check.

Write $z_h = \alpha h - \{\alpha h\} \in \mathbb{Z}^\ell$. Formula (5) gives a representative of (8) as

$$(\alpha h, g_0(n) M_{\alpha h}(g_1(n)), g_1(n)).$$

Right multiplication by $(-z_h, 1, 1) \in \Delta$ changes this to

$$(\{\alpha h\}, g_0(n) M_{\{\alpha h\}}(g_1(n)), g_1(n)). \tag{9}$$

The equality follows from the additive identity in (1). It is exact for every h, n; there is no illegal commutation of a fractional-part operation with a nonabelian product.

6 Fixed charts, unit completion and uniform bounds

Projection onto t gives a smooth map $\pi : H/\Delta \to V/\mathbb{Z}^\ell$. If a box $U \subset V$ has width less than one in each coordinate, projection is injective on U. On the corresponding open torus box there is an exact smooth trivialization

$$U \times K/\Lambda_K \longrightarrow \pi^{-1}(U \bmod \mathbb{Z}^\ell), \qquad (t, k\Lambda_K) \longmapsto (t, k)\Delta. \tag{10}$$

Indeed two representatives in U differ by an element of Δ only if their integer t displacement is zero; their k coordinates then differ precisely by right multiplication in Λ_K. Surjectivity follows by reducing the t coordinate to its unique lift in U. The formulas are smooth on this chart.

Choose finitely many fixed closed boxes I_j covering the closed fundamental cube $[-1/2, 1/2]^\ell$, with fixed slightly larger boxes U_j of widths less than one. They may extend beyond the cube's boundary; each U_j still projects injectively. Choose a smooth function θ_j on the torus which is zero on I_j modulo integers and $\pi/2$ outside a smaller open box with closure contained in U_j. It can be constructed by a fixed Euclidean bump in U_j, equal to one on I_j, followed by multiplication of its complement by $\pi/2$. The chart margins are fixed positive real numbers, not functions of N. At least one box contains $\{\alpha h\}$ for at least $|S|/J$ of the parameters, where J is the fixed number of boxes. Fix that box and call this subfamily S'.

On its chart define

$$W(t,(g,p)\Lambda_K) = \widetilde{F}(g\Gamma).$$

The projection $K/\Lambda_K \to G/\Gamma$ is well-defined and smooth, since the first coordinate of (4) is a homomorphism and maps Λ_K into Γ. Thus W is a smooth local unit vector. Define a character on the actual group (7) by

$$\eta'(0,p,1) = \bar{\eta}(p) \qquad (p \in P_m).$$

It is integral on $Z_* \cap \Delta$ by Lemma 2. Left multiplication by $(0,p_*,1) \in Z_*$ changes the chart's first group coordinate to p_*g, keeps t fixed, and hence multiplies W by $\mathrm{e}(\bar{\eta}(p_*))$. This uses equivariance on P_m, not merely on $T \cap P$.

The torus $Z_*/(Z_* \cap \Delta)$ acts freely on H/Δ: centrality makes the stabilizer at every point exactly $Z_* \cap \Delta$. It is compact by the rationality already proved. The quotient by Z_* has coordinates $V \times (G/P_m) \times P$, and the image of Δ has coordinates $\mathbb{Z}^\ell \times q(\Gamma) \times \Lambda$. Here $q(\Gamma)$ is discrete by LP04's Hall-coordinate quotient, so the image lattice is discrete and the quotient base is compact. Central complements and small coordinate neighborhoods give local lifts exactly as in Section 2. The finite local-section construction from that section therefore gives a global smooth unit vector W_0 with frequency η'. Set

$$F_j' = (\cos(\theta_j \circ \pi)W, \sin(\theta_j \circ \pi)W_0). \tag{11}$$

The first component is extended by zero outside the chart. It vanishes on a neighborhood of the chart boundary, so this extension is smooth. Outside the chart the second coefficient is one. Inside, the squared norm is $\cos^2\theta_j + \sin^2\theta_j = 1$. Both components have the same vertical frequency because Z_* fixes π. Therefore F_j' is a genuine unit-vector nilcharacter test, including outside the box.

For $h \in S'$, the representative (9) uses the unique lift $\{\alpha h\} \in I_j$. Its cutoff angle is zero, so (11) gives (3) for every integer n. The second component is exactly zero there, not an approximation.

Every chart used has compact closure in a fixed larger chart, and its fiber K/Λ_K is compact. The displayed projections and their derivatives are bounded on these sets. The finite bump functions, $\widetilde{F}$, W_0, and the fixed metrics have bounded derivatives. Taking the maximum over the finite list proves a finite Lipschitz bound depending only on the fixed groups, character and charts. Unbounded polynomial coefficients change only the orbit; they do not enter these derivative bounds. This also proves fixed finite dimension. If $\ell = 0$ the torus is a point, use one chart and $\theta = 0$; all arguments still apply. Trivial top groups or zero characters likewise cause no exception. This completes the ordinary proof of Theorem 1.

7 Internal domains and the LP04 consumer

All groups, charts, test functions and characters above are standard objects after fixing the selected dimension data and character. All algebraic identities hold for every real parameter and every group point. Taking representative sequences gives internal polynomial maps, including for unbounded limit coefficients. The standard tests extend to the whole ultrapower of the compact quotient. Their vertical laws then hold for every limit element of Z_* and every limit point, by transfer of the actual universal identity. We are not inferring all-limit equivariance from an identity checked only on standard points.

Finite pigeonhole over the fixed list of boxes preserves positive standard density of a limit parameter family. Equivalently apply the ordinary cardinality inequality to representatives and pass to the ultralimit. After choosing the resulting fixed box, the nilcharacter $\chi(h,n) = F_j'(O(h,n))$

is defined on all of ${}^*\mathbb{Z}^2$, not only on the dense subfamily. Its exact restriction is the fixed isometric inclusion of $\widetilde{\chi}_h$ for all limit integers n there.

Choose $\widetilde{F}$ by Section 2 before invoking LP04's comparison proposition. That proposition's top cancellation is unchanged because $\bar{\eta}|_T = \eta$. It produces the same strictly lower degree–rank comparison for $\chi_h \otimes \overline{\widetilde{\chi}_h}$. For clarity, unit norm gives the exact component recovery identity

$$(\chi_h)_a = \sum_b (\widetilde{\chi}_h)_b \big((\chi_h)_a \overline{(\widetilde{\chi}_h)_b}\big).$$

On the selected subfamily, replace $(\widetilde{\chi}_h)_b$ by the corresponding component of the single $\chi(h,n)$. The terms in parentheses are exactly the lower degree–rank sequences supplied by LP04. The sum is finite and its coefficients are fixed; no division or nonvanishing of an individual component is used. The zero padding in (3) changes none of these terms.

At $R = 1$, lower degree–rank means degree at most $m-1$. At $R > 1$ the subsequent rank iteration remains an upstream/downstream assembly obligation; we do not incorrectly replace it by an immediate ordinary degree drop. Likewise the frequency decomposition leading to (2), the initial correlation, and the remaining Theorem 7.1 induction premises are not established by this geometric join.

8 Bounded related-use and consistency audit

The search scope is the Section 12 construction and its immediate LP04 comparison and Theorem 7.1 consumers, published pp. 1308–1315. We also check the polynomial-product supplier B.4 at p. 1334. This is an authored bounded audit, not a claim of independent source-first coverage of GI.

1. The smaller degree–rank character domain versus the ordinary top group is repaired by the integral extension and the new test choice. The repeated final frequency on p. 1315 is explicitly restricted to P_m and uses $\bar{\eta}$. A favorable quotient-character review alone does not supply this extension.

2. The illustrative $H_{0,1}$ on p. 1312 restricts its first group coordinate to the petal subgroup, although the orbit contains the core $e_2^{\beta n}$. For nonzero generic β its n difference is outside that alleged group. The general p. 1314 definition uses full G_j; we use and prove that definition, not the defective illustration.

3. Zero or negative scaling parameters do not make every M_t invertible. The actual invertible automorphisms are the shears ρ_t, whose inverse and integer lattice preservation are proved here.

4. The lattice in p. 1314 is a subgroup of H, not of the original G. Every filtered lattice intersection and the exact right-coset reduction are checked above; neither discreteness alone nor a real endomorphism establishes the claim.

5. The cutoff on pp. 1312 and 1315 cannot alone be a unit-vector test or guarantee equality on the selected box. Fixed inner boxes, positive chart margins and (11) supply unit norm, literal equality and coefficient-independent Lipschitz bounds.

6. In particular a variable box of arbitrarily small width would not supply a fixed Lipschitz bound. The predetermined finite cover removes this possible parameter dependence at a controlled density loss.

7. The p. 1308 claim that pigeonhole puts the fractional parts in a short interval around zero is not valid for an arbitrary dense family. For $\alpha = 1/2$ and odd h, our fractional parts are all $-1/2$. Our fixed cover allows a box at any location, including across the chosen fundamental-cube boundary, and does not need a near-zero assertion.

8. Formula (12.11) is checked as right multiplication by an actual lattice element; the displayed identities do not silently change the object or introduce a different representative in the local test.

9. A lower rank comparison is not automatically a lower ordinary degree comparison. The consumer paragraph above preserves the exact LP04 conclusion and leaves the rank iteration to its actual proof.

Earlier printed statements and earlier frozen project packets are retained; this proposal does not relabel them as having contained the new argument. Any other current copies of the smaller-domain frequency claim or the illustrative filtration must be corrected or marked historical when this proof is incorporated. The attached occurrence record identifies these specific copies and the new canonical assertions for review.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics 176 (2012), 1231–1372, Section 12 and Corollary B.4. `https://doi.org/10.4007/annals.2012.176.2.11`.

[2] Project proof by /root/astra_v14_assertions_arithmetic, *GTZ-LP-04: the petal quotient and degree–rank construction*, lp04-001. Its exact manifest and proof hashes are recorded in the accompanying source-use record. Its dependencies and independent-review status remain separate.

Degree conversion and the finite rank iteration

Author proposal: /root/astra_v14_baseline_theta

7 November 2026

Status. This is a bounded author proof awaiting independent review. It proves the degree conversions and the assembly of the rank steps from the precisely stated one-step geometric/frequency interface. It does not prove the analytic sunflower or commutator hypotheses of GTZ Sections 10–11. The complete inverse theorem remains open. Every internal nilcharacter below has exact unit norm and vertical equivariance for all limit elements under the natural ultrapower interpretation. The main manuscript is not modified.

1 The actual degree-to-rank conversion

Let G/Γ be a connected simply connected nilpotent Lie group with lattice and rational degree filtration $(G_i)_{i\geq 0}$ supported on $0 \leq i \leq m$. Apply accepted bounded tautification first, so $G_0 = G_1 = G$ and polynomial orbits may be normalized at zero without changing the represented sequence. Write $\mathfrak{g}_i = \log G_i$. All the vector spaces here are rational for the rational structure determined by Γ.

For $(d, r) \in \mathrm{DR} = \{(d, r) \in \mathbb{N}^2 : 0 \leq r \leq d\}$ with $d > 0$ define $\mathfrak{h}_{(d,r)}$ to be the real span of all iterated Lie brackets of length $\ell \geq 1$ with entries $X_j \in \mathfrak{g}_{i_j}$, $i_j \geq 1$, for which

$$w := \sum_{j=1}^{\ell} i_j > d, \qquad \text{or} \qquad w = d \text{ and } \ell \geq \max(r, 1). \tag{1}$$

Take every bracketing and set $\mathfrak{h}_{(0,0)} = \mathfrak{g}$, $H_{(d,r)} = \exp \mathfrak{h}_{(d,r)}$. Only finitely many nonzero brackets occur, since their weight exceeds m when they can no longer contribute and every entry has positive weight.

Lemma 1. *These are connected rational groups forming a taut degree–rank filtration supported on (m, m). For each $d > 0$, $H_{(d,0)} = H_{(d,1)} = G_d$. Hence the original degree-polynomial orbits are exactly the degree–rank polynomial orbits for this refinement. A degree-m nilcharacter is a degree–rank (m, m) nilcharacter with the same function and with the top character restricted.*

Proof. All bracket spans in (1) are rational, because brackets of rational vectors are rational. Bracketing with $\mathfrak{g}_1$ raises weight by at least one, so each span is an ideal of $\mathfrak{g}$. It is therefore a Lie subalgebra, and its exponential is a connected normal subgroup, which is closed by the global exponential homeomorphism. Choose a rational Mal'cev basis for G/Γ compatible with its rational structure. Each bracket span has a rational basis in these coordinates. Its lower central series is rational, since brackets of rational vectors are rational. Choose one standard Q bounding the finitely many rational structure constants and basis coefficients. Green–Tao, *Annals of Mathematics* 175 (2012), Definition 2.5 and Proposition A.10, applied to this subgroup with its lower central filtration, supplies a Mal'cev basis relative to its actual intersection with Γ. That intersection is

therefore discrete and cocompact. This application uses Lie-algebra rationality in Definition 2.5; it does not assume the lattice conclusion.

The lexicographic nesting is immediate at a fixed d as r increases. If d increases, every new allowed weight was strictly above the old d, so its span is contained in the old one as well. For two positive indices $(d,r),(e,t)$, brackets either have weight greater than $d+e$, or weight exactly $d+e$ and length at least $\max(r,1)+\max(t,1)\geq\max(r+t,1)$. Thus the Lie bracket is in $\mathfrak{h}_{(d+e,r+t)}$. The BCH commutator and its higher terms lie there too, using the ideal property. At index zero use normality. This verifies every pair of indices, not only equal indices.

For $r=0$ or 1, every allowed bracket has weight at least d and thus lies in $\mathfrak{g}_d$. Conversely the length-one brackets with $i_1=d$ include all of $\mathfrak{g}_d$. This proves $H_{(d,0)}=H_{(d,1)}=G_d$. In particular $H_{(1,0)}=G$. If $d>m$ all brackets vanish. At degree m every nonzero bracket has length at most its weight m, which gives the asserted support.

The domain filtration on $\mathbb{Z}$ has $\mathbb{Z}$ at indices $(0,0),(1,0)$ and zero at all other indices, including $(1,1)$, as in Definition 6.17. The derivative criterion for an adapted map therefore asks for its ath differences to lie in $H_{(a,0)}$; these groups equal G_a. This proves the equality of the polynomial classes directly, matching published (6.8). The new top $H_{(m,m)}$ is a rational subgroup of G_m. Restricting the original vertical character is integral on its lattice and preserves all exact vertical identities and unit norm. No enlargement of its character domain is asserted. □

2 The total-degree conversion used in related applications

For a rational multidegree filtration $(G_a)_{a\in\mathbb{N}^q}$ supported on a finite downset J, first use bounded tautification to restrict to $H=\langle G_{e_1},\dots,G_{e_q}\rangle$. Let

$$K_0=H,\qquad \mathfrak{k}_i=\sum_{a\in J:\ |a|\geq i}\mathfrak{g}_a,\qquad K_i=\exp\mathfrak{k}_i\quad(i\geq 1).$$

Each $\mathfrak{g}_a$ is an ideal in $\mathfrak{g}_0$, so the sum is a rational ideal. It equals the Lie algebra of the connected generated group. The brackets satisfy $[\mathfrak{k}_i,\mathfrak{k}_j]\subseteq\mathfrak{k}_{i+j}$, because $[\mathfrak{g}_a,\mathfrak{g}_b]\subseteq\mathfrak{g}_{a+b}$ and $|a+b|=|a|+|b|$. The index-zero condition follows from normality. Also $K_1=H$ and K_i is trivial above $D=\max_{a\in J}|a|$. The rational subgroup lemma verifies every lattice intersection.

For the diagonal restriction $n\mapsto g(n,\dots,n)$ of a multivariable polynomial orbit, use its Taylor expansion. A factor with coefficient $u_a\in G_a$ has scalar exponent $\prod_j\binom{n}{a_j}$ of ordinary degree $|a|$. Such a factor is adapted to $K_\bullet$ since $u_a\in K_{|a|}$; its successive differences have the same one-parameter coefficient and the required scalar polynomial degree. The product is polynomial by B.4. This checks the actual diagonal map rather than treating the domain filtrations as identical. For a rectangle $J=\{a:a\leq d\}$, the only index of total weight $|d|$ is d, so $K_{|d|}=G_d$. A multidegree-d nilcharacter therefore restricts diagonally to a degree-$|d|$ nilcharacter with exactly the same top character. For a nonrectangular downset only the nilsequence degree assertion has been proved, and no unwarranted common top character is claimed.

3 The precise one-step interface

Fix $s\geq 3$, write $m=s-1$, and let $1\leq R\leq m$. Suppose f is bounded on $[N]$, H is a dense internal set of shifts, $A(h,n)$ is a single unit multidegree $(1,m)$ nilcharacter, and (χ_h) is an internal uniformly bounded family of unit degree–rank (m,R) nilcharacters. Suppose that for every retained h, $\Delta_h f$ has positive degree-at-most-$m-1$ correlation with $A(h,\cdot)\otimes\chi_h$.

The one-step premise used here is the following exact output, after restricting to a dense internal subset and fixing uniform standard bounds. It must be supplied by the corrected additive-quadruple result, the actual Section 10 frequency representation, the two conclusions of Theorem 11.1, accepted LP04 and the final Section 12 join:

1. There is a unit multidegree $(1, m)$ nilcharacter $Z(h, n)$, defined globally, and a family of unit-vector sequences $\zeta_h(n)$ such that $Z(h, n) = \iota\zeta_h(n)$ for retained h, with one fixed coordinate isometry ι. Zero padding is allowed.

2. The entire matrix-valued function $V_h = \chi_h \otimes \overline{\zeta_h}$ is a nilsequence of degree–rank at most $(m, R-1)$, with standard uniform geometry and Lipschitz bounds and internal parameter dependence.

The second clause states a rank drop; for $R > 1$ it does not state an ordinary-degree drop. The final Section 12 constructor must choose its test equivariant on the full ordinary top before invoking LP04's comparison. Equivariance only on $G_{(m,R)}$ does not by itself justify its larger multidegree top. Those obligations remain in the separately authored geometric join and are not hidden in this premise.

4 Recovering the next family without dividing by components

The unit-vector identity gives, for every component a and all relevant integers,

$$(\chi_h)_a(n) = \sum_b (\zeta_h)_b(n)\big((\chi_h)_a(n)\overline{(\zeta_h)_b(n)}\big). \tag{2}$$

The terms in parentheses are components of V_h. Replace the first factor by its corresponding coordinate in $Z(h, n)$ and form the unit nilcharacter $A'(h, n) = A(h, n) \otimes Z(h, n)$ on the product model. The product filtration has the same multidegree bound $(1, m)$.

Choose a scalar component and a degree-at-most-$m-1$ test witnessing the incoming correlation, with fixed bounds and a fixed positive threshold after the usual finite selection. Inserting (2) expresses it as a fixed finite sum of correlations of $\Delta_h f$ against components of $A'(h, \cdot)$, the same ordinary lower-degree test, and a component of V_h. At least one summand has a positive fixed correlation. Finite pigeonhole restricts to a dense subset with fixed component choices if necessary. No scalar component is inverted, or assumed nonzero or of modulus one.

On each fixed rational degree–rank model for V_h, apply the unit-character approximation construction to its top group at $(m, R-1)$. That group is central after tautification, since bracketing with $G_{(1,0)}$ raises the first coordinate above m. Its lattice is rational. Smooth local sections and a finite unit partition on the central torus produce exact-character unit-vector tests. The complex linear span of their components is a unital conjugation-stable algebra separating points; Stone–Weierstrass and a finite uniform net of the bounded Lipschitz test class give a finite approximation by components of such tests. This is the same explicit compact construction in the assembly's selection section, now on the stated degree–rank model. It uses centrality and rationality, not an identification of this top with the larger ordinary-degree top.

Choose the approximation error so that its correlation with the other fixed bounded factors is less than half the surviving threshold. Another finite selection supplies a unit degree–rank $(m, R-1)$ nilcharacter χ'_h for which $\Delta_h f$ has positive degree-at-most-$m-1$ correlation with $A'(h, \cdot)\otimes\chi'_h$. Finite choices and orbit choices are made on representatives and then passed to the ultralimit, retaining internal dependence and uniform bounds. Every test has its exact standard vertical identity before extension, so the all-limit convention is preserved. This is the output required by the next rank step.

5 Initialization, termination and the actual endpoint

Start with the degree-m family obtained from the preceding inverse order. Lemma 1 makes it a degree–rank (m, m) family with the same functions. Take $A = 1$, viewed as a constant unit multidegree $(1, m)$ nilcharacter. The original positive correlation is degree-at-most-$m-1$ correlation with the lower test equal to one.

Apply the one-step premise and the preceding recovery argument at $R = m, m-1, \ldots, 1$. There are exactly m steps, a fixed standard number. A finite sequence of positive standard density losses still leaves a positive standard density; the same holds for positive correlation and finite complexity bounds. Products of the successive multidegree $(1, m)$ nilcharacters retain that bound on their fixed product model.

At the last step the residual family has support at most $(m, 0)$. Its extra degree–rank axiom says $G_{(m,0)} = G_{(m,1)}$. But $(m, 1) > (m, 0)$, so the latter group is trivial. All groups of larger degree are also trivial. Thus the ordinary filtration $(G_{(i,0)})_i$ has degree at most $m-1$. The residual nilcharacter, viewed merely as a bounded nilsequence, can now be multiplied into the already present ordinary degree-at-most-$m-1$ correlation test. The product degree is still at most $m-1$.

We obtain one internal unit multidegree $(1, m) = (1, s-1)$ nilcharacter $A_{\text{final}}(h, n)$ such that $\Delta_h f$ has degree-at-most-$s-2$ correlation with it on a dense set of shifts. This is the exact Theorem 7.1 endpoint needed by the separately authored symmetry proof. It is conditional on the one-step premise being supplied at every input rank and every constructed family. Acceptance of the finite iteration alone does not establish those analytic premises or the full inverse theorem.

6 Source findings and bounded related-use scope

The actual final printed p. 1262 was inspected visually. In Example 6.11 the multidegree-to-degree group is printed with $|a| \le i$, which includes G_0 at every i and does not give bounded degree. For example $G_0 = G_1 = \mathbb{R}$, $G_2 = 0$ would give the group $\mathbb{R}$ at every index. The required rule is $|a| \ge i$, proved in Section 2 above.

The same example's degree-to-rank rule prints the equality of summed weights to the fixed bound d, rather than to the running index d'. The desired identity $H_{(d',0)} = G_{d'}$ then need not hold. For instance the linear filtration $G_0 = G_1 = \mathbb{R}$, $G_2 = 0$ with allowed degree bound $d = 2$ yields zero at $(1, 0)$ under that printed rule, whereas $G_1 = \mathbb{R}$. A nonconstant linear polynomial is then lost. Section 1 supplies the current-index rule and proves every filtration, lattice and polynomial assertion used here. The printed length condition is $\ell \ge \max(r', 1)$; that sign is not itself an error.

The bounded related-use scope is Example 6.11, the polynomial identity (6.8), Lemma 6.21's degree conversions, the initialization and rank-zero termination of Theorems 7.1–7.2, and the diagonal/total-degree conversions used in E.9 and the Section 13 assembly. Separate authors of the latter consumers have been notified. No claim is made here to have read every unrelated use of a filtration in the source or every historical correction. These formula errors are not evidence that the intended inverse theorem is false. Earlier source bytes and decisions are preserved.

Scaling symbols and symmetric multilinearisation with the factorial retained

Author proposal: /root/astra_v14_baseline_compiler

7 November 2026

Status and scope. This is a new proof proposal, not an independent acceptance. It supplies GTZ-LP05 and the scaling assertion E.8(v), including the divisibility used here. Its project input is the exact tautification theorem in [2], still pending independent review. It does not prove the remaining correction units or the inverse theorem. The unpublished erratum motivates the factorial correction but supplies no black-box inference. All source and project pins are in the accompanying contracts. Protocol v1.4's available/needed/connecting-argument distinctions are recorded there.

1 Conventions and preliminary constructions

We use the rational, connected, simply connected nilpotent groups, polynomial maps, finite-dimensional unit-sphere-valued nilcharacters and symbol equivalence of [1, Sections 3–6 and Appendix E]. For a vector-valued function u, $u^{\otimes r}$ means its ordinary tensor power if $r \geq 0$, with $u^{\otimes 0} = 1$, and $\overline{u}^{\otimes(-r)}$ if $r < 0$. Conjugation is always written explicitly. Equivalence at multidegree d means that every component of $u \otimes \overline{v}$ is a nilsequence with support in $\{a : a \leq d,\ a \neq d\}$; all inequalities on multiindices are coordinatewise. In the degree case this is degree at most $d-1$. We use the group law on symbols proved in published E.7 and E.8(i)–(iii). Their proofs use tensor products, conjugation, contraction of a unit vector with its conjugate, and a rational top-group quotient. With tautification, the top group is central, so those proofs apply to the present objects.

Initially all data are ordinary, and all integers d, k, q are fixed standard integers. The last section states exactly how the constructions pass to limits and parameter families. A multidegree nilcharacter on a limit subset comes with a representing global polynomial orbit and test function; we use that representation to extend it, and only then restrict back.

Apply [2, Theorem 1 and Corollary 2] to the given representation. We can work on a taut group and with an exactly equal represented sequence whose polynomial map g satisfies $g(0) = 1$. Positive groups and the vertical character are unchanged. For a degree filtration $G_0 = G_1$; for multidegree G_0 is generated by the G_{e_i}. In particular G_d is central when the support is contained below one positive d. Write $\mathfrak{g}_a = \log G_a$ with its rational structure induced by Γ.

Lemma 1 (Monomial coordinates)**.** *For multidegree support $a \leq d$, the normalised map has an expression*

$$g(n) = \prod_{0<a\leq d} \exp(X_a n^a), \qquad X_a \in \mathfrak{g}_a, \quad n^a = \prod_i n_i^{a_i}. \tag{1}$$

Fix any order refining total weight. For a degree-D map on $\mathbb{Z}^k$, replace $a \leq d$ by $0 < |a| \leq D$ and require $X_a \in \mathfrak{g}_{|a|}$. Every such ordered product is polynomial for the indicated filtration.

Proof. For a degree-D input, use the multidegree target $\mathfrak{g}_a = \mathfrak{g}_{|a|}$: coordinate derivatives are particular degree derivatives, so the input is polynomial for this target. Published Lemma B.9 gives ordered binomial monomials with coefficient at index a in G_a. Expand the finite BCH formula. A product of scalar binomial polynomials with labels $a_1, \dots, a_r$ has monomials n^b only for $b \le a_1 + \cdots + a_r$. Its Lie-bracket coefficient belongs to $\mathfrak{g}_{a_1+\cdots+a_r} \subseteq \mathfrak{g}_b$. Thus $\log g(n) = \sum_{0<b\le d} Y_b n^b$, $Y_b \in \mathfrak{g}_b$; terms outside the support vanish and the constant is zero. To obtain (1), process increasing total monomial weight: choose X_b to match the coefficient at n^b in the BCH logarithm of the partial ordered product. All commutators of previous factors have larger total monomial weight, and their coefficients at b belong to $\mathfrak{g}_b$. Consequently the adjustment is in $\mathfrak{g}_b$ and does not change any previously matched weight. The finite support makes this algorithm terminate. The argument with $\mathfrak{g}_{|b|}$ is identical. For the converse each factor is a polynomial map: in its one-parameter abelian subgroup an iterated difference of n^a vanishes if its coordinate order exceeds a, and otherwise the coefficient stays in $\mathfrak{g}_a \subseteq \mathfrak{g}_b$ at derivative index b. Apply the product theorem [1, Corollary B.4]. This use does not invoke E.8, E.9 or GI. □

We will use rational connected subgroups and rational central quotients. Their generated subgroups and central quotients are the geometric construction proved in [3, Section 2]. We spell out the additional rational-image check used below. For each rational G_a, choose finitely many elements $\gamma_{a,j}$ of $\Gamma \cap G_a$ whose logarithms span $\mathfrak{g}_a$. Such a choice exists: the real span of the logarithms of a nilpotent subgroup is a Lie algebra. Indeed conjugation by integer powers of any of its elements gives a polynomial in that integer, and finite differences extract its iterated brackets; BCH then gives the assertion. Its connected subgroup contains the lattice, so cocompactness forces it to be G_a, by the homogeneous-space argument of [3, Section 2]. The same polynomial/BCH argument gives the rational Lie structure $\operatorname{span}_{\mathbb{Q}} \log \Gamma$; all filtration spaces are rational for this structure.

For the scaling image below, the vectors $(q^{|a|} \log \gamma_{a,j}, \log \gamma_{a,j})$ exponentiate to the actual lattice elements $(\gamma_{a,j}^{q^{|a|}}, \gamma_{a,j})$. For the multilinear image, multiply each $\log \gamma_{a,j}$ by an integer N so that its rational second component in (8) lies in the lattice L constructed next. Its first component then exponentiates to $\gamma_{a,j}^N$. These finitely many lattice elements have logarithms spanning each image Lie algebra $\mathfrak{k}_b$. The finite-generator fact in [1, p. 1243] makes their generated discrete subgroup cocompact in K_b. Thus K_b is rational for the actual product lattice, not merely for an abstract rational form. This also checks $b = 0$.

The following explicit lattice choice records the extra finite-symmetry requirement needed below. In a rational nilpotent Lie algebra with a finite group P of rational automorphisms, choose a full rational lattice L_0 and replace it by $\sum_{\sigma\in P} \sigma L_0$. Clear rational bracket denominators and the finitely many BCH denominators by replacing L_0 by $L = ML_0$. Then $\mathrm{BCH}(L, L) \subseteq L$ and $-L = L$. The group $\exp L$ is discrete. It is cocompact: an adapted rational central-series basis, after integer scaling, lies in L; successive rounding of its coordinates gives a compact set of coset representatives. The same construction makes all specified rational filtration subspaces rational. If a rational linear functional on a specified rational central subspace is prescribed, increase M to make it integral on L intersected with that subspace. This gives a P-invariant lattice, not just a rational structure invariant under P. Each choice uses only fixed geometric data.

2 Scaling and divisibility on the actual orbit

Theorem 2 (Corrected E.8(v))**.** *Let $d \neq 0$, and let $\chi(n) = F(g(n)\Gamma)$ have multidegree at most d. For every standard integer q,*

$$[\chi(qn)] = q^{|d|}[\chi(n)]. \tag{2}$$

For degree $D > 0$ on $\mathbb{Z}^k$, the exponent is D. The equivalence is on the whole domain, hence on every indicated restriction. Its geometric and Lipschitz bounds are independent of the polynomial coefficients.

Proof. Use (1). Form the finite positive Rees Lie algebra

$$\mathcal{R} = \bigoplus_{0<a\le d} \mathfrak{g}_a t^a, \qquad [Xt^a, Yt^b] = [X,Y]t^{a+b},$$

where a component outside the downset is zero. This is a Lie algebra: it is a subalgebra of $\mathfrak{g} \otimes \mathbb{R}[t_1, \dots, t_k]$ modulo the ideal spanned by monomials outside the downset. Define the rational Lie map

$$A_q(Xt^a) = (q^{|a|}X, X) \in \mathfrak{g} \oplus \mathfrak{g}.$$

Its bracket identity is exact because $q^{|a|}q^{|b|} = q^{|a+b|}$, including negative and zero q. Let $\mathfrak{k}_b$ be the image of $\bigoplus_{a\ge b,\ a\neq 0} \mathfrak{g}_a t^a$, and put $K_b = \exp \mathfrak{k}_b$. These are rational, descending, and $[\mathfrak{k}_b, \mathfrak{k}_c] \subseteq \mathfrak{k}_{b+c}$. Also $\mathfrak{k}_0 = \sum_i \mathfrak{k}_{e_i}$, so the filtration is taut. The ambient group is K_0, with lattice $K_0 \cap \Gamma^2$. Every factor of

$$(g(qn), g(n)) = \prod_a \exp(n^a A_q(X_a t^a))$$

is polynomial at index a, and hence so is their product. The top group has Lie algebra $\mathfrak{k}_d = \{(q^{|d|}X, X) : X \in \mathfrak{g}_d\}$. For $c = q^{|d|}$, the restricted function

$$(x, y) \longmapsto F(x) \otimes \overline{F(y)^{\otimes c}} \tag{3}$$

is invariant under K_d: the two vertical phases are $c\eta(X)$ and $-c\eta(X)$. This calculation uses $F^{\otimes c}$, with the signed convention above, on the same second coordinate. The rational central quotient K_0/K_d has the quotient filtration with its top group zero. Thus (3) along the displayed orbit is a lower-degree nilsequence, which is precisely (2). Restriction and descent preserve a fixed Lipschitz bound since the geometric maps and compact quotient are fixed. For degree D use $\bigoplus_{0<|a|\le D} \mathfrak{g}_{|a|} t^a$ and the filtration by $|a| \ge j$. The top image is the same graph with exponent D; all other steps agree. At $q = 0$ the first orbit coordinate is 1, and (3) is the constant $F(1)$; no 0^0 or division is used. □

Corollary 3 (Divisibility, with the representative specified)**.** *For positive degree and every standard $Q \ge 1$, there is a nilcharacter ψ on the same domain with $[\chi] = Q[\psi]$.*

Proof. In (1) replace each X_a by $Q^{-|a|}X_a$ and call the resulting map g_Q. It is polynomial on the same group and lattice. With the same F put $\xi(n) = F(g_Q(n)\Gamma)$. Then $g_Q(Qn) = g(n)$ exactly, factor by factor in the same order. Theorem 2 gives $[\chi] = Q^D[\xi]$, where $D = |d|$ or the degree bound, and $\psi = \xi^{\otimes Q^{D-1}}$ gives the asserted equation. The test function still has unit norm; its frequency and finite target dimension change only by the displayed standard tensor power. In particular this is divisibility of the actual symbol, not division of an arbitrarily selected multilinear representative after construction. □

3 Raw multilinearisation and the factorial graph

Let $d = (d_1, \dots, d_k) \neq 0$, $D = |d|$, and divide $\{1, \dots, D\}$ into ordered blocks B_i of size d_i. For a subset J put $w(J) = (|J \cap B_i|)_i$ and $a! = \prod_i a_i!$; set $C = d!$. Construct

$$\widetilde{\mathfrak{g}} = \bigoplus_{\varnothing \neq J \subseteq [D]} \mathfrak{g}_{w(J)} \epsilon_J, \qquad [X\epsilon_J, Y\epsilon_L] = \begin{cases} [X,Y]\epsilon_{J\cup L}, & J \cap L = \varnothing, \\ 0, & J \cap L \neq \varnothing. \end{cases} \tag{4}$$

This is a subalgebra of $\mathfrak{g} \otimes \mathbb{R}[\epsilon_1, \dots, \epsilon_D]/(\epsilon_1^2, \dots, \epsilon_D^2)$; Jacobi therefore holds. There is no empty component: normalisation $g(0) = 1$ is what permits its removal. Put

$$\widetilde{\mathfrak{g}}_b = \bigoplus_{1_J \geq b} \mathfrak{g}_{w(J)} \epsilon_J. \tag{5}$$

These are descending rational Lie subalgebras, their bracket has the required index sum, and they vanish when any $b_i > 1$. The zero group is generated by the first-coordinate groups since each nonempty J contains a coordinate. Thus (5) is taut. Its top group is central. The block permutation group $P = \prod_i S_{d_i}$ preserves (4), permutes the groups in (5) by the same permutation of their indices, and acts trivially on the top copy $\mathfrak{g}_d \epsilon_{[D]}$. Choose a P-invariant lattice $\widetilde{\Gamma}$ as in Section 1, making

$$\widetilde{\eta}(\exp(X \epsilon_{[D]})) = \eta(\exp X) \tag{6}$$

integral on its top lattice. This is possible since η is integral on $\Gamma \cap G_d$ and hence rational on the rational top Lie algebra.

Choose a Lipschitz unit-sphere-valued function $\widetilde{F}$ with frequency $\widetilde{\eta}$. Here is the needed existence construction, with no invariance under P assumed. The central torus acts freely on $\widetilde{G}/\widetilde{\Gamma}$: its stabilizer is precisely the top lattice. On finitely many local trivialisations of this torus bundle, take the character (6) times a smooth base bump. Choose the bumps so that their supports cover the compact base and they vanish near the chart boundaries. The vector of these functions has no common zero; divide by its norm. This gives a globally Lipschitz unit vector of the required frequency, with a fixed finite dimension. It is the partition-of-unity construction underlying published (6.3).

For the coefficients of (1), define

$$\widetilde{g}(z) = \prod_{0 < a \leq d} \exp\left(a! \sum_{w(J)=a} z_J X_a \epsilon_J \right), \qquad z_J = \prod_{j \in J} z_j, \qquad \widetilde{\chi}(z) = \widetilde{F}(\widetilde{g}(z)\widetilde{\Gamma}). \tag{7}$$

Each factor is polynomial for (5). To see this without commuting distinct coefficients, fix a; the summands in its exponent commute because their Lie coefficients are the same X_a, so (7)'s factor is a product of the individual monomial factors. Then use product closure. Thus $\widetilde{\chi}$ has multidegree at most $(1, \dots, 1)$. Let Δn repeat n_i throughout B_i.

Lemma 4 (Raw diagonal identity)**.** *The diagonal $\widetilde{\chi}(\Delta n)$ is equivalent at multidegree d to $\chi(n)^{\otimes C}$. In particular for $k = 1$ the multiplier is $d!$.*

Proof. On the positive Rees algebra $\mathcal{R}$ of Section 2, set

$$T(X t^a) = \left(X, a! \sum_{w(J)=a} X \epsilon_J \right) \in \mathfrak{g} \oplus \widetilde{\mathfrak{g}}. \tag{8}$$

This is a rational Lie map. Indeed for each U with $w(U) = a + b$ the number of ordered disjoint decompositions $U = J \sqcup L$ with $w(J) = a, w(L) = b$ is $\prod_i \binom{a_i + b_i}{a_i}$. Multiplying by $a!b!$ gives $(a+b)!$. Thus $[T(Xt^a), T(Yt^b)] = T([X, Y]t^{a+b})$ exactly. If $a + b$ is outside the support, both sides vanish by $[\mathfrak{g}_a, \mathfrak{g}_b] \subseteq \mathfrak{g}_{a+b} = 0$.

Define $\mathfrak{k}_b = T(\bigoplus_{a \geq b, a \neq 0} \mathfrak{g}_a t^a)$, $K_b = \exp \mathfrak{k}_b$. As before these give a descending, rational, taut filtration on $K_0 \subseteq G \times \widetilde{G}$, with lattice $\Lambda = K_0 \cap (\Gamma \times \widetilde{\Gamma})$. They give the actual comparison orbit

$$(g(n), \widetilde{g}(\Delta n)) = \prod_a \exp\left(n^a T(X_a t^a)\right). \tag{9}$$

No asserted equality of top weights is being used to generate a non-descending filtration: every defining sum includes all $a \geq b$. The top is exactly

$$K_d = \{(\exp X, \exp(CX\epsilon_{[D]})) : X \in \mathfrak{g}_d\}. \tag{10}$$

It is rational and central. The function on the comparison nilmanifold is

$$\mathcal{F}(x, y) = F(x)^{\otimes C} \otimes \overline{\widetilde{F}(y)}. \tag{11}$$

Its two vertical phases on (10) are $C\eta(\exp X)$ and $-\widetilde{\eta}(\exp(CX\epsilon_{[D]})) = -C\eta(\exp X)$. Therefore (11) descends to a Lipschitz function F' on K_0/K_d with the projected lattice Λ' and quotient filtration. Writing k'_a for the projected $\exp T(X_a t^a)$ gives the corrected identity

$$\chi(n)^{\otimes C} \otimes \overline{\widetilde{\chi}(\Delta n)} = F'\left(\prod_a (k'_a)^{n^a} \Lambda'\right). \tag{12}$$

The right side has top group zero, so it has support strictly below d. For $k = 1$, (11) contains $F^{\otimes d!}$ and (12) contains $\chi^{\otimes d!}$. Dropping either factor would leave the uncancelled vertical phase $(1 - d!)\eta$ and would not prove the claimed equivalence.

The diagonal itself is a degree-d nilcharacter: coarsen (5) by taking the direct sum of components with $w(J) \geq a$ at index a. This is descending, rational and taut, its top is still the $J = [D]$ component, and (7) on the diagonal is polynomial for it. This verifies the exact symbol domain used in (12), without assuming that arbitrary linear maps preserve arbitrary multidegree filtrations. □

4 Literal symmetry and division before construction

The map $\widetilde{g}$ is equivariant under block permutations: $\widetilde{g}(\sigma z) = \sigma \widetilde{g}(z)$ after fixing the compatible action convention. An arbitrary $\widetilde{F}$ need not be invariant, so equivariance alone does not prove that $\widetilde{\chi}$ is symmetric. We supply the missing transition explicitly.

Let V be the finite-dimensional Hermitian target of $\widetilde{F}$ and $m = |P| = C$. Regard $\operatorname{Sym}^m V$ as the symmetric subspace of $V^{\otimes m}$ with its induced norm, and write $\odot$ for symmetrised tensor product. On the compact nilmanifold put

$$B(x) = \bigodot_{\sigma \in P} \widetilde{F}(\sigma x), \qquad F_{\mathrm{sym}}(x) = B(x)/\|B(x)\|. \tag{13}$$

The product of nonzero vectors in the symmetric algebra is nonzero: after a choice of basis this algebra is a polynomial ring over $\mathbb{C}$, an integral domain. Hence $\|B(x)\| > 0$ everywhere. Continuity and compactness give a positive minimum, so (13) is Lipschitz. Reindexing the finite product proves exact P-invariance. Since P fixes the top group, (13) has vertical frequency $m\widetilde{\eta}$. Consequently

$$\chi_{\mathrm{sym}}(z) = F_{\mathrm{sym}}(\widetilde{g}(z)\widetilde{\Gamma}) \tag{14}$$

is a literally block-symmetric multidegree-$(1, \ldots, 1)$ nilcharacter. On a block diagonal, $\sigma\widetilde{g}(\Delta n) = \widetilde{g}(\Delta n)$. All factors in (13) then equal the same unit vector v, and $v^{\odot m} = v^{\otimes m}$ has norm one. Thus the diagonal of (14) is exactly $\widetilde{\chi}(\Delta n)^{\otimes m}$ in the symmetric subspace, not merely a proportional vector with an uncontrolled scalar. Lemma 4 therefore gives

$$[\chi_{\mathrm{sym}}(\Delta n)] = Cm[\chi(n)]. \tag{15}$$

Theorem 5 (E.9 and E.10 with a symmetric output)**.** *For a multidegree-d nilcharacter χ there is a literally block-symmetric multidegree-$(1,\dots,1)$ nilcharacter $M(\chi)$ whose block diagonal is equivalent to χ. The one-variable case is the asserted symmetric E.9 output. The assertion holds on any given limit subset after restriction.*

Proof. If $d \neq 0$, first use Corollary 3 with $Q = Cm = C^2$ to choose the specified representative ρ with $[\chi] = Cm[\rho]$. Apply (4)–(14) to this ρ, including its actual polynomial coefficients, lattice, function and vertical frequency. Equation (15) now identifies its output diagonal with $[\chi]$. We do not divide the already symmetrised function and assume that its symmetry survives. If $C = 1$, all constructions and division remain valid. Coordinates with $d_i = 0$ have empty blocks; the normalised polynomial is independent of them. If all $d_i = 0$, the original sequence is a constant unit vector and that vector on the zero-variable domain gives the assertion exactly. □

5 The exact input to Section 13

Let $s \geq 2$ be fixed and let $\chi(h, n)$ be the multidegree-$(1, s-1)$ nilcharacter occurring immediately before published (13.4). First choose ρ with $[\chi] = s[\rho]$ by Corollary 3, in that multidegree symbol group. Then set $\widetilde{\chi} = M(\rho)$ from Theorem 5. It has s variables, multidegree $(1,\dots,1)$ and literal symmetry in its last $s-1$ variables. Its actual diagonal gives

$$[\chi(h,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)} = s[\widetilde{\chi}(h,n,\dots,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)}. \tag{16}$$

Here is the comparison of symbol groups in (16). For support $a \leq (1, s-1)$, give the same group the degree filtration generated by the G_a with $|a| \geq j$. Commutators satisfy the index law, rationality follows from the rational generated-subgroup lemma, and the orbit is polynomial by its monomial expansion. At total weight s the only possible index is $(1, s-1)$, so deleting that top index leaves degree at most $s-1$. Thus both multidegree equivalences used above become degree-s equivalences, with the same functions and the same input (h, n). This proves precisely (13.4), including its multiplier s.

The sequence $\Theta(n) = \widetilde{\chi}(n,\dots,n)$ is a degree-at-most-s nilcharacter by the same coarsening with the binary cube. Fixed linear pullbacks $n \mapsto n+h$ preserve that degree filtration and preserve lower-degree equivalence; one may either use the monomial expansion or published Corollary B.5 and E.8(vi) in its degree-only form. No full symmetry in all s variables is asserted here. Section 13's later symmetrisation argument is exactly what must address the first variable. Its uses of multilinearity require the separately proved replacement for Lemma 13.2; that result is not inferred from (16). Likewise the subsequent bias and density arguments remain separate.

6 Uniformity, limit interpretation, and audit boundary

All nilpotent groups, rational subspaces, finite tensor dimensions, integer powers and permutation groups above are fixed by the original geometric data, d, k, and the specified standard divisibility/scaling integers. They do not depend on orbit coefficients or interval length. The bounded-basepoint normalisation supplied by [2] has a uniform Lipschitz constant. Subgroup restrictions, rational quotients and metrics are fixed compact geometric maps. The lattice and the test function in (6)–(7) are selected once from those fixed data. The positive minimum in (13) is for that fixed test function on that fixed compact space, not an infimum over a varying family of independently chosen functions. For a varying original test function with bounded Lipschitz norm, the raw comparison (11) varies, but the auxiliary $\widetilde{F}$ and (13) can still be fixed because their frequency and group are fixed.

Every coefficient change is a finite BCH expression or a rational scalar multiple in a fixed Lie algebra. Apply it coordinatewise to a representing sequence to obtain a limit map. The finite formulas, exact identities, positive fixed lower bound in (13), and uniform Lipschitz estimates pass to the natural ultrapowers used in published Appendix A. Where geometric data range over a standard finite list, perform this construction for each member of that list. No selection depending on an unbounded interval length or uncontrolled pointwise choice is introduced. All vertical identities here have that natural-ultrapower meaning: they hold for every limit point and every limit top-group element, by transfer of the exact ordinary identity. An identity tested only on standard points would not suffice; no such restriction is used here.

The affected-use audit in this unit covers published E.8(v),(vii), E.9, E.10, and (13.4) through the definition of Θ. It also checks the literal symmetry used for the last $s-1$ variables at those sites. It does not assert completeness of all corrections to GTZ. The arbitrary-test-function symmetry transition and the factorial omissions are preserved as findings with proposed repairs awaiting independent mathematical review.

References

[1] B. Green, T. Tao and T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$-norm, Annals of Mathematics 176 (2012), 1231–1372. DOI: 10.4007/annals.2012.176.2.11. Authoritative final PDF pinned in contracts.

[2] Project proof, Tautification and the shifted nilcharacter construction, project author theta, tautification-003, Theorem 1 and Corollary 2. Pending independent review.

[3] Project proof, local Lemma 13.2, author theta, lemma13-2-002, Section 2. Pending independent review. Exact version pinned in contracts.

The plain additive-quadruple implication on a finite interval

Author proposal: /root/astra_v14_baseline_theta

7 November 2026

Scope. This local proof supplies the exact plain correlation statement of GTZ Proposition 8.4, p. 1277, needed by the Section 13 join. It does not use Proposition 8.3 or infer plain bias from lower-degree bias. The argument is the elementary two-step Cauchy–Schwarz pattern; the finite domains and constants are written explicitly. No source error in Proposition 8.4 is asserted. This author packet requires independent review.

1 Precise finite assertion

Put $J = \{1, \dots, N\}$, $I = \{-N, \dots, N\}$, and suppose $H \subseteq I$ has $|H| \geq \alpha |I|$ for fixed $\alpha > 0$. Let $f : J \to \mathbb{C}$ be 1-bounded and extend it by zero to $\mathbb{Z}$. For each $h \in H$ let $\chi_h : \mathbb{Z} \to \mathbb{C}$ be globally bounded by a fixed $B > 0$, and assume

$$\left| \mathbb{E}_{n \in J} f(n+h)\overline{f(n)}\chi_h(n) \right| \geq \delta > 0.$$

The values outside J are part of the data and are not truncated.

Theorem 1. *Write $a = \alpha\delta/3$ and $\eta = a^4/4$. There are at least*

$$\frac{\eta}{2B^4}(4N+1)(2N+1)^2$$

distinct additive quadruples $(h_1, h_2, h_3, h_4) \in H^4$, $h_1 + h_2 = h_3 + h_4$, for which

$$\left| \mathbb{E}_{n \in J} \chi_{h_1}(n)\chi_{h_2}(n + h_1 - h_4)\overline{\chi_{h_3}(n)\chi_{h_4}(n + h_1 - h_4)} \right| \geq \eta/2.$$

In particular their number is a positive fixed multiple of N^3.

2 Two exact Cauchy–Schwarz steps

For each $h \in H$ choose a scalar c_h of modulus one so the given average multiplied by c_h is real and at least δ. Set $u_h = c_h\chi_h$ on H, and $u_h = 0$ for every $h \notin H$. The functions remain defined on all integers. Then

$$\mathbb{E}_{h \in I}\mathbb{E}_{n \in J} f(n+h)\overline{f(n)}u_h(n) \geq \alpha\delta.$$

Let $M = J + I = \{1 - N, \dots, 2N\}$, so $|M| = 3N$. Set $m = n + h$. The preceding average is exactly

$$3\mathbb{E}_{m \in M} f(m)\mathbb{E}_{h \in I}\overline{f(m-h)}u_h(m-h).$$

Terms with $m - h \notin J$ vanish because f was zero-extended. Cauchy–Schwarz in m, using $|f| \leq 1$, therefore yields

$$\mathbb{E}_{m \in M} \left| \mathbb{E}_{h \in I}\overline{f(m-h)}u_h(m-h) \right|^2 \geq a^2. \tag{1}$$

Expand the square and put $h' = h + k$, $n = m - h$. Let $K = \{-2N, \dots, 2N\}$. The expanded left side equals

$$\frac{N}{|M|}\frac{|K|}{|I|}\mathbb{E}_{k\in K}\mathbb{E}_{n\in J}\overline{f(n)}f(n-k)T(k,n), \qquad T(k,n) = \mathbb{E}_{h\in I}u_h(n)\overline{u_{h+k}(n-k)}.$$

This is an equality with the following domain checks. The factor $\overline{f(n)}$ forces $n \in J$, and then $m = n + h \in M$ automatically for $h \in I$. The condition $h + k \in I$ is supplied by $u_{h+k} = 0$ outside $H \subseteq I$. Thus no additional cutoff in $n-k$ is inserted into T. Its values there are legitimate because the original functions are globally defined. The remaining factor $f(n - k)$ stays in the bounded weight.

The prefactor is positive and at most two. By (1), the absolute value of the displayed average without prefactor is at least $a^2/2$. Cauchy–Schwarz in (k, n), dropping only the bounded weight $\overline{f(n)}f(n - k)$ by an inequality, gives

$$\mathbb{E}_{k\in K}\mathbb{E}_{n\in J}|T(k,n)|^2 \geq a^4/4 = \eta.$$

Expand this square with independent $h, h' \in I$. It is the average over $(k, h, h') \in K \times I^2$ of

$$C(k,h,h') = \mathbb{E}_{n\in J}u_h(n)u_{h'+k}(n-k)\overline{u_{h'}(n)u_{h+k}(n-k)}.$$

The average is real and at least η, while $|C| \leq B^4$. Consequently at least a fraction $\eta/(2B^4)$ of triples have $|C| \geq \eta/2$. Indeed the complementary contribution to the average of $|C|$ is at most $\eta/2$, and the contribution of the selected fraction is at most that fraction times B^4.

Such a triple has $h, h', h + k, h' + k \in H$, since otherwise $C = 0$. Set

$$(h_1, h_2, h_3, h_4) = (h, h' + k, h', h + k).$$

It is an additive quadruple and $h_1 - h_4 = -k$. The map is injective, with inverse $h = h_1$, $h' = h_3$, $k = h_4 - h_1$. The factors c_h contribute only a constant phase of modulus one to C, so they do not change its absolute value. This proves the theorem, including the exact unweighted average on J.

3 Vector families, internal limits and actual consumer

For vector-valued χ_h of fixed finite dimension D, a positive Euclidean correlation implies that some component has magnitude at least $\delta/\sqrt{D}$. Finite pigeonhole retains a component on a subfamily of density at least α/D. Apply the scalar theorem there. The scalar output is a component of the full tensor product, so it implies plain vector bias. A finite list of dimensions can first be padded into one fixed dimension. For a bounded f with a fixed bound other than one, divide it by that bound, changing δ by a fixed factor only.

Apply the finite assertion to representatives of an internal family, with uniform standard bounds, density and correlation. Choices of components and phases can be made for each finite family and then passed to the ultralimit. All constants above are positive standard numbers, so the selected quadruples have positive standard density on the N^3 scale and the correlation is positive by a standard amount. This is exactly the plain Proposition 8.4 output.

In Section 13 the input to this theorem is the entire bounded product that includes the selected lower nilcharacter. Keeping it inside χ_h throughout the calculation preserves that lower information. The theorem requires no degree bound and introduces no lower-degree test factor. It therefore supplies the stronger plain statement that cannot be replaced merely by the degree-biased LP03 output.

The bounded application audit checks the original J, the enlarged $M = J + I$, the parameter range K, the exact inverse reindexing and the global domain of χ_h. A possible artificial cutoff $1_J(n-k)$ is unnecessary: $f(n-k)$ is removed by the second Cauchy–Schwarz inequality, not by claiming an equality after erasing its support. This resolves the domain question without classifying the published Proposition 8.4 as false or defective. The modified Proposition 7.3, its nilsequence-specific factorization and the remaining Section 13 hypotheses are outside this theorem's scope.

The Section 13 correlation and symmetry join

Author proposal: /root/astra_v14_baseline_compiler

7 November 2026

Status. This is a separate author proposal following LP05-001. It does not alter that frozen proof or claim a complete inverse theorem. Its project premises and unchanged published inputs are explicit below. Mathematical acceptance, related-use completeness and incorporation require independent decisions. The signed identities here retain the actual unconjugated α in published (13.10).

1 Target and exact inputs

Fix a standard integer $s \geq 3$ and an unbounded limit integer N. Write $I_t = [-tN, tN] \cap {}^*\mathbb{Z}$, $J = [N]$, $r = s-1$ and $m = s-2$. All geometric data and finite dimensions are standard. Vertical identities hold for every limit point and every limit top-group element under the natural ultrapower extension, as in the project tautification and LP13 units. Tensor products have their Hilbert tensor norms. Every nilcharacter is unit-vector-valued; a scalar coordinate need not have modulus one.

Theorem 1 (Theorem 7.4 on the consumed domain)**.** *Let f be bounded on J, extended by zero outside J, and let $H \subseteq I_1$ be a dense limit subset. Suppose that $\chi(h,n)$ is a globally represented multidegree-$(1, s-1)$ nilcharacter and $\Delta_h f$ has degree-at-most-$(s-2)$ correlation with $\chi(h,\cdot)$ on J for every $h \in H$, with GTZ's correlation convention. Then there are a globally represented degree-at-most-s unit nilcharacter Θ and a bounded globally represented nilsequence $\Psi(h,n)$ supported on*

$$\mathcal{J} = \{(i,j) : i + j \leq s-1\} \cup \{(i, s-i) : 2 \leq i \leq s\} \tag{1}$$

such that, on $I_1 \times J$, each component of $\chi(h,n)$ is a bounded finite linear combination of components of

$$\Theta(n+h) \otimes \overline{\Theta(n)} \otimes \Psi(h,n). \tag{2}$$

All families and bounds are standard in the limit sense, independently of the interval length. No effective numerical constants are asserted.

We use the following exact premises. The project LP05-001 supplies a unit nilcharacter $\widetilde{\chi}$ of cube multidegree $(1, \ldots, 1)$, literally symmetric in its last r variables, with

$$[\chi(h,n)]_{\mathrm{Symb}_s({}^*\mathbb{Z}^2)} = s[\widetilde{\chi}(h,n,\ldots,n)]_{\mathrm{Symb}_s({}^*\mathbb{Z}^2)}. \tag{3}$$

The project LP13 theorem supplies multilinearity of the symbol of $\widetilde{\chi}$ under fixed integer linear substitutions, in each slot. The project tautification-003 Theorem 3 supplies the uniform shift tensor of a positive-degree nilcharacter, lowering degree by one in the ordinary degree case. It applies to limit shifts and limit families.

The unchanged published inputs are the definitions and elementary symbol arithmetic E.3, E.7 and E.8(i)–(iii),(vi); nilcharacter approximation and selection E.5/E.6 and Appendix A; Proposition

8.4 with its plain-correlation hypothesis; the converse Proposition 5.6; and E.11/E.13. Their exact source applications are recorded separately. In particular Proposition 8.4 is a distinct input. The weaker degree-$(s-2)$ biased-quadruple conclusion of Proposition 7.3 alone would not retain the lower factors needed here. No corrected assertion is supplied by citing the unpublished erratum.

For an ordered list of linear forms define

$$T(L_1; L_2, \dots, L_s) = [\widetilde{\chi}(L_1, L_2, \dots, L_s)]_{\mathrm{Symb}_s}.$$

The semicolon separates the distinguished first slot. We use the notation n^j in a slot list for j repeated entries, not a power of n. All formulas between such symbols are on the full indicated limit lattice until a box is explicitly named.

2 Plain correlation and retention of the lower factors

By E.5/E.6 in the ordinary one-dimensional degree ordering and Appendix A selection, after restricting to a still dense internal subset of H, choose degree-at-most-$(s-2)$ unit nilcharacters φ_h so that $\Delta_h f$ plainly correlates with $\chi(h,\cdot)\otimes\varphi_h$. The family is internal, with a standard finite geometry list, bounded dimensions and common Lipschitz bounds. Here the selection is into the countable increasing complexity presentation of the nilcharacter class. An internal image contained in that union is contained in a standard stage by Appendix A compactness; otherwise nested nonempty internal sets of parameters outside successive stages give an image outside the union. Positive correlations can likewise be made uniform by A.6. Pad targets by zero coordinates to one fixed dimension. Outside the retained set use a fixed constant unit nilcharacter, and keep all parameter indicators separately. This preserves the unit norm needed for the identities below. Conjugate the selected lower nilcharacter if needed to match the published correlation convention. A finite pigeonhole also fixes a scalar coordinate of $\chi(h,\cdot)\otimes\varphi_h$ retaining plain correlation on a dense set. Thus the scalar version of Proposition 8.4 suffices; the full unit vectors remain available behind these coordinates.

Apply published Proposition 8.4 to this actual family. Its hypotheses are plain correlation on J, internal dependence, bounded functions and a dense subset of I_1; all hold. Its conclusion gives bias for many additive quadruples (h_1,h_2,h_3,h_4) in that subset, with tensor signs $+,+,-,-$ and the shift $n+h_1-h_4$ on the second and fourth factors. Parameterise

$$(h_1,h_2,h_3,h_4) = (h_0+a, h_0+b, h_0+a+b, h_0).$$

This is an injective correspondence with the retained quadruples. $h_0 \in I_1$ and $a,b \in I_2$. Pigeonhole in h_0 to obtain one fixed h_0 and a dense set of pairs $(a,b) \in I_2^2$. Define the global unit nilcharacter

$$\tau(a,b,n) = \chi(h_0+a,n)\otimes\chi(h_0+b,n+a)\otimes\overline{\chi(h_0+a+b,n)}\otimes\overline{\chi(h_0,n+a)}. \tag{4}$$

It has total degree at most s, not degree three for general s.

For any coordinate of a unit nilcharacter v, the exact identity

$$v_j(n+a) = \sum_k (v_j(n+a)\overline{v_k(n)})v_k(n) \tag{5}$$

follows from $\sum_k |v_k(n)|^2 = 1$. The bracket is a coordinate of the shift tensor. Apply the project shift theorem to φ_{h_0+b} and φ_{h_0}, with shift a. These brackets have degree at most $s-3$, uniformly in all parameters. Conjugate (5) when the shifted factor is conjugated. Thus the original bias implies degree-at-most-$(s-3)$ bias of

$$\tau(a,b,n)\otimes\varphi_{h_0+a}(n)\otimes\varphi_{h_0+b}(n)\otimes\overline{\varphi_{h_0+a+b}(n)}\otimes\overline{\varphi_{h_0}(n)}. \tag{6}$$

Selection gives internal scalar degree-at-most-$(s-3)$ tests with common bounds and finite geometries. A standard finite choice of coordinates retains a positive density of the pairs. Phase-align their scalar averages by a bounded parameter scalar, and set the test equal to zero on other pairs. These factors are constant in n, so they have degree zero. We obtain fixed scalar coordinates t of τ and u, v, w, z of the displayed φ factors with

$$\left|\mathbb{E}_{a,b\in I_2,n\in J} t(a,b,n)u_a(n)v_b(n)\overline{w_{a+b}(n)z(n)}p_{a,b}(n)\right| \gg 1. \tag{7}$$

The scalar $p_{a,b}$ is a uniformly bounded internal lower-degree family. Finite coordinate selection, rather than division by a coordinate, justifies this passage even if individual coordinates vanish.

3 The three Cauchy–Schwarz eliminations

Every average in this section has $n \in J$. All bounds are standard. Applying Cauchy–Schwarz in (a, n) to remove the bounded factor $u_a\overline{z}$ in (7), and expanding the squared inner average, gives

$$\left|\mathbb{E}_{a,b,b'\in I_2,n\in J} t(a,b,n)\overline{t(a,b',n)}v_b(n)\overline{v_{b'}(n)w_{a+b}(n)}w_{a+b'}(n)p_{a,b}(n)\overline{p_{a,b'}(n)}\right| \gg 1. \tag{8}$$

Put $c = a + b + b'$. Keep $b, b' \in I_2$, take $c \in I_6$, and retain $\mathbf{1}_{c-b-b'\in I_2}$. This exact indicator is independent of n. The normalized average changes by the factor $|I_2|/|I_6|$, comparable to one. No three-term sum of elements of I_2 is asserted to lie in I_3. Pigeonhole in c to fix $c_0 \in I_6$. Set

$$\alpha(b,b',n) = \tau(c_0-b-b',b,n)\otimes\overline{\tau(c_0-b-b',b',n)}, \qquad V_b(n) = v_b(n)w_{c_0-b}(n). \tag{9}$$

In particular the second factor of V_b is not conjugated: its sign follows from the last two w factors in (8). Absorb the parameter cutoff into the scalar lower test. Equation (8) becomes, for a fixed coordinate a_0 of the actual α,

$$\left|\mathbb{E}_{b,b'\in I_2,n\in J} a_0(b,b',n)V_b(n)\overline{V_{b'}(n)}q_{b,b'}(n)\right| \gg 1. \tag{10}$$

The functions V_b need only be bounded; no inverse or unit norm for them is used. Cauchy–Schwarz in (b', n) removes $\overline{V_{b'}}$ and produces the product $a_0(b_1,b',n)\overline{a_0(b_2,b',n)}V_{b_1}\overline{V_{b_2}}$ with the product of the corresponding q tests. Cauchy–Schwarz in (b_1, b_2, n) removes the remaining V factors. Its expanded four-term product is

$$a_0(b_1,b_1',n)\overline{a_0(b_2,b_1',n)a_0(b_1,b_2',n)}a_0(b_2,b_2',n). \tag{11}$$

Its scalar test is the product of four q tests with the same signs, still degree at most $s-3$, internal and uniformly bounded. Pigeonhole in (b_2, b_2'), calling the fixed values $(b_0, b_0') \in I_2^2$. The resulting expression is a coordinate of the global unit nilcharacter

$$\alpha'(b,b',n) = \alpha(b,b',n)\otimes\overline{\alpha(b_0,b',n)}\otimes\overline{\alpha(b,b_0',n)}\otimes\alpha(b_0,b_0',n). \tag{12}$$

Hence $\alpha'(b,b',\cdot)$ is degree-at-most-$(s-3)$ biased on J for many $(b,b') \in I_2^2$. To see the last assertion quantitatively, if the final average has modulus at least $\delta > 0$ and each inner average is at most B, a proportion at least $\delta/(2B)$ has modulus at least $\delta/2$. All selections and parameter cutoffs above depend internally on the parameters. Products and conjugates do not increase the ordinary nilsequence degree or destroy the fixed finite complexity bounds.

4 Exact symbols and the lower-slot remainder

Constants such as h_0, c_0, b_0, b_0' are limit integers. Give the original multidegree group its total-degree filtration generated by all components whose total weight is *at least* the index, as verified in LP05. Apply the uniform project ordinary-degree shift theorem on the full slot domain to a constant translation in the relevant coordinate. Its difference has degree at most $s-1$, and stays lower-degree under any fixed integer linear pullback. Thus a constant added to a slot can be removed in the degree-s symbol. A slot consisting only of a constant becomes zero. LP13 additivity gives $T(\dots,0,\dots) = 2T(\dots,0,\dots)$, so that symbol is zero. This argument does not choose an implicit subgroup basepoint for a fixed-slot orbit. The coarsening uses $|a| \geq j$; the opposite inequality printed in Example 6.11 is not used.

From (3),(4), linearity in the first slot, and last-block symmetry,

$$\begin{aligned}[\tau(a,b,n)] &= s\big(T(b;(n+a)^r) - T(b;n^r)\big)\\ &= s\sum_{j=1}^{r}\binom{r}{j}T(b;a^j,n^{r-j}).\end{aligned} \tag{13}$$

In the first line $(n+a)^r$ denotes repeated slots. The leading term is $srT(b;a,n^m)$. Let

$$R_\tau(a,b,n) = s\sum_{j=2}^{r}\binom{r}{j}T(b;a^j,n^{r-j}). \tag{14}$$

Every summand has at most $s-3$ slots equal to n. Substitute $a = c_0 - b - b'$ in (9), remove c_0 as just justified, and expand the j copies of $-b-b'$ by LP13. The leading symbol of α is

$$-sr\,T(b-b';b+b',n^m). \tag{15}$$

This has the order of the two distinguished entries dictated by (13). Interchanging them would require an unproved first-slot symmetry and is not permitted. The remainder is the difference of the two substitutions in (14), each fully expanded into entries in $\{b,b',n\}$ and still with at most $s-3$ copies of n.

Apply the mixed difference (12). At the level of degree-s symbols, constant slots vanish; a monomial with no b or no b' cancels. In (15) the pure terms $T(b;b,n^m)$ and $T(b';b',n^m)$ therefore disappear. There is a globally represented unit nilcharacter β such that

$$[\alpha'(b,b',n)] = -sr\big(T(b;b',n^m) - T(b';b,n^m)\big) + [\beta(b,b',n)]. \tag{16}$$

Here $[\beta]$ is an explicit finite integer linear combination of symbols $T(v_1;\dots,v_s)$, with every $v_i \in \{b,b',n\}$ and at most $s-3$ slots equal to n. More precisely, expand (14) after the two substitutions $(a,b) = (-b-b',b)$ and $(-b-b',b')$, subtract the second, then delete the monomials missing b or missing b'. Tensor the corresponding actual pullbacks of $\widetilde{\chi}$ with their signed integer multiplicities. This constructs β and proves (16). Thus the needed remainder assertion is stronger than merely a separate bound on its degree in n.

Define the actual unit nilcharacter

$$\sigma(b,b',h_1,\dots,h_m,n) = \bigotimes_{\omega\in\{0,1\}^m} \mathcal{C}^{|\omega|}\alpha'\left(b,b',n+\sum_{j=1}^{m}\omega_j h_j\right), \tag{17}$$

where $\mathcal{C}$ is conjugation. Every shift has its own h_j. It has total degree at most s by linear pullback and tensor arithmetic. Apply the alternating sum to (16). In a term with fewer than m slots equal

to n, expand each such slot into n and the available h_j. If a subset S of the shift labels is used, its coefficient contains

$$\sum_{\omega:\,\omega_j=1\ (j\in S)} (-1)^{|\omega|} = 0 \quad \text{unless } S=[m].$$

There are too few slots to use all m labels, so every β term vanishes. For the leading term there are exactly $m!$ surviving bijections of labels to slots, each with coefficient $(-1)^m$. Last-block symmetry identifies their values. With the minus sign in (16), the exact result is

$$[\sigma] = (-1)^{s-1}s!\big(T(b;b',h_1,\dots,h_m) - T(b';b,h_1,\dots,h_m)\big). \tag{18}$$

The unused coordinate n on the right is still a coordinate of the same domain. Formula (18) keeps the actual definitions (9),(12),(17). If one instead conjugates α and α' explicitly, its coefficient changes sign; silently using the other object is not an identification.

5 From correlation to a zero symbol on the correct box

By Proposition 5.6, the positive degree-at-most-$(s-3)$ bias obtained after (12) implies a positive $U^m[J]$ norm for a scalar coordinate of α', for many (b,b'). Appendix A uniformity gives a standard positive lower bound on a dense internal set. Equivalently the average of the vector $U^m[J]^{2^m}$ quantities is $\gg 1$. Since the dimension is standard finite, at least one fixed scalar coordinate contributes $\gg 1$. For $s=3$, this is the converse at degree zero and U^1, so no extra one-step test has been inserted.

Expand the interval Gowers norm using its zero-extension definition and the same normalization as GTZ. One can choose the auxiliary cyclic group of order a fixed multiple of N greater than $2(m+1)N$. For a nonzero cube all vertices lie in J, so its differences have unique lifts $h_j \in I_1$, and no lifted sum can wrap around that cyclic group. The normalizing denominator is a positive standard constant multiple of the count of such cubes. Thus the expansion gives a coordinate of (17) with a positive standard average, up to a fixed positive normalizing factor, on the convex polytope

$$\Omega = \{(b,b',h,n) : b,b' \in I_2, \quad 1 \le n+\omega\cdot h \le N \text{ for all } \omega\in\{0,1\}^m\}. \tag{19}$$

In particular $\Omega \subseteq I_8^{s+1}$, and $\left|\mathbb{E}_{x\in I_8^{s+1}}\mathbf{1}_\Omega(x)\sigma(x)\right| \gg 1$. All defining inequalities in (19) are affine linear; their number depends only on s. The selection of biased pairs is used only for a lower bound on the nonnegative norm average, not as an extra arbitrary cutoff in (19).

Published E.11 asks for dense subprogressions of a positive interval. Partition I_8^{s+1} into its finitely many sign orthants and coordinate-zero hyperplanes. Since σ is bounded, the latter have normalized mass $O_s(1/N)$ and contribute $o(1)$. By the triangle inequality one open orthant has positive standard weighted average. For its signs $\varepsilon_i \in \{-1,1\}$, reflect $x_i = \varepsilon_i y_i$. The reflected polytope is convex and the reflected nilcharacter $\sigma_\varepsilon(y) = \sigma(\varepsilon y)$ has degree at most s. Take every progression in E.11 to be the full positive interval $[8N]$. The hypothesis is plain bias, hence bias of degree strictly below s. E.11, with dimension $s+1$, degree $s \ge 3$ and scale $8N$, gives $\sigma_\varepsilon \in \mathrm{Nil}_{<s}(I_8^{s+1})$. Reflecting back gives

$$[\sigma]_{\mathrm{Symb}_s(I_8^{s+1})} = 0. \tag{20}$$

Only one orthant was used to obtain bias; E.11 itself supplies the conclusion on the full symmetric box. No extrapolation from positive points is implicit.

Pull (18),(20) back along the fixed integer linear map

$$(h,n) \longmapsto (b,b',h_1,\dots,h_m,n_{\mathrm{aux}}) = (h,n,n,\dots,n,0).$$

It maps I_8^2 into I_8^{s+1}. Lower-degree nilsequences pull back with the same degree, so

$$s!\big(T(h;n^r) - T(n;h,n^{r-1})\big) = 0 \quad \text{in } \mathrm{Symb}_s(I_8^2). \tag{21}$$

The sign in (18) can be removed because both sides equal zero. Apply E.13 on this symmetric box, with dimension two, degree $s \geq 3$ and the fixed positive integer $s!$. Its torsion-free conclusion gives

$$T(h;n^r) = T(n;h,n^{r-1}) \quad \text{in } \mathrm{Symb}_s(I_8^2). \tag{22}$$

We restrict to $I_1 \times J$ only after this use. E.13 is not being applied directly to an asymmetric rectangle.

6 Exact finite-component factorization

Set $\Theta(n) = \widetilde{\chi}(n,\dots,n)$, the same object as in LP05. It is a unit degree-at-most-s nilcharacter. Expand $[\Theta(n+h)] - [\Theta(n)]$ by LP13. Terms with exactly one h equal

$$T(h;n^r) + rT(n;h,n^{r-1}) = sT(h;n^r) = [\chi(h,n)] \tag{23}$$

on $I_1 \times J$, by (22),(3). Let $R(h,n)$ be the tensor product of the actual pullbacks of $\widetilde{\chi}$ for all subsets of the s slots of cardinality at least two, putting h in those slots and n elsewhere. There is no permutation of the distinguished first slot in this definition. Every factor has multidegree at most $(i, s-i)$ for its subset size i: coarsen the cube filtration by counting the coordinates assigned to h and n, using all weights at least the index. The polynomial orbit, rationality and top frequency are as in the LP05 block coarsening. Their product is therefore supported on the union of these downsets, contained in $\mathcal{J}$.

By the symbol equality just proved, the unit-vector expression

$$W(h,n) = \chi(h,n) \otimes \overline{\Theta(n+h)} \otimes \Theta(n) \otimes R(h,n) \tag{24}$$

is a degree-at-most-$(s-1)$ nilsequence on $I_1 \times J$. By the definition of the restricted nilsequence class, it has a global representation whose restriction is (24); no global equality beyond this rectangle is claimed. Such a degree filtration gives multidegree support $i + j \leq s-1$ by assigning the group at (i,j) its total-degree group. Choose that representation and set $\Psi = \overline{R} \otimes W$. It has support in (1) by the product filtration and is uniformly bounded.

Finally tensor (24) with $\Theta(n+h) \otimes \overline{\Theta(n)} \otimes \overline{R}$. Contract the matching pairs using their unit norms. Each contraction is a fixed linear map between fixed finite tensor spaces. The result is exactly χ, so (2) holds component by component with fixed bounded coefficients. This does not divide by any scalar coordinate or replace symbol equality by an unproved scalar equality. It proves Theorem 1.

7 Bounds, source differences and review boundary

All finite coordinate selections, tensor dimensions, permutation counts and parameter boxes depend only on fixed degree and geometric data. Appendix A selection/uniformity supplies the finite family of lower tests before the averaging manipulations. The three Cauchy–Schwarz steps use common

standard bounds, and each pigeonhole retains a standard positive amount. Fixed parameter values may be unbounded limit integers; the project shift theorem supplies uniform constructions for exactly that case. All algebraic identities are finite and transfer coordinatewise, with the all-limit vertical convention retained. E.11 and E.13 have the exact degree, dimension, progression and symmetric-box domains stated above.

The proof retains, rather than erases, the published factorial history. Theorem 7.4 on both pp. 1271 and 1315 prints the stricter hypothesis "$< s-2$-correlates", whereas the preceding Theorem 7.1 supplies degree-at-most-$(s-2)$ correlation. The target here deliberately assumes the latter, and Section 2 proves the required selection and shift step under that weaker hypothesis. The printed Theorem 7.4 is not a premise. In addition to the repeated shift index and alternating-derivative sign in (13.14)–(13.17), the first two slots in printed Lemma 13.4 are interchanged relative to (13.3),(13.10). This changes the sign in (13.13). Formula (18) is the resulting coefficient for the actual unconjugated objects. The published simplification in Lemma 13.3 also retains an a where cancellation requires b; its tail has degree at most $s-3$ in n. The scalar lower-factor bookkeeping and the I_6 substitution above give their own exact formulas. These source discrepancies are not counterexamples to the final zero relation. Their required related uses and current representations must be checked independently before a complete correction claim.

This packet proves a local composition relative to its stated project and published premises. It does not reprove E.11/E.13's equidistribution inputs, Proposition 5.6's published converse sources, Proposition 8.4, the frequency analysis, or the final finite-family inverse-theorem assembly. Those are direct dependencies or separate consumers, not silently discharged here.

Same-step linearization using the increment polynomial group

Project proof draft: /root/astra_v14_baseline_theta

7 November 2026

Status. This is a proposed local replacement for the same-step linearization construction in GTZ Appendix C. It is not an independent acceptance. It uses the bounded normalization and tautification proved in the separately frozen project unit [2], and the published algebraic polynomial-map results specified below. It does not assume GI(s).

1 The required conclusion

Let G/Γ be a connected simply connected nilpotent Lie group modulo a lattice, with a rational degree filtration $G_\bullet$ of degree at most $s \geq 1$. Every G_i is connected and rational. Let a scalar polynomial nilsequence be $F(g(n)\Gamma)$, with F complex-valued, 1-bounded and Lipschitz. The polynomial g is adapted to $G_\bullet$ in the right-difference convention $\partial_u g(n) = g(n+u)g(n)^{-1}$.

Theorem 1. *For this fixed filtered nilmanifold there is a fixed s-step nilmanifold $\widetilde{G}/\widetilde{\Gamma}$ with a fixed smooth metric, and a constant C depending only on the original fixed geometric data, with the following property. For every polynomial g and every such F, there are $a \in \widetilde{G}$ and a 1-bounded Lipschitz function $\widetilde{F} : \widetilde{G}/\widetilde{\Gamma} \longrightarrow \mathbb{C}$ such that*

$$F(g(n)\Gamma) = \widetilde{F}(a^n\widetilde{\Gamma}) \quad (n \in \mathbb{Z}), \qquad \mathrm{Lip}(\widetilde{F}) \leq C\,\mathrm{Lip}(F).$$

Fixed smooth metric comparisons are included in C. The same construction applies to limit polynomial maps and bounded Lipschitz limit functions, with fixed nilmanifold and uniform constants. Consequently a finite-family polynomial inverse theorem of degree s implies the finite-family linear inverse theorem of step s, without changing the input Gowers norm or losing correlation.

If $\mathrm{Lip}(F) = 0$, the constant test function has a constant extension and the displayed bound is understood literally. A point nilmanifold can be used for a constant sequence. Step at most s is the meaning of s-step throughout.

By [2, Theorem 1], restrict to a fixed taut representation with $G_0 = G_1 = G$ and replace the polynomial by h, where $h(0) = 1$. The exact identity is

$$g(0) = q\gamma, \qquad h(n) = q^{-1}g(n)\gamma^{-1}, \qquad F_q(x\Gamma) = F(qx\Gamma), \qquad F_q(h(n)\Gamma) = F(g(n)\Gamma).$$

Here q lies in a fixed compact set and $\mathrm{Lip}(F_q)$ is bounded by a fixed multiple of $\mathrm{Lip}(F)$. The underlying taut filtered nilmanifold is fixed independently of g. It therefore suffices to prove the theorem for this normalized h and F_q. In what follows all geometric data refer to this fixed taut representation.

2 Polynomial groups and their Lie algebras

For $i \geq 0$ put

$$P_i = \mathrm{poly}(\mathbb{Z}_{\mathbb{N}} \longrightarrow G_{\bullet+i}).$$

The zeroth group of the shifted filtration is G_i; it need not be taut. The group operation is pointwise multiplication. By [1, Corollary B.4 and Proposition B.6], these are groups with

$$[P_i, P_j] \subseteq P_{i+j}, \qquad P_i \triangleleft P_0. \tag{1}$$

Those results concern abstract filtered groups and require no equality of the first two groups. Their conditions hold because the shifted groups inherit nesting and the commutator law from $G_\bullet$.

Here is an explicit finite-dimensional description, including the smooth real shift needed below. Let $\mathfrak{g}_i = \log G_i$ and define

$$V_i = \left\{ X(t) = \sum_{j=0}^{s-i} X_j t^j : X_j \in \mathfrak{g}_{j+i} \right\} \quad (0 \leq i \leq s), \qquad V_i = 0 \quad (i > s).$$

Use pointwise Lie bracket. Bracketing coefficients gives

$$[V_i, V_j] \subseteq V_{i+j}. \tag{2}$$

For instance, a bracket of a degree-a coefficient in $\mathfrak{g}_{a+i}$ and a degree-b coefficient in $\mathfrak{g}_{b+j}$ belongs to $\mathfrak{g}_{a+b+i+j}$ and has polynomial degree $a+b$. Thus V_i is a finite-dimensional nilpotent Lie algebra for every i. Its simply connected group is obtained by the terminating BCH formula, or equivalently by pointwise exponentiation.

Lemma 2. *The pointwise exponential identifies that group with P_i. In particular P_i is connected and simply connected, and its smooth topology can be described either by the coefficients of* $\log p(t)$ *or by the Taylor coefficients in [1, Lemma B.9].*

Proof. If $i > s$, the zeroth group of the shifted filtration is trivial, so both P_i and V_i are trivial. Fix $0 \leq i \leq s$ and put $m = s - i$. We use the connectedness of every G_j stated in the setup.

First recall explicitly why the global exponential coordinates used here are smooth. For a connected nilpotent Lie group H, let $Q_a(x) = x^a$, where a is a positive integer. In left tangent trivializations, differentiating the a factors gives

$$DQ_a(x) = \sum_{k=0}^{a-1} \mathrm{Ad}(x)^{-k}.$$

The infinitesimal adjoint action lowers the lower central series. An exponential neighborhood generates H, so $\mathrm{Ad}(x)$ induces the identity on each successive quotient of that series. Consequently $DQ_a(x)$ induces multiplication by a on each quotient and is invertible. The usual local Lie exponential is smooth with derivative the identity at zero. Given $X \in \mathfrak{h}$, choose a so large that X/a lies in this local chart. The identity

$$\exp_H(Y) = Q_a(\exp_H(Y/a))$$

then proves that the exponential has invertible derivative near X. For simply connected nilpotent H, its global bijectivity is the exponential homeomorphism recalled in [1, Section 3]. Its local smooth inverses therefore agree to give a smooth global inverse. This argument uses the local Lie

exponential, differentiation of group operations and the inverse function theorem. It does not infer smoothness from the word *homeomorphism*.

By [1, Proposition B.6], P_i is a group under pointwise multiplication. By [1, Lemma B.9], its elements have the unique increasing-order Taylor expansion

$$p(n) = \prod_{j=0}^{m} u_j^{\binom{n}{j}}, \qquad u_j \in G_{i+j}, \qquad n \in \mathbb{Z}.$$

The coefficients are recovered by the triangular formulas

$$u_0 = p(0), \qquad u_j = \left(\prod_{k<j} u_k^{\binom{j}{k}} \right)^{-1} p(j).$$

The order of the displayed inverse product is part of the formula. Transfer the product manifold structure of $\prod_{j=0}^{m} G_{i+j}$ to P_i using these coordinates. Each evaluation at an integer is smooth. Evaluation of a product or inverse followed by the recovery formulas proves smoothness of the group operations. The recovered coefficients lie in the prescribed embedded subgroups; smoothness into those subgroups follows from their local embedded charts. The same formulas identify this topology with the pointwise topology.

Evaluation at $0, \dots, m$ is an injective group homomorphism $P_i \longrightarrow G^{m+1}$. Hence P_i is nilpotent, with class at most the class of G. Each closed connected subgroup of a simply connected nilpotent group is simply connected. In exponential coordinates it is the exponential of its Lie subalgebra: the finite nilpotent BCH formula makes that exponential image a connected subgroup, and uniqueness of the connected subgroup with that Lie algebra gives the assertion. Thus the product manifold above, and therefore P_i, is connected and simply connected. This step does not assert that P_0 has class at most s.

At the identity, differentiating the Taylor parametrization gives

$$(X_0, \dots, X_m) \longmapsto \left(n \longmapsto \sum_{j=0}^{m} \binom{n}{j} X_j \right), \qquad X_j \in \mathfrak{g}_{i+j}.$$

The binomial-to-monomial matrix is triangular with nonzero diagonal. A contribution to degree b comes only from $j \geq b$, and $\mathfrak{g}_{i+j} \subseteq \mathfrak{g}_{i+b}$. Descending triangular inversion preserves these same coefficient spaces. This identifies the tangent space with exactly V_i. Evaluation maps are smooth homomorphisms, so their derivatives preserve Lie brackets. The finitely many evaluations just used determine every tangent vector. They therefore identify the bracket with the pointwise bracket on V_i.

Uniqueness of one-parameter subgroups now gives

$$\mathrm{ev}_n(\exp_{P_i} X) = \exp_G(X(n)) \qquad (n \in \mathbb{Z}).$$

Apply the smooth-global-exponential argument above to P_i. Its inverse assigns to p exactly the Lie polynomial whose values are $\log_G(p(n))$. Conversely every $X \in V_i$ gives an element of P_i. This proves the pointwise exponential identification and the smooth logarithmic coefficient coordinates. The group law is determined by multiplication in G at the integer arguments. Under the ordinary nilpotent BCH identification it is the pointwise BCH law. The coordinate spaces and finite evaluation maps depend only on the fixed filtration and i. □

For real u, define T^u on V_i by $T^u X(t) = X(t+u)$. Coefficient expansion and nesting show that this preserves V_i. It is a smooth one-parameter group of Lie algebra automorphisms, since the bracket is pointwise. Exponentiating defines a smooth automorphism group on P_i, which at integer u is the ordinary shift of polynomial sequences. Its infinitesimal action is ordinary differentiation, and

$$\frac{d}{dt} V_i \subseteq V_{i+1}. \tag{3}$$

Indeed the coefficient of t^j in $X'(t)$ belongs to $\mathfrak{g}_{j+1+i}$.

3 The lattice and the same-step semidirect product

Define

$$\Lambda_i = \{p \in P_i : p(n) \in \Gamma \text{ for every } n \in \mathbb{Z}\}.$$

It is a subgroup by pointwise multiplication. We give the needed lattice argument to avoid relying on the disputed step estimate in Appendix C.

Lemma 3. *Λ_i is a discrete cocompact subgroup of P_i. Integer shifts preserve Λ_i.*

Proof. It is enough to consider $i \le s$ and put $d = s - i$; for larger i both groups are trivial. Discreteness follows from evaluation at $0, \dots, d$. A lattice-valued polynomial close enough to the identity at these finitely many points equals the identity at all those points, since Γ is discrete. Taylor uniqueness then makes it the identity polynomial.

For cocompactness, use induction to write

$$p = \gamma_j h_j c_j,$$

where $\gamma_j \in \Lambda_i$, c_j ranges over a fixed compact subset of P_i, and h_j is the identity at $0, \dots, j-1$. For $j = 0$ take $\gamma_0 = c_0 = 1$ and $h_0 = p$. Taylor expansion shows $h_j(j) \in G_{j+i}$. Choose a fixed compact left fundamental set for the lattice $G_{j+i} \cap \Gamma$ in G_{j+i} and write $h_j(j) = \lambda_j v_j$, where λ_j is in that lattice and v_j is in the compact set. Define

$$\ell_j(t) = \lambda_j^{\binom{t}{j}}, \qquad v_j^*(t) = v_j^{\binom{t}{j}}.$$

Both are in P_i, by the same one-parameter difference check as above; $\ell_j \in \Lambda_i$ because $\binom{n}{j}$ is an integer for every $n \in \mathbb{Z}$. Now put

$$\gamma_{j+1} = \gamma_j \ell_j, \qquad h_{j+1} = \ell_j^{-1} h_j (v_j^*)^{-1}, \qquad c_{j+1} = v_j^* c_j.$$

These formulas preserve the factorization and make h_{j+1} the identity at $0, \dots, j$. The possible v_j^* form a compact subset of P_i, so the possible c_{j+1} remain in a fixed compact set. At $j = d+1$, Taylor uniqueness gives $h_{d+1} = 1$. Thus $p = \gamma c$ with c in a fixed compact set, which proves left cocompactness; inversion gives right cocompactness. Finally an integer shift preserves both polynomiality and all integer lattice values, and its inverse does also. □

Now use the *increment* group P_1, not P_0, and set

$$\widetilde{G} = P_1 \rtimes_T \mathbb{R}, \qquad (p,u)(q,v) = (p\, T^u q, u+v), \qquad \widetilde{\Gamma} = \Lambda_1 \rtimes_T \mathbb{Z}. \tag{4}$$

The real action just constructed makes $\widetilde{G}$ a connected simply connected Lie group, diffeomorphic to $P_1 \times \mathbb{R}$. Its Lie algebra is $V_1 \rtimes \mathbb{R}\tau$, where $[\tau, X] = X'$. Give this Lie algebra the filtration

$$\mathfrak{l}_0 = \mathfrak{l}_1 = V_1 \oplus \mathbb{R}\tau, \qquad \mathfrak{l}_j = V_j \quad (j \ge 2).$$

Equations (2) and (3) prove $[\mathfrak{l}_i, \mathfrak{l}_j] \subseteq \mathfrak{l}_{i+j}$ for all indices. In particular $\mathfrak{l}_{s+1} = 0$. Thus $\widetilde{G}$ is nilpotent of step at most s. This explicit shifted-filtration calculation is the same-step bound used in the theorem.

The subgroup $\widetilde{\Gamma}$ is discrete, because its clock coordinate is integral and Λ_1 is discrete. It is cocompact as well. Given (p, u), left-multiply by $(1, -\lfloor u \rfloor)$, then left-reduce the resulting polynomial coordinate modulo Λ_1 using Lemma 3. The resulting representative lies in the product of a fixed compact set in P_1 and $[0, 1]$. This proves left cocompactness; inversion proves right cocompactness. The operations used for left reduction belong to $\widetilde{\Gamma}$ by the integer-shift assertion of the lemma.

4 The orbit identity and fixed compact fiber

Let $\theta : \widetilde{G}/\widetilde{\Gamma} \longrightarrow \mathbb{R}/\mathbb{Z}$ be the map induced by the clock coordinate, and set $K = \theta^{-1}(0)$. It is a fixed compact submanifold canonically identified with P_1/Λ_1: every element of the fiber can be written $(p, 0)\widetilde{\Gamma}$, and two such representatives agree precisely when their polynomial coordinates differ on the right by an element of Λ_1. Evaluation at zero gives a smooth map

$$\pi : K \longrightarrow G/\Gamma, \qquad \pi((p, 0)\widetilde{\Gamma}) = p(0)\Gamma. \tag{5}$$

It is well-defined because $\lambda(0) \in \Gamma$ for $\lambda \in \Lambda_1$. Since K is compact, π is Lipschitz for fixed smooth metrics.

The normalized polynomial $h \in P_0$ has $h(0) = 1$. Define its right increment by

$$b(t) = h(t)^{-1} h(t + 1).$$

The left increment $h(t + 1)h(t)^{-1}$ lies in P_1 directly from the iterated-difference definition. Proposition B.6 gives $P_1 \triangleleft P_0$, so conjugating that left increment by $h(t)^{-1}$ shows $b \in P_1$. This verifies the orientation of the increment actually used, rather than substituting a left increment into a right-telescoping product.

Put $a = (b, 1) \in \widetilde{G}$. For every integer n define $b_n(t) = h(t)^{-1} h(t+n)$. The exact cocycle identity

$$b_{n+m}(t) = b_n(t)\, b_m(t + n)$$

and $b_0 = 1$ prove, for positive and negative integers alike,

$$a^n = (b_n, n).$$

Modulo $\widetilde{\Gamma}$, right multiplication by $(1, -n)$ gives $a^n\widetilde{\Gamma} = (b_n, 0)\widetilde{\Gamma} \in K$. Consequently

$$\pi(a^n\widetilde{\Gamma}) = b_n(0)\Gamma = h(n)\Gamma, \qquad n \in \mathbb{Z}. \tag{6}$$

The last equality uses the normalization $h(0) = 1$.

5 Test functions, uniformity and the inverse-theorem join

The function $F_q \circ \pi$ is 1-bounded on K and has a Lipschitz constant bounded by a fixed multiple L of $\mathrm{Lip}(F)$. Extend its real and imaginary parts separately by the explicit McShane formula

$$E_u(x) = \inf_{y \in K} \big(u(y) + L\, d(x, y)\big).$$

This equals u on K and is L-Lipschitz: the triangle inequality proves the Lipschitz assertion, and the original Lipschitz inequality proves equality at points of K. Clip each extension to $[-1, 1]$,

combine them as a complex function, and apply metric projection onto the closed unit disk. This projection Q is 1-Lipschitz: the nearest-point inequalities for the disk give

$$\langle x - Qx, Qy - Qx \rangle \le 0, \qquad \langle y - Qy, Qx - Qy \rangle \le 0,$$

and therefore $\|Qx - Qy\|^2 \le \langle Qx - Qy, x - y \rangle \le \|Qx - Qy\| \|x - y\|$. It also fixes the original values on K. Thus the result $\widetilde{F}$ is 1-bounded, agrees with $F_q \circ \pi$ on K, and is at most $\sqrt{2}L$-Lipschitz. Equation (6) proves the exact equality in Theorem 1.

The nilmanifold, lattice, compact fiber, evaluation map and all metric constants depend only on the fixed original filtered nilmanifold and s. The possibly unbounded polynomial coefficients occur only in the allowed translation element a and in the bounded representative q affecting the test function. They do not alter the geometry or its Lipschitz bound. For limit data the same algebraic formulas and one fixed bounded representative section give limit choices of b and a. Apply the explicit extension formula to each representative test function. Its uniform bound and Lipschitz constant give a bounded Lipschitz limit function on the same standard nilmanifold. The ordinary orbit identities hold at every member of the defining sequences and therefore at every limit integer.

Suppose now that a polynomial inverse theorem supplies, for fixed s and δ, a finite family of rational degree-s filtered nilmanifolds, a fixed Lipschitz bound, and a positive correlation lower bound against 1-bounded test functions. Apply the theorem to each member of that finite family, taking the maximum of its finitely many fixed metric constants. There is still a finite family, now of step at most s with linear orbits, and the correlation is unchanged because the represented sequences are equal pointwise. No comparison of Gowers norms is involved in this join. This proves the final assertion of Theorem 1.

The construction also covers a point input group: P_1 is trivial and the clock group yields a circle with a constant test function, or it may be replaced by a point. For $s = 1$ the increment group consists of constant polynomials in an abelian group and the shift action is trivial, so $\widetilde{G}$ is abelian, as the required bound predicts.

6 Comparison with the failed full-group construction

The original Appendix C used $P_0 \rtimes \mathbb{R}$. For the taut abelian degree-s filtration on $\mathbb{R}$, the commutator of its clock with $p(t) = \frac{1}{2}\binom{t}{s}$, repeated s times, is the nonzero constant $1/2$. That is a commutator of length $s+1$, so that group is not s-step. This adverse example is recorded in the separate finding packet. The present construction uses P_1 and its increment b, reducing the polynomial degree available in the clock action by one. Its Lie filtration above verifies the required step bound for all inputs, not only this example. It changes the construction, while retaining the same-step linear conclusion needed by GI(s).

The finite-family polynomial inverse theorem is still a separate premise. This unit does not prove its rank reduction, symmetrization, or ultralimit-to-finite-family extraction. Those dependencies remain explicit in the theorem assembly.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics **176** (2012), 1231–1372. Only the specified algebraic polynomial-map and Lie-group facts are used; the Appendix C step claim is replaced here.

[2] Project proof, *Tautification and the shifted nilcharacter construction*, draft 003, 7 November 2026. The exact author manifest and independent scope decision are pinned in the accompanying application record.

A normalized local converse for the GI assembly

Author proposal: /root/astra_v14_baseline_theta

7 November 2026

Scope. This author proof, awaiting independent review, derives the precise correlation-to-Gowers-norm implication used by the assembly. It imports the published compact cube constraint, Green–Tao, *Linear equations in primes*, Proposition 11.5, p. 1799, with its definitions in 11.4 and proof in Appendix E. It uses the accepted local increment linearization for polynomial tests. It does not import Proposition 11.2's averaging-domain transition, assume an inverse theorem, or claim to reprove the cube constraint.

1 Available geometric assertion and needed conclusion

Fix an integer $d \geq 1$ and a fixed d-step nilmanifold $M = G/\Gamma$ with a smooth metric. The available cube constraint says there is a compact subset $\Sigma \subseteq M^{\{0,1\}^{d+1}\setminus\{0\}}$ and a continuous map $P : \Sigma \to M$ such that, for every $g \in G$, $x \in M$, $n \in \mathbb{Z}$ and $h \in \mathbb{Z}^{d+1}$, the nonzero vertices $(g^{n+\omega\cdot h}x)_{\omega\neq 0}$ lie in Σ and

$$g^n x = P((g^{n+\omega\cdot h}x)_{\omega\neq 0}).$$

No bound on g or the integers is imposed. In our application G is connected simply connected and at most d-step; these satisfy the published nilmanifold hypothesis.

Theorem 1. *For fixed d, M, L and every $\kappa > 0$ there is $c > 0$ such that every 1-bounded $v : [N] \to \mathbb{C}$ and 1-bounded L-Lipschitz test $F : M \to \mathbb{C}$ satisfying*

$$|\mathbb{E}_{n\in[N]} v(n)F(g^n x)| \geq \kappa$$

obey $\|v\|_{U^{d+1}[N]} \geq c$. The bound is uniform in N, g, x, F. The same conclusion holds for a fixed finite family of rational degree-at-most-d polynomial models, using the exact same-step increment linearization and taking a finite maximum of its bounds.

2 A finite tensor approximation uniform in the test

The algebra generated by coordinate functions that are Lipschitz on M separates points on the compact set Σ, contains constants and is closed under conjugation. Complex Stone–Weierstrass gives, for every fixed continuous F and $\epsilon > 0$, an approximation

$$F(P(z)) = \sum_{a=1}^{A} c_a \prod_{\omega\neq 0} H_{\omega,a}(z_\omega) + R(z), \qquad |R(z)| \leq \epsilon,$$

where the factors are 1-bounded Lipschitz functions. Scaling each factor into the unit disk is absorbed into the finite coefficients c_a; no coefficient is discarded. By Arzela–Ascoli the class

$\|F\|_\infty \le 1$, $\mathrm{Lip}(F) \le L$ is compact in the uniform norm. A finite uniform net, approximated as above, makes A, $\sum_a |c_a|$ and all factor bounds depend only on M, L, ϵ. Choose a net radius $\epsilon/2$ and approximate its members to $\epsilon/2$ to retain the displayed total error.

For each orbit, put $b_{\omega,a}(t) = H_{\omega,a}(g^t x)$. The cube constraint gives

$$F(g^n x) = \sum_a c_a \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h) + O(\epsilon) \tag{1}$$

for every integer n, h, uniformly. The factors are 1-bounded; their dependence on the orbit will no longer matter in the analytic estimate.

3 Normalized smooth averaging and a uniform Fourier bound

Choose once and for all a smooth $\sigma : \mathbb{R} \to [0,1]$ supported on $[-1,2]$ and equal to one on $[0,1]$. Define

$$S_N = \sum_{h \in \mathbb{Z}} \sigma(h/N), \qquad \mu_N(h) = \sigma(h/N)/S_N.$$

Then $N \le S_N \le 3N + 1 \le 4N$, and μ_N is a probability measure with finite support. Average (1) against the product probability measure $\prod_j \mu_N(h_j)$ on all integer h. The error remains at most ϵ, with no missing normalizing scalar. Thus

$$F(g^n x) = \sum_a c_a B_a(n) + O(\epsilon), \quad B_a(n) = S_N^{-(d+1)} \sum_{h \in \mathbb{Z}^{d+1}} \prod_j \sigma(h_j/N) \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h).$$

Take $Q = C_d N$ with a fixed integer $C_d > 10(d+2)$, and identify representatives of $\mathbb{Z}/Q\mathbb{Z}$ with a centered interval. The sampled cutoff $w_N(h) = \sigma(h/N)$ has support strictly inside that interval. Write its discrete Fourier expansion as $w_N(h) = \sum_r \widehat{w}_N(r) e(rh/Q)$ with normalized coefficients. There is a bound

$$\sum_{r \in \mathbb{Z}/Q\mathbb{Z}} |\widehat{w}_N(r)| \le C'_d \tag{2}$$

independent of N. To prove it, the zero coefficient and every coefficient are at most $S_N/Q = O_d(1)$. For $r \neq 0$, summation by parts twice gives

$$|1 - e(r/Q)|^2 |\widehat{w}_N(r)| \le Q^{-1} \sum_h |\Delta^2 w_N(h)| \le C/(NQ).$$

The final bound follows from the second derivative of the fixed smooth cutoff, with only $O(N)$ possibly nonzero terms. For the distance $r_* \in [1, Q/2]$ to zero modulo Q, $|1 - e(r/Q)| \ge 4r_*/Q$. Thus $|\widehat{w}_N(r)| \le C_d/r_*^2$ and summation proves (2). Smooth zero extension of σ ensures that the same finite-difference bound holds at its support boundaries.

4 The exact Gowers estimate

Zero-extend v to $\mathbb{Z}/Q\mathbb{Z}$. Extend each bounded integer function $b_{\omega,a}$ to the cyclic group so that it agrees on all integer arguments that occur when $n \in [N]$ and $h_j \in [-N, 2N]$. Those arguments lie between $-(d+1)N$ and $(2d+3)N$; the choice of C_d places the whole interval strictly inside the centered representative range. There is therefore no conflicting cyclic value. The normalized correlation with B_a is exactly

$$\frac{Q}{N} \left(\frac{Q}{S_N} \right)^{d+1} \mathbb{E}_{n,h \in \mathbb{Z}/Q\mathbb{Z}} v(n) \prod_j w_N(h_j) \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h).$$

The prefactor is bounded above by a constant depending only on d. This identity comes from an all-integer smooth average; no hard box average is replaced by a larger summation domain.

Expand the product of cutoffs by Fourier series. The sum of absolute values of its coefficients is at most $(C'_d)^{d+1}$. In each term, the factor $e(\sum_j r_j h_j/Q)$ can be absorbed into the nonzero vertex functions. For each j, choose an index $k \neq j$, which exists since $d+1 \geq 2$, and use $h_j = (n+h_j+h_k)-(n+h_k)$. Multiply the functions at vertices e_j+e_k and e_k by the corresponding opposite linear phases. Their absolute bounds remain one. The Gowers–Cauchy–Schwarz inequality for the full $(d+1)$-cube gives the absolute bound $\|v\|_{U^{d+1}(\mathbb{Z}/Q\mathbb{Z})}$ for every such term. Complex conjugations are absorbed into the vertex functions, leaving the norm and absolute bounds unchanged. Hence

$$|\mathbb{E}_{[N]} v B_a| \leq K_d \|v\|_{U^{d+1}(\mathbb{Z}/Q\mathbb{Z})} \leq K_d \|v\|_{U^{d+1}[N]}.$$

The last inequality uses the prescribed normalized interval norm: the cyclic norm is the interval norm multiplied by the cyclic norm of the interval indicator, which is at most one. If necessary enlarge C_d also to exceed 2^{d+1} so this single ambient group realizes that interval normalization.

Combining the finite tensor approximation with this estimate yields

$$|\mathbb{E}_{[N]} v(n) F(g^n x)| \leq K_d \Big(\sum_a |c_a| \Big) \|v\|_{U^{d+1}[N]} + \epsilon.$$

Choose $\epsilon < \kappa/2$. This proves Theorem 1 with a positive bound depending only on d, M, L, κ. A fixed bound other than one for v or F is handled by dividing by that bound first. This proof is complex-valued throughout; it does not need a real-valued dual-norm convention.

5 Polynomial input and source comparison

For the actual assembly input, a scalar polynomial nilsequence has degree $d = s-2 \geq 1$ on fixed rational geometry. The accepted increment linearization001 represents it exactly on a fixed group of step at most d, with a fixed Lipschitz multiplier and unchanged norm bound. Applying the theorem there and taking finite maxima over a finite family gives precisely the positive normalized $U^{s-1}[N]$ norm required by the last induction step. This is a converse implication and has no inverse-theorem premise.

The literal published proof of 11.2 on p. 1801 defines an average over $h \in [N]^{d+1}$ with $\sigma(h_j/N)$, while $\sigma = 1$ on $[0,1]$. Its later passage from (11.6) to (11.7) widens the shift domain to a cyclic group with the same cutoff supported on $[-1,2]$. That changes the set of summands. The local proof above instead starts with the normalized all-integer smooth average and tracks its exact cyclic prefactor. Whether an earlier source correction already supplied this interpretation has not been established. The observation is a bounded proof-transition issue, not a counterexample to Proposition 11.2 or the converse theorem.

The related-use scope is the normalization of the 11.2 average, its dual-norm estimate, Corollary 11.6, and the present GI converse application. Published 11.5 remains an explicit unchanged supplier whose source-level application must be independently accepted. No claim of whole-source acceptance of Appendix E or of the full GI assembly follows from this local analytic proof.